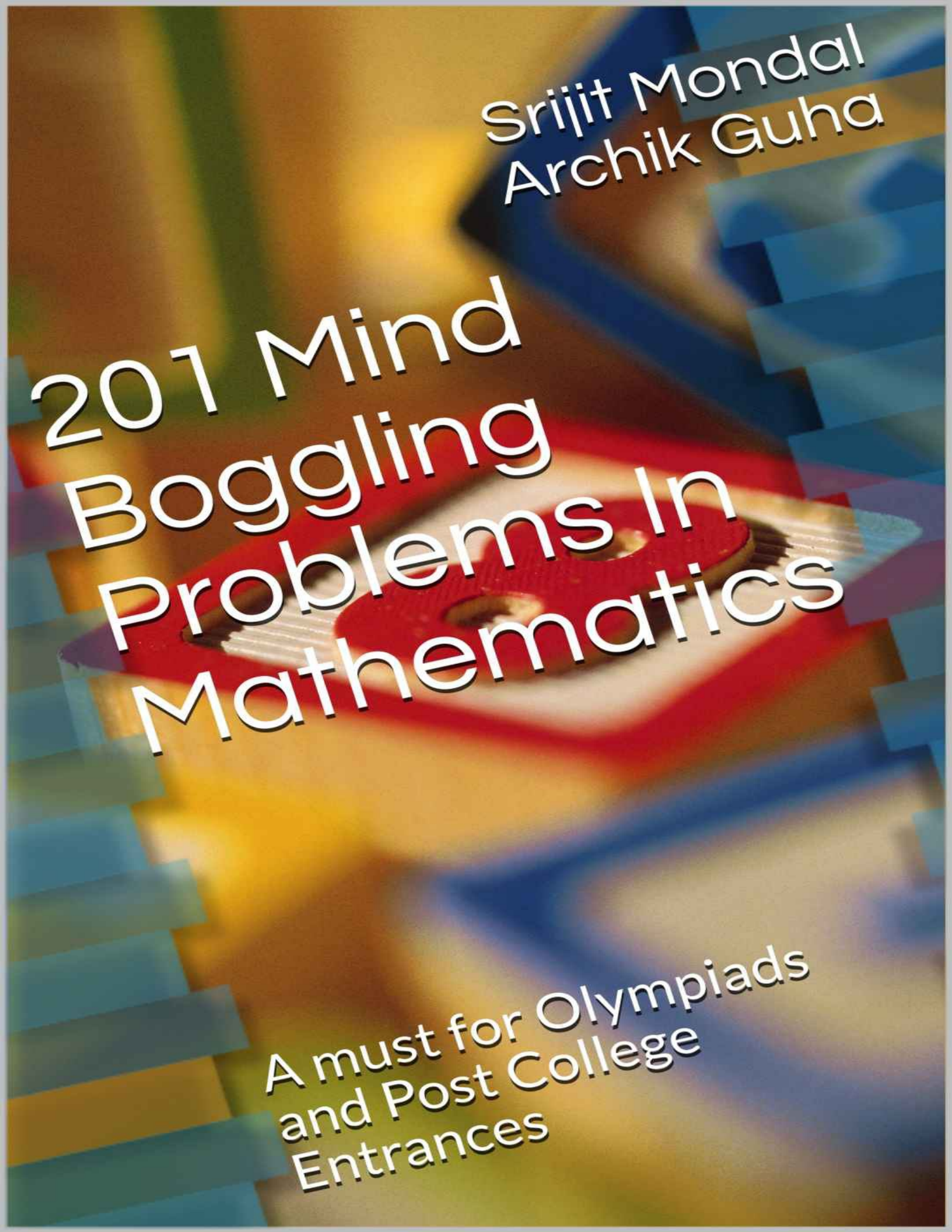


The background of the book cover is an abstract composition of various geometric shapes, including rectangles and squares, in shades of blue, yellow, and orange. A prominent red gear with several teeth is positioned in the center, partially overlapping the text. The overall effect is a vibrant and dynamic visual.

Srijit Mondal  
Archik Guha

# 201 Mind Boggling Problems In Mathematics

A must for Olympiads  
and Post College  
Entrances



**1.** Let  $a_1, a_2, \dots, a_{100}$  be real numbers, each less than one, satisfy

$$a_1 + a_2 + a_3 + \dots + a_{100} > 1$$

(I) Let  $a_0$  be the smallest integer  $n$  such that

$$a_1 + a_2 + \dots + a_n > 1$$

Show that all the sums  $a_{n_0}, a_{n_0} + a_{n_0-1}, \dots, a_{n_0} + a_{n_0-1} + a_{n_0-2} + \dots + a_1$  are positive.

(II) show that there exists two integers  $p$  and  $q$ ,  $p < q$  such that the numbers

$$a_q, a_q + a_{q-1}, \dots, a_q + \dots + a_p \\ a_p, a_p + a_{p+1}, \dots, a_p + \dots + a_q$$

are all positive.

Ans:

PART 1: clearly  $a_1 + a_2 + \dots + a_{n_0-1} < 1, a_1 + \dots + a_{n_0-2} < 1, \text{etc.}$  since  $a_{n_0}$  is the smallest integer with  $a_1 + a_2 + \dots + a_n > 1$  so

$$a_{n_0} > 1 - a_1 + a_2 + \dots + a_{n_0-1} > 0$$

and  $a_{n_0} + a_{n_0-1} > 1 - a_1 + \dots + a_{n_0-2} > 0$  etc. this way

all sums  $a_{n_0}, a_{n_0} + a_{n_0-1}, \dots, a_{n_0} + a_{n_0-1} + a_{n_0-2} + \dots + a_1$  are positive.

PART 2 : for convenience i replace  $n_0$  by  $q$

Let us consider the maximum among the sums  $(a_q, a_q + a_{q-1}, \dots, a_q + \dots + a_1)$

$[a_q < 1$  and  $a_q + \dots + a_1 > 1$  so  $a_q$  is not the maximum]

Let that maximum sum be  $S = a_q + \dots + a_{q-r}$

now by our assumption  $S > a_q + \dots + a_{q-r+1}$  implying  $a_{q-r} > 0$

$S > a_q + \dots + a_{q-r+2}$  implying  $a_{q-r} + a_{q-r+1} > 0$

this way  $a_{q-r} + a_{q-r+1} + a_{q-r+2}, \dots, a_{q-r} + \dots + a_{q-r+r}$  are all positive.

replacing  $q - r$  by  $p$  we get the result.

2. A troop *5meters* long starts marching. A soldier at the end of the file steps out and immediately turns around and marches back at the same speed. As soon as he reaches the end of the file, the troop stops marching, and it is found that the troop has moved by exactly *5meters*. What distance has the soldier traveled?

Ans: Let  $x$  and  $y$  be the speed (measured in meter per second) of the troop and the soldier passing the troop respectively. The time it takes the soldier passing the column is seconds and on the way back to the end of the column he spends seconds. The column has moved 5 meters in seconds while the soldier has marched from the tail to the head and back to the tail of the column. In other words

$$5/(y - x) + 5/(x + y) = 5/x \text{ or } 2xy = y^2 - x^2$$

$$\text{or } (y/x)^2 - 2y/x - 1 = 0 \text{ solving } y/x = 1 + 2^{1/2}$$

now the soldier had travelled at  $y$  speed for  $5/x$  time.

$$\text{HENCE distance covered by him } y * (5/x) = 5y/x = 5(1 + 2^{1/2})$$

3. Let  $P$  be an interior point of the triangle  $\triangle ABC$ . assume that  $AP, BP, CP$  meet the opposite sides of  $BC, CA, AB$  are  $D, E, F$  respectively show that

$$\frac{AF}{FB} + \frac{AE}{EC} = \frac{AP}{PD}$$

Ans: BY **Ceva's Theorem**

$$(AF/FB) * (BD/DC) * (CE/AE) = 1$$

$$\text{or, } AF/FB = (DC/BD) * (AE/CE) \dots \dots \dots \text{eq1}$$

$$LHS = AF/FB + AE/EC = [AE/EC] * [DC/BD + 1] \dots [USING 1]$$

$$= [AE/EC] * [(DC + BD)/BD] = [AE/EC] * [BC/BD] \dots \dots \dots \text{eq2}$$

IN TRIANGLE APE,

$$AE/\sin APE = AP/\sin PEA \text{ or } AE = AP * (\sin APE/\sin PEA) \dots \dots \dots \text{eq3}$$

IN Tr. ECB

$$BC/\sin BEC = EC/\sin EBC \text{ or } BC/EC = \sin BEC/\sin EBC \dots \dots \text{eq4}$$

IN Tr. PBD,

$$PD/\sin PBD = BD/\sin BPD \text{ or } 1/BD = (\sin PBD/\sin BPD)/PD \dots \dots \text{eq5}$$

Using 3,4,5, in 2

$$(AE/BD) * (BC/EC) = (AP/PD) * [\sin APE/\sin PEA] * [\sin PBD/\sin BPD] * [\sin BEC/\sin EBC]$$

$$= [AP/PD] * (\sin APE/\sin APE) * [\sin BEC/\sin(\pi - BEC)] \dots [\text{since } PEA = \pi - BEC \text{ and } APE = BPC]$$

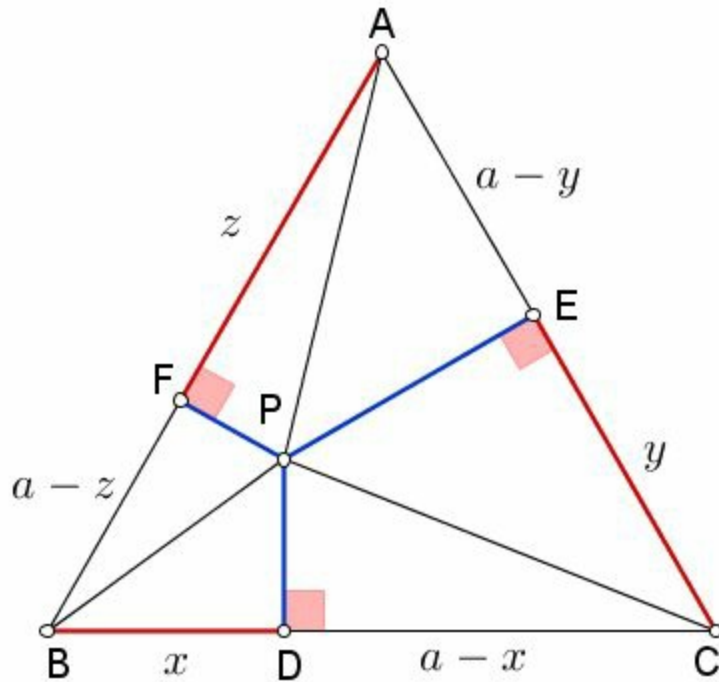
$$= AP/PD = RHS$$

4. Inside equilateral triangle  $ABC$  an arbitrary point  $P$  is taken from which perpendiculars  $PD, PE, PF$  are dropped on  $BC, CA, AB$  respectively. P.T. the ratio

$(PD + PE + PF)/(BD + CE + AF)$  is unique irrespective of  $P$  and find its value.

Ans: By Carnot's theorem:  $x^2 + y^2 + z^2 = (a - x)^2 + (a - y)^2 + (a - z)^2$

, where we get 
$$x + y + z = \frac{3a}{2}$$



5.  $ABC$  is a triangle right angled at  $A$ ,  $D$  be any point on side  $AB$ .  $DE$  drawn parallel to  $BC$  meets  $AC$  at  $E$ .  $F$  is the foot of perpendicular from  $E$  to  $BC$ .

If  $AD = p$ ,  $DB = q$ ,  $BF = r$ ,  $EF = s$ ,  $AE = t$  show that

$$(p + q)/t = (pr + st)/(rt - ps).$$

Ans:

$$\begin{aligned} \theta &= \angle BEA \\ c &= \angle DEA \\ \phi &= \angle FBE \\ \frac{x_1}{x_5} + \frac{x_2}{x_5} \\ &= \frac{x_1 + x_2}{x_5} \\ &= \tan \theta \\ &= \tan(\phi + c) \end{aligned}$$

$$\begin{aligned}
&= \frac{\tan \alpha + \tan \phi}{1 - \tan \alpha \tan \phi} \\
&= \frac{\frac{x_1}{x_5} + \frac{x_4}{x_3}}{1 - \frac{x_1 \cdot x_4}{x_5 \cdot x_3}} \\
&= \frac{x_1 x_3 + x_4 x_5}{x_3 x_5 - x_1 x_4}
\end{aligned}$$

6.  $AB$  is a chord of circle  $C$

(a) find a point  $P$  on the circumference  $C$  such that  $PA \cdot PB$  is maximum

(b) find a point  $P$  on the circumference  $C$  which maximizes  $PA + PB$ .

Ans: let  $AB$  subtend  $x$  angle in centre of circle, let  $r = \text{radius}$

let  $S = PA \cdot PB = (ry) \cdot (rz)$  where  $y = \text{angle subtended by } PA$ ,  $z = \text{angle subtended by } PB$

so  $S = (r^2)(y)(x - y)$

differentiating above w.r.t  $y$  and equating to 0 we get  $y = x/2$  second derivative is always negative

so clearly  $P$  is mid point of arc  $AB$ .

Second part is similar.

7. There are 1000 doors  $D_1, D_2, \dots, D_{1000}$  and 1000 people  $P_1, P_2, P_3, \dots, P_{1000}$ . Initially all doors were closed. Person  $P_1$  goes in and opens all doors.  $P_2$  goes in and opens all  $D_2, D_4, \dots, D_{1000}$ ...and so on...  $P_m$  goes in and opens  $D_m, D_{2m}, \dots, D_{nm}$ ...leaving all other doors untouched. Finally  $P_{1000}$  goes in and opens  $D_{1000}$  if it was closed and closes if it was open. At the end, how many doors remain open?

Ans: no. of doors whose state has changed an odd number of times = no. of positive integers between 1 and 1000 which have odd number of divisors = no.

of perfect squares between 1 and 1000= $\lfloor \sqrt{1000} \rfloor = 31$ . Thus, 31 is the final answer.

[ if  $b$  be a positive integer and its prime factorisation be  
 $b = (p_1^{a_1})(p_2^{a_2}) \dots (p_k^{a_k})$

where  $p_i$ 's are the prime factors of  $b$  and  $a_i$ 's are their respective powers then number of divisors of  $b$  is  $g = (a_1 + 1)(a_2 + 1) \dots (a_k + 1)$

if  $b$  is square then  $a_i$ 's are even so  $g$  is odd, hence  $b$  has odd number of divisors.

Conversely if  $b$  has odd number of divisors,  $g$  must be odd, so each of  $(a_1 + 1), \dots, (a_k + 1)$  must be odd so each of the  $a_i$ 's should be even. So  $b$  is a square.]

8. Suppose there are  $k$  teams in a round robin tournament, each team plays against all other teams and no game ends in a draw. Suppose that the  $i$ th team loses  $l(i)$  games and wins  $w(i)$  games. Show that:-

$$\sum_{i=1}^k l(i)^2 = \sum_{i=1}^k w(i)^2$$

Ans: Since team  $i$  plays  $(k - 1)$  games and loses  $l(i)$  games, and there can be no draw, we have  $w(i) = (k - 1 - l(i))$ ...now we are to prove that

$$\sum_{i=1}^k l(i)^2 = \sum_{i=1}^k (k - 1 - l(i))^2$$

Now  $RHS$  becomes

$$k((k - 1)^2) + LHS - 2(k - 1) \sum_{i=1}^k l(i)$$

..now in a round robin league, each

match is counted as a win and a loss both..so  $\sum_{i=1}^k l(i)$  = total number of matches

played =  $\binom{k}{2} = k(k - 1)/2$ ...thus cancelling out, we have  $LHS = RHS$ .

9. Suppose that  $P(x)$  is a polynomial of degree  $n$  such that,

$$P(k) = \frac{k}{k+1} \text{ for } k = 0, 1, 2, 3, \dots, n$$

Find the value of  $P(n+1)$ .

Ans: Let  $Q(x) = (x+1)P(x) - x$ .

Then the polynomial  $Q(x)$  vanishes for  $k = 0, 1, 2, \dots, n$  that is,

$$(x+1)P(x) - x = a * x * (x-1) * (x-2) * \dots * (x-n)$$

To find  $a$  we set  $x = -1$  and get,

$$1 = a(-1)^{n+1}(n+1)! \text{ thus,}$$

$$P(x) = \frac{(-1)^{n+1}x(x-1)(x-2)\dots(x-n)/(n+1)! + x}{x+1}$$

And ,

$$P(n+1)$$

$$= 1 \text{ for odd } n$$

$$= \frac{n}{n+2} \text{ for even } n.$$

10. If  $n$  is a positive integer greater than 1, such that  $3n+1$  is a perfect square, then show that  $n+1$  is the sum of three perfect squares.

Ans: Let  $3n + 1 = p^2$  or  $n = (p - 1)(p + 1)/3$  since  $n$  is integer, either 3 divides  $p - 1$  or  $p + 1$

CASE 1: 3 divides  $p - 1$  i.e,  $p - 1 = 3r$

then  $n = r(3r + 2)$  or  $n + 1 = 3r^2 + 2r + 1 = r^2 + r^2 + (r + 1)^2$

CASE 2: 3 divides  $p + 1$ ,  $p + 1 = 3s$

then  $n + 1 = s(3s - 2) + 1 = s^2 + s^2 + (s - 1)^2$

hence proved.

11. Show that the number 11....1 with  $3^n$  digits is divisible by  $3^n$ .

Ans: we assume that 1111..... $3^n$  digits is divisible by  $3^n$  so  
 $[10^{3^n} - 1]/9 = k * 3^n$  [why?]

so  $[10^{3^{n+1}} - 1]/9 = [(10^{3^n} - 1)/9] * (3r)$  [for some integral r]

hence  $[10^{3^{n+1}} - 1]/9 = (k * 3^n) * 3r$  which is divisible by  $3^{n+1}$

1111..... $3^n$  digits is divisible by  $3^n$  by induction

*hence proved*

12. Let  $a_1, a_2, \dots, a_n$  be  $n$  numbers such that each  $a_i$  is either 1 or  $-1$  If

$$a_1 a_2 a_3 a_4 + a_2 a_3 a_4 a_5 + \dots + a_n a_1 a_2 a_3 = 0$$

prove that 4 divides  $n$ .

Ans: Let the sum be denoted by  $S$ . Initially  $S = 0$  which is divisible by 4.

Suppose we change the sign of any one of  $a_i$ 's, (i.e, we make it positive if it was negative or vice-versa). Since  $a_i$  occurs in exactly four of the terms, we consider the following cases:

a) All four terms were negative (i.e, were equal to  $-1$  by question)

Then these terms added to  $-4$ . But sign change of the  $a_i$  makes sum of these terms 4.

So new sum  $S' = S - (-4) + 4 = S + 8 = 8$

b) Three were positive, one negative

In a similar fashion, new sum  $= S + 4 = 4$

c) Two positive, two negative:

New sum  $= S$

d) One negative, three positive:

New sum  $= S - 4 = -4$

e) All positive:

New sum  $= S + 8$

We observe from the above cases, that after the change of signs, new sum  $S'$  remains divisible by 4 (that is, **the sum modulo 4 is invariant**)

We keep on changing signs of different  $a_i$ 's until all terms become 1 and the sum equals  $n$ .

But the sum will still be divisible by 4

**So 4 divides  $n$ .**

13. Suppose  $k$  and  $n$  are positive integers greater than or equal to 1. Show that  $(nk)!$  is divisible by  $(k!)^n$ .

Ans: And here is my solution:-

We have:-

$(nk)! = \frac{(nk)!}{((n-1)k)!} \cdot \frac{((n-1)k)!}{((n-2)k)!} \cdot \dots \cdot k!$  since cancelling out terms from *RHS* gives *LHS*. There are  $n$  terms on *RHS*. Now, we multiply and divide by  $(k!)^n$

$(nk)! = ((k!)^n) \binom{nk}{k} \cdot \binom{(n-1)k}{k} \dots \binom{k}{k} = ((k!)^n) \cdot L$ , where  $L$  is a positive integer since the binomial coefficient is a positive integer. Thus  $(nk)! = ((k!)^n)L$ ...which implies that  $(nk)!$  is divisible by  $(k!)^n$ .

14. For  $x > 0$ , show that  $(x^n - 1)/(x - 1) \geq nx^{(n-1)/2}$  where  $n$  is a positive integer.

Ans: applying **AM-GM inequality** to the quantities  $1, x, x^2, \dots, x^{n-1}$

$$\frac{1 + x + \dots + x^{n-1}}{n} \geq (1 \cdot x \cdot x^2 \cdot \dots \cdot x^{n-1})^{\frac{1}{n}}$$

$$\text{or } \frac{1 + x + \dots + x^{n-1}}{n} \geq x^{(1+2+\dots+n-1)/n}$$

$$\text{or } \frac{1 + x + \dots + x^{n-1}}{n} \geq x^{n(n-1)/2n}$$

$$\text{or } \frac{x^n - 1}{x - 1} \geq nx^{(n-1)/2}.$$

15. consider the circle  $C$  whose equation is  $(x - 2)^2 + (y - 8)^2 = 1$  and the parabola  $P$  with the equation  $y^2 = 4x$  find the minimum value of the length of the segment  $AB$  as  $A$  moves on the circle  $C$  and  $B$  moves on the parabola  $P$ .

Ans: Any point on  $y^2 = 4x$  is  $H(t^2, 2t)$ .

Centre of  $(x - 2)^2 + (y - 8)^2 = 1$  is  $G(2, 8)$ . Distance between  $GH$  is  $[(2 - t^2)^2 + (8 - 2t)^2]^{1/2}$

Let  $z = (2 - t^2)^2 + (8 - 2t)^2$ .  $GH$  is minimum if  $z$  is minimum.

So differentiating,  $\frac{dz}{dt} = 2(2 - t^2)(-2t) - 2(8 - 2t)(2)$ , equating to 0 we get  $t = 2$ .

Now  $\frac{d^2z}{dt^2} = 12t^2 > 0$  so  $z$  is minimum at  $t = 2$

So  $H$  is  $(4, 4)$  so  $GH = 2 \cdot (5^{1/2})$

**So Required shortest distance =  $2 \cdot (5^{1/2}) - \text{radius of circle} = 2 \cdot (5^{1/2}) - 1$ .**

16. Given  $f : \mathbb{R} \rightarrow \mathbb{R}$  and  $g : \mathbb{R} \rightarrow \mathbb{R}$  are two quadratic polynomials with rational coefficients. Suppose  $f(x) = 0$  and  $g(x) = 0$  have a common irrational solution. Prove that  $f(x) = rg(x)$  for all  $x$  where  $r$  is a rational number.

Ans: Irrational roots occur in conjugate pairs.....now as the degree of both the polynomials is 2 so if they have one common root then both the irrational roots of the polynomial are equal.....now let  $g(x) = a(x - c)(x - d)$  and  $f(x) = b(x - c)(x - d)$ .....as both the polynomials have rational coefficients so  $a$  and  $b$  both are rationals.....

.now  $\frac{f(x)}{g(x)} = \frac{a}{b} =$  **a rational number r** [since  $a, b$  are rationals]

Hence.....  $f(x) = rg(x)$  where  $r$  is a rational no.

17. Consider the parabola  $y^2 = 4x$ . Let  $P(a, b)$  be any point inside it, so that  $b^2 < 4a$ . Let  $F$  be the focus of the parabola.

Find the point  $Q$  on the parabola so that  $FQ + QP$  is minimum. Also show normal to parabola bisects angle  $FQP$ .

Ans: Point  $P(a, b)$  and parabola  $y^2 = 4x$  with focus  $F(1, 0)$ .

Point  $Q(\lambda^2, 2\lambda)$  on this parabola.

$$FQ + QP = \sqrt{(\lambda^2 - 1)^2 + (2\lambda)^2} + \sqrt{(\lambda^2 - a)^2 + (2\lambda - b)^2}$$

$$f(\lambda) = \lambda^2 + 1 + \sqrt{(\lambda^2 - a)^2 + (2\lambda - b)^2}$$

$f(\lambda)$  has a minimum if  $f'(\lambda) = 0$ .

$$\lambda + \frac{(\lambda^2 - a)\lambda + (2\lambda - b)}{\sqrt{(\lambda^2 - a)^2 + (2\lambda - b)^2}} = 0$$

Solving:  $\lambda = \frac{b}{2}$ .

Point  $Q(\frac{b^2}{4}, b)$ .

$PQ \parallel$  x-axis

Let  $QG$  be the tangent to the parabola at  $Q$  intersecting  $x$ -axis at  $G$ . Let  $QH$  be the normal intersecting  $x$ -axis at  $H$ . (draw figure)

Clearly  $PQ \parallel HF$  so  $\angle PQH = \angle FHQ$  and  $\angle PQF = \angle FQO$

also  $\angle FHQ + \angle FQH = \angle QFO$

We see that  $y^2 = 4x$  implies  $-\frac{dx}{dy} = y/2$  so slope of normal at  $Q(b^2/4, b)$  is  $-b/2$

$$\tan(\angle PQH) = -b/2 \text{ or } \tan(\angle QHF) = \tan(\pi - \angle PQH) = b/2$$

so  $\angle QHF = b/2$  also slope of  $QF = b/2$

Angle between  $FQ$  and  $HQ$  is  $\angle FQH = \arctan\left|\frac{\frac{4b}{b^2-4} - (-\frac{b}{2})}{1 + \frac{4b}{b^2-4}(-\frac{b}{2})}\right| = \arctan(b/2)$   
 so  $\angle FQH = \angle PQH$   
 so  $QH$  bisects  $\angle FQP$ .

18. If  $a, b, c$  are positive numbers, show that

$$\frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} + \frac{a^2 + b^2}{a + b} \geq a + b + c$$

Ans: We have  $(b - c)^2 \geq 0$  or  $b^2 + c^2 \geq 2bc$  or  $2(b^2 + c^2) \geq 2bc + b^2 + c^2 = (b + c)^2$

$$\text{so } \frac{2(b^2 + c^2)}{(b + c)} \geq b + c$$

applying similar inequalities for pairs  $c, a$  and  $a, b$  and adding and simplifying we get

$$\frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} + \frac{a^2 + b^2}{a + b} \geq a + b + c.$$

19. Let  $a, b, c, d$  be positive reals such that  $abcd = 1$   
 Prove that  $(1+a)(1+b)(1+c)(1+d) \geq 16$ .

Ans: Applying **AM-GM** on 1 and  $a$  we get  $(1+a) \geq 2a^{1/2}$

similarly  $1+b \geq 2b^{1/2}$ ,  $1+c \geq 2c^{1/2}$ ,  $1+d \geq 2d^{1/2}$

Multiplying and using  $abcd = 1$  we get  $(1+a)(1+b)(1+c)(1+d) \geq 16$

20. Let  $E$  be the ellipse with centre at origin  $O$  whose major and minor axes are of length  $2a$  and  $2b$  respectively.

Let  $\theta$  be the acute angle at which  $E$  is cut with the circle with centre at origin (i.e.  $\theta$  is the acute angle of intersection of their tangents at the point of intersection). Prove that maximum value of  $\theta$  is  $\arctan\left(\frac{a^2 - b^2}{2ab}\right)$ .

Ans: equation of tangent of ellipse at  $(x, y)$  is  $Xx/a^2 + Yy/b^2 = 1$  with slope  $-(x/a^2)/(b^2/y) = m$  (say)

tangent of the circle is of form  $Xx + Yy = t^2$  (assuming radius of circle =  $t$ ) with slope  $x/y = n$  (say)

$$\tan\theta = |(m - n)/(1 + mn)| = \left| \frac{\left(\frac{x}{y}\right)\left(1 - \frac{b^2}{a^2}\right)}{\left(1 + \frac{(x^2)(b^2)}{(y^2)(a^2)}\right)} \right|$$

So

for  $\theta$  to be maximum  $\tan\theta$  should be maximum. after rearrangement

$$\tan\theta = \left| \frac{(a^2 - b^2)}{\frac{(a^2 y^2 + b^2 x^2)}{xy}} \right| \text{ let this be eq. 1}$$

above is maximum if its denominator is minimum.

By AM-GM,  $\frac{(a^2)y^2 + (b^2)(x^2)}{2} \geq abxy$  or  $\frac{(a^2 y^2 + b^2 x^2)}{xy} \geq 2ab$

putting this minimum value in eq. 1 we get maximum of  $\tan\theta$  is  $\frac{a^2 - b^2}{2ab}$

hence maximum value of  $\theta$  is  $\arctan\left(\frac{a^2 - b^2}{2ab}\right)$ .

21. show that if  $n > 2$  then  $(n!)^2 > n^n$

Ans: Proceed by induction

True for  $n = 3$

Let's assume its true for  $n = k$

To show,  $\{(k+1)!\}^2 > (k+1)^{k+1}$

Now,  $\{(k+1)!\}^2 = (k+1)^2 \{k!\}^2 > (k+1)^2 k^k$

To show,  $(k+1)^2 k^k > (k+1)^{k+1}$

$\iff k^k > (k+1)^{k-1}$

$\iff k^{\frac{1}{k-1}} > (k+1)^{\frac{1}{k}}$

Consider the function  $f(x) = x^{\frac{1}{x-1}}$

$$f'(x) = f(x) \left( \frac{x-1-x\ln x}{x(x-1)^2} \right)$$

Now  $g(x) = x - x \ln x - 1 \implies g'(x) = -\ln x < 0 \forall x > 1$

So  $g(x) < g(1) = 0$

Hence  $f'(x) < 0 \forall x > 1$

Hence  $f$  is decreasing, so  $k^{\frac{1}{k-1}} > (k+1)^{\frac{1}{k}}$

Hence we're done.

22. Let  $\theta_1, \theta_2, \theta_3, \dots, \theta_{10}$  be any values in the closed interval  $[0, \pi]$ . Show that  $F = (1 + \sin^2 \theta_1)(1 + \cos^2 \theta_1)(1 + \sin^2 \theta_2)(1 + \cos^2 \theta_2) \dots (1 + \sin^2 \theta_{10})(1 + \cos^2 \theta_{10}) \leq \left(\frac{9}{4}\right)^{10}$ .

What is the maximum value attainable by  $F$  and at what values of  $\theta_1, \theta_2, \theta_3, \dots, \theta_{10}$  is the maximum value attained?

Ans: **First part.**

Pick up one pair of sin and cosine from the function  $F$ .

Let us have  $(1 + \sin^2 \theta_1)(1 + \cos^2 \theta_1)$  as one pair. Now when we expand this we get

$$1 + \sin^2 \theta_1 + \cos^2 \theta_1 + \sin^2 \theta_1 \cos^2 \theta_1 = 2 + \sin^2 \theta_1 \cos^2 \theta_1$$

now using **AM-GM** inequality we may write ,

$$\begin{aligned} \sin^4 \theta_1 + \cos^4 \theta_1 &\geq 2 \sin^2 \theta_1 \cos^2 \theta_1 \\ \Rightarrow (\sin^2 \theta_1 + \cos^2 \theta_1)^2 - 2 \sin^2 \theta_1 \cos^2 \theta_1 &\geq 2 \sin^2 \theta_1 \cos^2 \theta_1 \\ \Rightarrow 4 \sin^2 \theta_1 \cos^2 \theta_1 &\leq 1 \\ \Rightarrow \sin^2 \theta_1 \cos^2 \theta_1 &\leq \frac{1}{4} \\ \Rightarrow 2 + \sin^2 \theta_1 \cos^2 \theta_1 &\leq \frac{1}{4} + 2 = \frac{9}{4} \end{aligned}$$

So, each pair of  $\sin^2 \theta_i, \cos^2 \theta_i, i \in \{1, 2, \dots, 10\}$  is less than equal to  $\frac{9}{4}$ .

Hence product of 10 such pairs is less than equal to  $\left(\frac{9}{4}\right)^{10}$ .

**Second part.**

From the first we can say that the maximum value attainable by  $F$  is  $\left(\frac{9}{4}\right)^{10}$ . So thus we have to find out for which values of  $\theta_i, i \in \{1, 2, \dots, 10\}$ ,

$$\sin^2 \theta_i \cos^2 \theta_i = \frac{1}{4}$$

now from here using some trigonometric equation we can say that  $\theta_i = \frac{\pi}{4}$  or

$$\theta_i = \frac{3\pi}{4} \text{ as } \theta_i \in [0, \pi].$$

23. Let  $x$  be a positive number. A sequence  $\{x_n\}$  of real numbers is defined as follows:

$$x_1 = \frac{1}{2}\left(x + \frac{5}{x}\right), x_2 = \frac{1}{2}\left(x_1 + \frac{5}{x_1}\right), \dots, \text{ and in general,}$$

$$x_{n+1} = \frac{1}{2}\left(x_n + \frac{5}{x_n}\right) \text{ for all } n \geq 1.$$

$$(a) \text{ Show that, for all } n \geq 1, \frac{x_n - \sqrt{5}}{x_n + \sqrt{5}} = \left(\frac{x - \sqrt{5}}{x + \sqrt{5}}\right)^n$$

$$(b) \text{ Hence find } \lim_{n \rightarrow \infty} x_n.$$

Ans: **First part**

Induction on  $n$

See for  $n = 1$  it is obvious.

Let for some  $n > 1$

$$\frac{x_n - \sqrt{5}}{x_n + \sqrt{5}} = \left(\frac{x - \sqrt{5}}{x + \sqrt{5}}\right)^n$$

Then we would show it true for  $n + 1$

then take  $x_n = y$

and define  $x_{n+1} = y_1$  and  $y_1 = 1/2(y + 5/y)$

Then since  $y_1$  is same as  $x_1$  except that in  $y_1$  the  $x$  of  $x_1$  has been replaced by  $y$  so it obeys the condition which we need to prove (From Induction hypothesis)

Then we are precisely done!!

## **Part 2**

See  $(x_n)$  is a monotonically decreasing sequence since  $x > 0$  and it is bounded (below by 0) so limit exists.

Now let  $m$  be the limit.

Then

$$m = \frac{1}{2}\left(m + \frac{5}{m}\right)$$
$$\Rightarrow m = \sqrt{5}.$$

24. For real numbers  $x, y, z$  show that

$$|x| + |y| + |z| \leq |x + y - z| + |y + z - x| + |z + x - y|.$$

Ans: For quantities  $(x + y - z)$  and  $(y + z - x)$  applying triangle inequality, we get

$$|(x + y - z) + (y + z - x)| \leq |x + y - z| + |y + z - x| \text{ or}$$
$$2|y| = |x + z - x| + |y + z - x|$$

Obtaining two similar inequalities for  $(x + y - z)$ ,  $(z + x - y)$  and  $(z + x - y)$ ,  $(y + x - z)$  and adding the three inequalities thus obtained we get

$$|x| + |y| + |z| \leq |x + y - z| + |y + z - x| + |z + x - y|.$$

25. Show that number of ways in which four distinct integers can be chosen

from  $1, 2, \dots, n \geq 7$ , such that no two are consecutive is  $\binom{n-3}{4}$ .

Ans: Let us assume the four integers be  $a > b > c > d$  since  $a, b, c, d$  are not consecutive we have

$$a - 3 > b - 2 > c - 1 > d$$

Now  $a - 3, b - 2, c - 1, d$  may or may not be consecutive. We can choose them from  $n - 3$  terms.

$$\text{So Total number of ways to choose} = \binom{n-3}{4}.$$

26. Show that if  $n > 2$  then  $(n!)^2 > n^n$ .

Ans: Proceed by induction

True for  $n = 3$

Let's assume its true for  $n = k$

To show,  $\{(k+1)!\}^2 > (k+1)^{k+1}$

Now,  $\{(k+1)!\}^2 = (k+1)^2 \{k!\}^2 > (k+1)^2 k^k$

To show,  $(k+1)^2 k^k > (k+1)^{k+1}$

$$\iff k^k > (k+1)^{k-1}$$

$$\iff k^{\frac{1}{k-1}} > (k+1)^{\frac{1}{k}}$$

Consider the function  $f(x) = x^{\frac{1}{x-1}}$

$$f'(x) = f(x) \left( \frac{x-1-x \ln x}{x(x-1)^2} \right)$$

Now  $g(x) = x - x \ln x - 1 \implies g'(x) = -\ln x < 0 \forall x > 1$

So  $g(x) < g(1) = 0$

Hence  $f'(x) < 0 \forall x > 1$

Hence  $f$  is decreasing, so  $k^{\frac{1}{k-1}} > (k+1)^{\frac{1}{k}}$

Hence we're done.

27. Let  $\{C_n\}$  be an infinite sequence of circles lying in the positive quadrant of the  $XY$ -plane, with strictly decreasing radii and satisfying the following conditions. Each  $C_n$  touches both the  $X$ -axis and the  $Y$ -axis. Further, for all  $n \geq 1$ , the circle  $C_{n+1}$  touches the circle  $C_n$  externally. If  $C_1$  has radius 10 cm,

then show that the sum of the areas of all these circles is  $\frac{25\pi}{3\sqrt{2}-4}$  sq cm.

Ans: Let  $R_n$  is the radius of the circle  $C_n$ . From all the given tangency conditions we have

$$\sqrt{2}R_n - \sqrt{2}R_{n+1} = R_n + R_{n+1} \implies \frac{R_{n+1}}{R_n} = \frac{\sqrt{2}-1}{\sqrt{2}+1} = (\sqrt{2}-1)^2 = 3-2\sqrt{2} = \lambda.$$

[Say]

Hence the required sum,

$$S = \pi(R_1^2 + R_2^2 + R_3^2 + \dots) = \pi(R_1^2 + \lambda^2 R_1^2 + \lambda^4 R_1^2 + \dots) = \frac{\pi R_1^2}{1-\lambda^2} \quad [\text{Since } |\lambda| \leq 1.]$$

$$\implies S = \frac{25\pi}{3\sqrt{2}-4} \text{ sq cm. [Putting values of } R_1 \text{ and } \lambda.]$$

28. Suppose that the equations  $ax^2 - 2bx + c = 0$ ,  $bx^2 - 2cx + a = 0$  and  $cx^2 - 2ax + b = 0$  only have positive roots. Show that  $a = b = c$ .

Ans: The equations have positive (hence real) roots. So their discriminants are positive or zero.

So,  $4b^2 \geq 4ac$  or  $b^2 \geq ac$  similarly  $c^2 \geq ab$ ,  $a^2 \geq bc$

Multiplying the inequalities we get,  $a^2 b^2 c^2 \geq a^2 b^2 c^2$  This is possible iff  $b^2 = ca, c^2 = ab, a^2 = bc$

from the first two equalities  $b^2 - c^2 = a(c - b)$  or  $(b - c)(a + b + c) = 0$

Given roots of equation are positive hence sum of roots are positive, So  $\frac{2b}{a}, \frac{2c}{b}, \frac{2a}{c}$  are positive which is possible iff  $a, b, c$  are of same sign.

So  $a + b + c$  cannot equal 0. So  $b = c$  Similarly,  $c = a$

Hence  $a = b = c$ .

29. Let  $x$  and  $n$  be positive integer such that  $1 + x + x^2 + \dots + x^{n-1}$  is a prime. Then show that  $n$  is also a prime number.

Ans: Let  $n = ab$  then,

$$\begin{aligned} &1 + x + x^2 + \dots + x^{n-1} \\ &= (1 + x + x^2 + \dots + x^{a-1}) + x^a(1 + x + \dots + x^{a-1}) + \dots + x^{a(b-1)}(1 + x + \dots + x^{a-1}) \\ &= (1 + x + x^2 + \dots + x^{a-1})(1 + x^a + x^{2a} + \dots + x^{a(b-1)}) \end{aligned}$$

As  $x \geq 1$  both the factors are  $> 1$  hence contradiction.

$n$  has to be a prime.

30. Find the maximum and minimum values of  $f(x) = x^2 - x \sin(x)$  in the closed interval  $[0, \frac{\pi}{2}]$ .

Ans:

$$f(x) = x(x - \sin x) \text{ so } f'(x) = (x - \sin(x)) + x(1 - \cos(x))$$

Clearly since  $\cos(x) \leq 1$  and  $x \geq \sin(x)$  in  $[0, \frac{\pi}{2}]$  so  $f' \geq 0$  or  $f$  increases in  $[0, \frac{\pi}{2}]$

So  $f$  is maximum at  $\frac{\pi}{2}$  and minimum at 0

So, **required maximum value** =  $f(\frac{\pi}{2}) = \frac{\pi^2}{4} - \frac{\pi}{2}$

**Minimum value** =  $f(0) = 0$

31. Suppose  $p$  is a prime number such that  $\frac{p-1}{4}$  and  $\frac{p+1}{2}$  are also primes. Show that  $p = 13$ .

Ans: This is a pretty simple problem, clearly  $p = 4k + 1$  for some  $k$ ...

now by the given condition,  $\frac{p-1}{4}$  and  $\frac{p+1}{2}$  are both primes, implying that both  $k$  and  $2k + 1$  are primes. So the problem basically reduces to finding all primes  $k$  such that  $k, 2k + 1, 4k + 1$  are all primes.

For  $k = 2$ , clearly this doesn't hold. Now for any  $k > 3$ , we must have,  $k = 3t + 1$  or  $k = 3t + 2$ . In the first case, we have  $3 | 2k + 1$ , so it cannot be prime, and in second case,  $3 | 4k + 1$ , another contradiction. So,  $k = 3$  is the only solution, which gives  $p = 4k + 1 = 13$  as the only solution.

32. (a) Prove that for any odd integer  $n$ ,  $n^4$  when divided by 16 always leaves remainder 1.

(b) Hence or otherwise show that we cannot find integers  $n_1, n_2, \dots, n_8$  such that

$$n_1^4 + n_2^4 + \dots + n_8^4 = 1993.$$

Ans: (a) For any odd integer  $n$  we have,

$$n \equiv \pm 1, \pm 3, \pm 5, \pm 7 \pmod{16}.$$

$$\implies n^2 \equiv 1, 9 \pmod{16}.$$

$$\implies n^4 \equiv 1 \pmod{16}.$$

(b) Suppose  $n_1^4 + n_2^4 + \dots + n_8^4 \equiv r \pmod{16}$ .

For even integer  $n$ ,  $16|n^4$ .

Hence for all integer  $n$ , we have  $n^4 \equiv 0, 1 \pmod{16}$ .

Hence  $r \leq 8$ . But  $1993 \equiv 9 \pmod{16}$ .

CONTRADICTION!

33. Show that there is one value of  $x$  for which  $x^{\frac{1}{3}} + x^{\frac{1}{2}} = 1$ .

Ans: Let  $f(x) = x^{\frac{1}{3}} + x^{\frac{1}{2}}$

Clearly  $f(0) = 0$ ,  $f(1) = 2$  and  $f$  is continuous in interval  $[0, 1]$ . So by intermediate value theorem  $f$

takes all values between 0 and 2. So  $f$  assumes the value 1 too.

So there is a real root of  $f$  in the interval  $[0, 1]$ .

**Hence Proved.**

34. A vessel contains  $x$  gallons of wine and another  $y$  gallons of water. From each vessel,  $z$  gallons are taken and transferred to the other. From resulting mixture,  $z$  gallons are again taken out and transferred into the other. After second transfer, quantity of wine in each vessel remains equal as it was after

first transfer prove that  $z(x + y) = xy$ .

Ans: Let vessel  $A$  contain wine and vessel  $B$  contain water initially..

After first transfer,  
vessel  $A$  contains  $(x - z)$  gallons wine.  
vessel  $B$  contains  $z$  gallons wine.

So, fraction of wine in  $A = (x - z)/x$   
fraction of wine in  $B = z/y$

During second transfer, quantity of wine pulled out from:-

$$A - > \frac{x - z}{x} \cdot z$$

$$B - > \frac{z}{y} \cdot z$$

Now from  $B$ ,  $\frac{z^2}{y}$  gallon is pulled out and the wine which is pulled from  $A$ , i.e.,  $\frac{x - z}{x} \cdot z$  gallon, is poured in  $B$ , and resulting mixture contain same quantity of wine as in first transfer, i.e.,

$$z + \frac{x - z}{x} \cdot z - \frac{z^2}{y} = z$$

Simplifying we get  $\boxed{z(y + x) = xy}$ .

35. If a circle intersects the hyperbola  $y = \frac{1}{x}$  at four distinct points  $(x_i, y_i)$ ,  $i = 1, 2, 3, 4$

Prove that  $x_1 x_2 = y_3 y_4$ .

Ans: Let the circle be,  $x^2 + y^2 + 2gx + 2fy + c = 0$

Since it intersects the hyperbola  $y = \frac{1}{x}$  at four points, it passes through

$$(x, 1/x)$$

$$\text{So } x^2 + \frac{1}{x^2} + 2gx + 2\frac{h}{x} + c = 0 \text{ or } x^4 + 2gx^3 + cx^2 + 2hx + 1 = 0$$

Clearly product of roots of above equation = 1 **[By, Vieta's Theorem]**

Since it passes through  $(x_1, 1/x_1), (x_2, 1/x_2), (x_3, 1/x_3), (x_4, 1/x_4)$   
 $x_1, x_2, x_3, x_4$  are roots of the equation.

$$\text{So } x_1x_2x_3x_4 = 1 \text{ or } x_1x_4 = \frac{1}{x_3x_4}$$

$$\text{so } \boxed{x_1x_2 = y_3y_4}.$$

36. Show that there is no real constant  $c > 0$  such that  $\cos(\sqrt{x+c}) = \cos(\sqrt{x})$  for all  $x \geq 0$ ,  $x$  being real.

Ans: Let there be a  $c$  such that,

$$\cos(\sqrt{x+c}) = \cos(\sqrt{x}) \text{ or } \sqrt{x+c} = 2n\pi \pm \sqrt{x} \quad [n = 0, 1, 2, \dots]$$

$$\text{or } x+c = 4n^2\pi^2 \pm 4n\pi\sqrt{x} + x \text{ or } \\ c = 4n\pi[n\pi \pm \sqrt{x}]$$

**RHS** of above is independent of  $x$  iff  $n = 0$  which contradicts  $c > 0$ . For all other  $n$ ,  $c$  depends on  $x$ .

Hence, there is no real constant  $c > 0$  such that  $\cos(\sqrt{x+c}) = \cos(\sqrt{x})$  for all  $x \geq 0$ ,  $x$  being real.

37.  $x$  red balls,  $y$  black balls,  $z$  white balls are to be arranged in a row. Suppose two balls are of same color and are indistinguishable. Given  $x + y + z = 30$  prove that number of possible arrangements is maximum when  $x = y = z = 10$ .

Ans: Number of ways where  $x + y + z$  things can be arranged with  $x$  of one type,  $y$  of second,  $z$  of third is

$$N = \frac{(x + y + z)!}{x!y!z!}$$

Now  $x + y + z = 30$  being constant,  $N$  is maximum if  $x!y!z!$  is minimum, which happens at  $x = y = z = 10$  according to question.

Let us assume the contrary, there are  $x', y', z'$  not all equal to 10 for which  $x'!y'!z'! < 10!10!10!$

Now three cases may arise **WLOG**,

i)  $x' > 10, y' > 10, z' < 10$  ii)  $x' > 10, y' = 10, z' < 10$  iii)  $x' > 10, y' < 10, z' < 10$

Let us consider 1st case.  $x' = 10 + r, y' = 10 + s, z' = 10 - r - s$  for positive integers  $r, s$

Clearly  $x' + y' + z' = 30$

Then  $(10 + r)!(10 + s)!(10 - r - s)! < 10!10!10!$

which implies

$$[(10 + 1)(10 + 2) \dots (10 + r)][(10 + 1)(10 + 2) \dots (10 + s)] < (10 - 1)(10 - 2) \dots (10 - r - s)$$

But this is contradictory since, **each term of LHS of above > each terms of RHS of above**

and the terms are positive.

So  $x'!y'!z'! > (10!)^3$  for all  $r, s$

Similar contradiction can be shown in other two cases.

**So possible arrangement is maximum when  $x = y = z = 10$ .**

38. Consider the equations in  $x$  and  $y$ :

$x + y + axy = a$ ,  $x - 2y - xy^2 = 0$  where  $a$  belongs to set of reals.

Show that these equations admit real solutions in  $x$  and  $y$ .

Ans:

$$x + y + axy = a \text{ implies } x = \frac{a - y}{1 + ay} \dots\dots\dots(i)$$

Also  $x - 2y - xy^2 = 0$ , substituting value of  $x$  from (i) we get,

$$\frac{a - y}{1 + ay} + 2y - \frac{a - y}{1 + ay} y^2 = 0 \quad \text{Simplifying, } y^3 + ay^2 + y + a = 0$$

Above is a cubic equation and will be satisfied for at least one real  $y$ .

If  $y$  is real,  $x = \frac{a - y}{1 + ay}$  is also real.

Hence, **The equations have real solutions.**

39. Suppose  $S = \{0, 1\}$  with the following addition and multiplication rules,

$$0 + 0 = 1 + 1 = 0$$

$$0 + 1 = 1 + 0 = 1$$

$$0 * 0 = 0 * 1 = 1 * 0 = 0$$

$$1 * 1 = 1$$

A system of polynomials is defined with coefficients in  $S$ . The sum and product of two polynomials in the system are usual sum and

product, respectively, where for the addition and multiplication of coefficients the above mentioned rules apply. For example, in the system,

$$(x + 1) * (x^2 + x + 1) = x^3 + (1 + 1)x^2 + (1 + 1)x + 1 = x^3 + 0x^2 + 0x + 1 = x^3 + 1$$

Show that in the system  $(x^3 + x + 1)$  is not factorizable, that is, one cannot write

$$x^3 + x + 1 = (ax + b)(cx^2 + dx + e)$$

Where  $a, b, c, d$  and  $e$  are elements of  $S$ .

Ans: Suppose,

$$\begin{aligned} x^3 + x + 1 &= (ax + b)(cx^2 + dx + e) \\ &= (a * c)x^3 + (b * c + a * d)x^2 + (b * d + a * e)x + b * e \end{aligned}$$

Now, comparing the coefficients,

$$a * c = 1 \implies a = 1, c = 1 \dots (1)$$

$$b * c + a * d = 0 \dots \dots \dots (2)$$

$$b * d + a * e = 1 \dots \dots \dots (3)$$

$$b * e = 1 \implies b = 1, e = 1 \dots (4)$$

Now, from (2),

$$1 * 1 + 1 * d = 0 \implies 1 + 1 * d = 1 + 1 \implies d = 1$$

Hence,  $b * d + a * e = 1 * 1 + 1 * 1 = 0$  but we have  $b * d + a * e = 1$

Hence, contradiction!!!

40. Consider the squares of an  $8 \times 8$  chessboard filled with the numbers 1 to 64 (see the figure below). If we choose 8 squares with the property that there is exactly one from each row and exactly one from each column, and add up the numbers in the chosen squares, show that the sum obtained is always 260

|    |    |    |    |    |    |    |    |
|----|----|----|----|----|----|----|----|
| 1  | 2  | 3  | 4  | 5  | 6  | 7  | 8  |
| 9  | 10 | 11 | 12 | 13 | 14 | 15 | 16 |
| 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 |
| 25 | 26 | 27 | 28 | 29 | 30 | 31 | 32 |
| 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
| 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 |
| 49 | 50 | 51 | 52 | 53 | 54 | 55 | 56 |
| 57 | 58 | 59 | 60 | 61 | 62 | 63 | 64 |

Ans: Basically we are choosing the numbers  
 $(8c_1 + 1), (8c_2 + 2), (8c_3 + 3), \dots, (8c_8 + 8)$

Where  $c_1, c_2, \dots, c_8$  belongs to the set  $\{0, 1, 2, 3, 4, 5, 6, 7\}$

Hence,

$$\sum_{i=0}^7 8c_i + (i + 1)$$

$$= \sum_{i=0}^7 8i + 36$$

$$= 224 + 36 = 260$$

41. Let  $f(x)$  be a real valued function of variable  $x$  such that  $f'(x)$  takes both positive and negative values and  $f''(x) > 0 \forall x$ . Show that there is a real number  $p$  such that  $f(x)$  is an increasing function  $\forall x \geq p$ .

Ans: Since  $f''(x) > 0$ , so  $f'(x)$  is strictly increasing.

Now  $f'(x)$  assumes both positive and negative values, hence there is exactly one root of  $f'(x) = 0$ , say  $x = p$  (since  $f'$  strictly increases)

Now  $f'$  is increasing hence  $f'(x) \geq f'(p) = 0$  when  $x \geq p$

Hence  $f(x)$  is an increasing function when  $x \geq p$ .

42. Let  $a_1, a_2, \dots, a_n$  be  $n$  numbers such that each  $a_i$  is either 1 or  $-1$  If

$$a_1 a_2 a_3 a_4 + a_2 a_3 a_4 a_5 + \dots + a_n a_1 a_2 a_3 = 0.$$

Ans: Let the sum be denoted by  $S$ . Initially  $S = 0$  which is divisible by 4.

Suppose we change the sign of any one of  $a_i$ 's, (i.e, we make it positive if it was negative or vice-versa). Since  $a_i$  occurs in exactly four of the terms, we consider the following cases:

a) All four terms were negative (i.e, were equal to  $-1$  by question)

Then these terms added to  $-4$ . But sign change of the  $a_i$  makes sum of these

terms 4.

So new sum  $S' = S - (-4) + 4 = S + 8 = 8$

b) Three were positive, one negative

In a similar fashion, new sum  $= s + 4 = 4$

c) Two positive, two negative:

New sum  $= S$

d) One negative, three positive:

New sum  $= S - 4 = -4$

e) All positive:

New sum  $= S - 8$

We observe from the above cases, that after the change of signs, new sum  $S'$  remains divisible by 4 (that is, **the sum modulo 4 is invariant**)

We keep on changing signs of different  $a_i$ 's until all terms become 1 and the sum equals  $n$ .

But the sum will still be divisible by 4

**So 4 divides  $n$ .**

43. At time 0, a particle is at point 0 on real line. At time 1, the particle divides into two and instantaneously after division, one particle moves one unit to the left and other moves one unit to the right. When two particles meet, they destroy each other leaving nothing behind. How many particles will be there after time  $2^{11} + 1$ ?

Ans: With smaller examples say at times 2 and 4, the particle number is 2 trivially and at times  $3 (= 2^1 + 1)$  and  $5 = (2^2 + 1)$ , it is 4.

Suppose we make hypothesis that there are 2 particles at positions  $\pm 2^k$  on

number line after time  $= 2^k$

Then consider the point  $2^k$  on the number line as origin. In next  $2^k$  time the particle at  $2^k$  will move the same way as it had moved in its previous  $2^k$  time, hence become 2 particles one of which will be situated at  $2^k$  unit right other at  $2^k$  unit left from the  $2^k$  point on number line. Same will happen for the particle at  $-2^k$

But for the particle at  $2^k$ , the position at  $2^k$  units left from it is the origin. Same is true for the position of  $2^k$  units right for the particle at  $2^k$ . Hence among the total of four particles as described above, two will destroy.

So after  $2^k + 2^k = 2^{k+1}$  time also particles will be 2 so our hypothesis was correct.

So at  $2^k + 1$  time these two particles divide into two each, resulting in four particles.

So **number of particles after time  $2^{11} + 1 = \boxed{4}$** .

44. or natural number  $n$  let  $a_n = n^2 + 20$ . If  $d_n$  is the g.c.d of  $a_n$  and  $a_{n+1}$  prove that  $d_n$  divides 81.

Ans:

$$a_{n+1} = (n+1)^2 + 20 = n^2 + 2n + 21$$

$$\text{So } \gcd(a_n, a_{n+1}) = \gcd(n^2 + 20, n^2 + 2n + 21)$$

$$= \gcd(n^2 + 20, n^2 + 2n + 21 - (n^2 + 20))$$

$$= \gcd(n^2 + 20, 2n + 1) = \gcd(2(n^2 + 20) - n(2n + 1), 2n + 1)$$

$$= \gcd(40 - n, 2n + 1) = \gcd(2(40 - n) + 2n + 1, 2n + 1)$$

$$= \gcd(81, 2n + 1)$$

Hence  $d_n$  divides 81.

45. Using calculus, sketch the graph of following function on plain paper.

$$f(x) = \frac{5 - 3x^2}{1 - x^2}.$$

Ans: **Clearly**

i)  $f$  cuts  $y$ -axis at  $(0, 5)$  and  $x$ -axis at  $(\pm \frac{\sqrt{5}}{\sqrt{3}}, 0)$ .

ii)  $f(-x) = f(x)$  implying it is even function and symmetric about  $Y$ -axis

iii) It has infinite discontinuity at  $x = \pm 1$

iv)  $f'(x) = \frac{(1 - x^2)(-6x) - (5 - 3x^2)(-2x)}{(1 - x^2)^2}$  which has root at  $x = 0$

v)  $f'(x)$  is positive in right and negative in left of  $x = 0$  So  $f$  increases at  $x > 0$ , decreases at  $x < 0$ , minimum at  $x = 0$

vi)  $f''(x) = \frac{(1 - x)(1 + x)(12x^2 + 4)}{(1 - x^2)^4}$  Clearly  $f'' > 0$  at  $(-1, 1)$  and  $f'' < 0$  otherwise. So  $f$  is concave up at  $(-1, 1)$  concave down otherwise.

vii) Let  $y = f(x)$  then  $y(1 - x^2) + 5 - 3x^2 = 0$  Since it is a degree two equation hence have at most two asymptotes.

viii) equating Coefficient of  $y$  to 0 we get  $1 - x^2 = 0$  so  $x = \pm 1$  so those are the only asymptotes.

Using above conditions we trace the curve.

46. Let,  $P_1, P_2, \dots, P_n$  be polynomials of  $x$ , each having all integer coefficients, such that

$$P_1 = P_1^2 + P_2^2 + \dots + P_n^2.$$

Assume that  $P_1$  is not the zero polynomial. Show that  $P_1 = 1$  and  $P_2 = P_3 = \dots = P_n = 0$ .

Ans:

$$P_1 - P_1^2 = P_2^2 + P_3^2 + \dots + P_n^2 \geq 0$$

$$\Rightarrow P_1 \geq P_1^2$$

$$0 \leq P_1 \leq 1$$

But a polynomial can never be bounded,

Because after a certain time [after  $\text{Max}(\text{min})\{r_i\}$ , where  $r_i$ 's are the root of a polynomial] polynomial becomes monotonic, unless,

1.) Polynomial is constant

2.) Polynomial becomes constant after a certain point

But case 2 is not possible, because in that case  $G(x) = P(x) - c = 0$  has infinitely many roots, hence  $G(x)$  becomes an identity  $\Rightarrow P(x) = c$  [for all  $x$ ]

$$\text{Hence, } 0 \leq P_1 = c \leq 1$$

But,  $P_1$  has integer coefficients,

$$P_1 = 1 \text{ [} P_1 \text{ is not a zero polynomial]}$$

$$P_1 = P_1^2 + P_2^2 + \dots + P_n^2$$

$$\Rightarrow P_2^2 + \dots + P_n^2 = 0$$

$$\Rightarrow P_2 = \dots = P_n = 0$$

47. Let  $P(x) = x^4 + ax^3 + bx^2 + cx + d$ , where  $a, b, c, d$  are integers. The sums of the pairs of roots of  $P(x)$  are given by 1, 2, 5, 6, 9, 10. Find  $P(\frac{1}{2})$ .

Ans: Let the roots of  $P(x)$  be  $x_1, x_2, x_3, x_4$

Wlog,  $x_1 \leq x_2 \leq x_3 \leq x_4$

Now, the largest sum is  $x_4 + x_3 = 10$

and second largest sum is  $x_4 + x_2 = 9$

And, smallest sum is  $x_1 + x_2 = 1$

Second smallest sum is  $x_1 + x_3 = 2$

But we can't say anything about  $(x_1 + x_4)$  and  $(x_2 + x_3)$

So, two cases arise,

Case-1:

$$x_1 + x_4 = 5, x_2 + x_3 = 6$$

$$x_4 + x_3 = 10, x_2 + x_4 = 9, x_1 + x_2 = 1, x_1 + x_3 = 2$$

Solving these we get,

$$x_1 = -\frac{3}{2}, x_2 = \frac{5}{2}, x_3 = \frac{7}{2}, x_4 = \frac{13}{2}$$

But this case is not possible as  $d = x_1 x_2 x_3 x_4 = -\frac{1365}{16}$  [not an integer]

Case 2:

$$x_1 + x_4 = 6, x_2 + x_3 = 5$$

$$x_4 + x_3 = 10, x_2 + x_4 = 9, x_1 + x_2 = 1, x_1 + x_3 = 2$$

Solving these we get,

$$x_1 = -1, x_2 = 2, x_3 = 3, x_4 = 7$$

And The polynomial becomes,

$$P(x) = (x - 7)(x - 3)(x - 2)(x + 1)$$

$$P\left(\frac{1}{2}\right) = -\frac{13}{2} * \frac{5}{2} * \frac{3}{2} * \frac{3}{2} = -\frac{585}{16}.$$

48. Let  $P$  be a point  $(3, 4)$  and  $Q$  be  $(x, \sqrt{25 - x^2})$ . If  $M(x)$  be slope of  $PQ$  find  $\lim_{x \rightarrow 3} M(x)$ .

Ans: The required slope  $M(x) = \frac{\sqrt{25 - x^2} - 4}{x - 3}$

$$\text{So } \lim_{x \rightarrow 3} M(x) = \lim_{x \rightarrow 3} \frac{\sqrt{25 - x^2} - 4}{x - 3}$$

$$= \lim_{x \rightarrow 3} -\frac{2x}{-2\sqrt{25 - x^2}} \text{ [by L hospital rule]}$$

$$= -3/4.$$

49. Show that for all  $n = 1, 2, 3, \dots$

$$\frac{1}{2} * \frac{3}{4} * \frac{5}{6} \dots * \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}.$$

Ans: Let  $P(n) : \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \dots \frac{2n-1}{2n} \leq \frac{1}{\sqrt{3n+1}}, n \in \mathbb{N}$

$$P(1) : \frac{1}{2} \leq \frac{1}{2} \text{ which is true.}$$

Let  $P(k)$  be true

$$P(k) : \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \dots \frac{2k-1}{2k} \leq \frac{1}{\sqrt{3k+1}}$$

$$P(k+1) : \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \dots \frac{2k-1}{2k} \cdot \frac{2k+1}{2k+2} \leq \frac{1}{\sqrt{3k+1}} \cdot \frac{2k+1}{2k+2}$$

$$\text{Now } \frac{1}{\sqrt{3k+1}} \cdot \frac{2k+1}{2k+2} \leq \frac{1}{\sqrt{3k+4}} \iff (2k+1)^2(3k+4) \leq (2k+2)^2(3k+1)$$

$$\iff 3(2k+1)^2 \leq (3k+1)(4k+3) \iff 0 \leq k \text{ which is true.}$$

50. Let,  $f(x, y) = xy$  where  $x \geq 0$  and  $y \geq 0$ . Prove that the function  $f$  satisfies the following property:

$$f(\lambda x + (1 - \lambda)x', \lambda y + (1 - \lambda)y') > \min\{f(x, y), f(x', y')\}$$

for all  $(x, y)$  not equal to  $(x', y')$  and for all  $\lambda \rightarrow (0, 1)$ .

Ans: Denote  $\lambda$  by  $k$  and  $1 - \lambda$  by  $l$ . Hence  $k + l = 1$ .

Without loss of generality we assume  $xy < x'y'$ .

$$(\lambda x + (1 - \lambda)x')(\lambda y + (1 - \lambda)y') = (kx + lx')(ky + ly').$$

$$= k^2xy + l^2x'y' + kl(x'y + xy') > k^2xy + l^2xy + klxy \left( \frac{x'}{x} + \frac{y'}{y} \right).$$

$$\geq k^2xy + l^2xy + 2klxy \left( \sqrt{\frac{x'y'}{xy}} \right). \quad [\text{AM-GM}]$$

$$> k^2xy + l^2xy + 2klxy = xy(k + l)^2 = xy = \min\{xy, x'y'\}.$$

51. The sum of squares of the digits of a three-digit positive number is 146, while the sum of the two digits in the unit's and the ten's place is 4 times the digit in the hundred's place. Further, when the number is written in the reverse order, it is increased by 297. Find the number.

Ans: Let  $c$  is unit's digit,  $b$  is ten's digit and  $a$  is the hundred's digit.

Then from the given information we get

$$a^2 + b^2 + c^2 = 146 \dots (i)$$

$$b + c = 4a \dots (ii)$$

$$100a + 10b + c + 297 = 100c + 10a + a \dots (iii)$$

$$b + c = 4a$$

$$\text{From (iii) we get } 99(c - a) = 297$$

$$\Rightarrow (c - a) = 3$$

Now from (ii) we get

$$\Rightarrow b + 3 = 3a$$

$$\Rightarrow b = 3(a - 1)$$

So  $b$  is a multiple of 3,  $b$  can be 0, 3, 6, 9.

For  $b = 0$

not possible as

$$b = 0 \Rightarrow a = 1 \Rightarrow c = 4$$

$$\text{then } a^2 + b^2 + c^2 \neq 146$$

Similarly for  $b = 3 \Rightarrow a = 2 \Rightarrow c = 5$

&  $b = 6 \Rightarrow a = 3 \Rightarrow c = 6$  are not possible.

For  $b = 9 \Rightarrow a = 4 \Rightarrow c = 7$

$$a^2 + b^2 + c^2 = 146$$

So the only solution is 497.

52. Consider the equation  $x^3 + Gx + H = 0$ , where  $G$  and  $H$  are complex numbers. Suppose that this equation has a pair of complex conjugate roots. Show that both  $G$  and  $H$  are real.

Ans: Let three roots of the equation  $x^3 + Gx + H = 0$

are  $\alpha, \beta, \gamma$  [ Let  $\alpha, \beta$  are complex conjugates]

Now  $\alpha\beta\gamma = -H \dots$  (i)

$\alpha + \beta + \gamma = 0 \dots$  (ii)

$$\alpha\beta + \beta\gamma + \gamma\alpha = G \dots \text{(iii)}$$

From (ii) we get

$$\alpha + \beta + \gamma = 0 \text{ [ } \alpha, \beta \text{ are complex conjugates so they are real]}$$

$$\Rightarrow \gamma = \text{real}$$

Now as  $\gamma = \text{real}$

$$\beta\gamma + \gamma\alpha = \gamma(\beta + \alpha)$$

$$= \text{real} \times \text{real}$$

$$= \text{real} \dots \text{(iv)}$$

$$\alpha, \beta \text{ are complex conjugates so } \alpha\beta = \text{real} \dots \text{(v)}$$

$$\text{From (iv) \& (v) we get } \alpha\beta + \beta\gamma + \gamma\alpha = \text{real} + \text{real} = \text{real}$$

$$\Rightarrow G = \text{real} \text{ [from (iii)]}$$

Now  $\alpha\beta$  is real and  $\gamma$  is real

$$\text{so } \alpha\beta\gamma = \text{real}$$

$$\Rightarrow H = \text{real}.$$

53. Describe the set of all real numbers  $x$  which satisfy  $2^{\log_{2x+3} x} < 1$ .

$$\text{Ans: } 2^{\log_{2x+3} x} < 1 \Rightarrow \log_{2x+3} x < \frac{1}{2} \Rightarrow \frac{\ln x}{\ln(2x+3)} < \frac{1}{2} \dots (1)$$

Hence we have,  $x > 0$ .

$$\Rightarrow 2x + 3 > 3 \Rightarrow \ln(2x + 3) > 0.$$

Therefore from (1) we get,

$$2 \ln x < \ln(2x + 3)$$

$$\begin{aligned} \Rightarrow \ln(x^2) &< \ln(2x+3) \\ \Rightarrow x^2 &< 2x+3 \Rightarrow x^2 - 2x - 3 < 0 \Rightarrow (x+1)(x-3) < 0 \Rightarrow -1 < x < 3. \end{aligned}$$

But  $x > 0$ . Hence  $0 < x < 3$  or in other words,  $x \in (0, 3)$ .

54. Suppose that all roots of the polynomial equation

$x^4 - 4x^3 + ax^2 + bx + 1 = 0$  are positive real numbers. Show that all the roots of the polynomial are equal.

Ans: Let us assume the roots to be  $x_1, x_2, x_3, x_4$ .

Clearly we have  $x_i > 0$  for all  $i$ .

Also from the polynomial we have,

$$x_1 + x_2 + x_3 + x_4 = 4 \text{ and } x_1 x_2 x_3 x_4 = 1.$$

Since  $x_i$ 's are positive applying  $AM - GM$  on  $x_i$ 's we have,

$$\frac{x_1 + x_2 + x_3 + x_4}{4} \geq (x_1 x_2 x_3 x_4)^{\frac{1}{4}}.$$

Evaluating, we get the equality case, when  $AM = GM$ . Hence all  $x_i$ 's are equal.

55. Let  $J = \{0, 1, 2, 3, 4\}$ . For  $x, y$  in  $J$  define  $x \oplus y$  to be the remainder of the usual sum of  $x$  and  $y$  after division by 5 and  $x \odot y$  to be the remainder of the usual product of  $x$  and  $y$  after division by 5. For example  $4 \oplus 3 = 2$  while  $4 \odot 3 = 2$ . Find  $x$  and  $y$  in  $J$ , satisfying the following equations simultaneously:

$$(3 \odot x) \oplus (2 \odot y) = 2, (2 \odot x) \oplus (4 \odot y) = 1.$$

Ans: Using directly the standard product and sum signs and considering we are in  $\mathbb{Z}/5\mathbb{Z}$ , the system is :

$$\begin{aligned} 3x + 2y &= 2 \\ 2x + 4y &= 1 \end{aligned}$$

Adding the two lines gives  $y = 3$  and multiplying the first line by two gives then  $x = 2$

Hence the answer :  $\boxed{(x, y) = (2, 3)}$ .

56. Show that every circle passing through  $(2, 0)$  and  $(-2, 0)$  is orthogonal to the circle  $x^2 + y^2 - 5x + 4 = 0$ .

Ans: Let equation of the family of circles through  $(2, 0)$  and  $(-2, 0)$  be

$$x^2 + y^2 + 2g_1x + 2f_1y + c_1 = 0$$

It passes through  $(2, 0)$  and  $(-2, 0) \implies$

$$4 + 4g_1 + c_1 = 0, 4 - 4g_1 + c_1 = 0 \text{ solving } c_1 = -4, g_1 = 0 \text{ whatever be } f_1$$

Comparing  $x^2 + y^2 - 5x + 4 = 0$  with general equation of circle  $x^2 + y^2 + 2g_2x + 2f_2y + c_2 = 0$  we get,  $g_2 = -5/2, f_2 = 0, c_2 = 4$

Condition for orthogonality is  $2g_1g_2 + 2f_1f_2 = c_1 + c_2, \dots\dots\dots(i)$

Putting respective values separately in **LHS** and **RHS** of expression (i) we see that condition for orthogonality is satisfied.

**HENCE:** Every circle passing through  $(2, 0)$  and  $(-2, 0)$  is orthogonal to the circle  $x^2 + y^2 - 5x + 4 = 0$ .

57. Suppose  $x_1 = \tan^{-1}2 > x_2 > x_3 > \dots$  are positive real numbers satisfying

$$\sin(x_{n+1} - x_n) + 2^{-(n+1)} \sin x_n \sin x_{n+1} = 0 \text{ for } n \geq 1$$

Find  $\cot x_n$ . Also show that  $\lim_{n \rightarrow \infty} x_n = \frac{\pi}{4}$ .

Ans: Given,  $\sin(x_{n+1} - x_n) + 2^{-(n+1)} \sin x_n \sin x_{n+1} = 0$

$$\Rightarrow \sin x_{n+1} \cos x_n - \sin x_n \cos x_{n+1} = -\frac{1}{2^{n+1}} \sin x_n \sin x_{n+1}$$

$$\Rightarrow \cot x_n - \cot x_{n+1} = -\frac{1}{2^{n+1}} \text{ [dividing by } \sin x_n \sin x_{n+1}]$$

Hence,  $\cot x_{n+1} - \cot x_n = \frac{1}{2^{n+1}}$

$$\cot x_n - \cot x_{n-1} = \frac{1}{2^n}$$

.....

.....

$$\cot x_2 - \cot x_1 = \frac{1}{2^2}$$

adding,

$$\cot x_n - \cot x_1 = \frac{1}{2^2} + \dots + \frac{1}{2^n}$$

$$\Rightarrow \cot x_n - \cot(\tan^{-1}2) = \frac{1}{2^2} + \dots + \frac{1}{2^n}$$

$$\Rightarrow \cot x_n = \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^n} = 1 - \frac{1}{2^n}$$

So  $\lim_{n \rightarrow \infty} \cot x_n = 1$

or  $\boxed{\lim_{n \rightarrow \infty} x_n = \frac{\pi}{4}}$ .

58. Consider the set of points

$$S = \{(x, y) : x, y \text{ are non-negative integers } \leq n\}$$

Find the number of squares that can be formed with vertices belonging to  $S$  and sides parallel to coordinate axes.

Ans: Let us start with the squares of sides 1 square unit.

Least possible value of top right corner coordinate is  $(1, 1)$  Shifting continuously to right from  $(1, 1)$ , possible values of  $(i, j)$  are  $(2, 1)$ ,  $(3, 1)$ , ...,  $(n, 1)$ . Moving continuously upwards from  $(1, 1)$  possible points are  $(1, 2)$ , ...,  $(1, n)$  so there are  $n^2$  squares of size 1 sq. units

For 4 sq unit squares, starting from  $(2, 2)$  we have  $(3, 2)$ ,  $(4, 2)$ , ...,  $(n, 2)$  on right, and  $(2, 3)$ ,  $(2, 4)$ ,  $(2, n)$  upwards. So there are  $(n - 1)^2$  squares of 4 sq. units

Similarly for 9, 16, ...,  $n^2$  sq unit squares, they are respectively  $(n - 2)^2$ ,  $(n - 3)^2$ , ...,  $1^2$  in number.

$$\text{So total squares} = \sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

59. Prove that,  $\int_0^\pi \left| \frac{\sin(nx)}{x} \right| dx > \frac{2}{\pi} \left( 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right)$

Ans: Let  $nx = t \implies dx = \frac{dt}{n}$

$$\text{So } \int_0^\pi \left| \frac{\sin(nx)}{x} \right| dx = \int_0^{n\pi} \left| \frac{\sin t}{t} \right| dt$$

$$\text{or } \int_0^\pi \left| \frac{\sin(nx)}{x} \right| dx = \sum_{r=1}^n \int_{(r-1)\pi}^{r\pi} \left| \frac{\sin t}{t} \right| dt$$

$$\text{Let } \mathbf{I} = \int_{(r-1)\pi}^{r\pi} \left| \frac{\sin t}{t} \right| dt$$

$$\begin{aligned} \text{or } \mathbf{I} &= \int_0^\pi \left| \frac{\sin(r-1)\pi + p}{(r-1)\pi + p} \right| dp \\ &= \int_0^\pi \frac{\sin p}{(r-1)\pi + p} dp \end{aligned}$$

Now,  $(r-1)\pi + p \leq r\pi$  in  $0 \leq p \leq \pi$

$$\text{So } \mathbf{I} = \int_{(r-1)\pi}^{r\pi} \left| \frac{\sin t}{t} \right| dt \geq \frac{1}{r\pi} \int_0^\pi (\sin t) dt = \frac{2}{r\pi}$$

$$\text{or } = \int_0^{n\pi} \left| \frac{\sin x}{x} \right| dx \geq \frac{2}{\pi} \sum_{r=1}^n \frac{1}{r}$$

$$\text{or } \int_0^\pi \left| \frac{\sin(nx)}{x} \right| dx \geq \frac{2}{\pi} \left( 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} \right).$$

60. i) If  $k$  and  $l$  are positive integers such that  $k$  divides  $l$ , show that for every positive integer  $m$ ,  $1 + (k+m)l$  and  $1 + ml$  are relatively prime.

ii) Consider the smallest number in each of  $\binom{n}{r}$  subsets (of size  $r$ ) of  $S = 1, 2, \dots, n$ . Show that arithmetic mean of the numbers thus obtained is  $\frac{n-1}{r-1}$ .

Ans: i)

$$\gcd[1 + (k+m)l, 1 + ml] = \gcd[1 + (k+m)l - (1 + ml), 1 + ml] = \gcd(kl, 1 + ml)$$

Let  $l = kr$  (since  $k$  divides  $l$ )

$$\text{So } \gcd(kl, 1 + ml) = \gcd(k^2r, 1 + kmr)$$

The prime factors of  $k^2r$  and  $kr$  are same. Primes dividing  $k^r$  divides  $mkr$  as well. Hence no primes dividing  $k^2r$  divides  $mkr + 1$

Hence  $\gcd(k^2r, mkr + 1) = 1$

So,  $1 + (k + m)l$  and  $1 + ml$  are relatively prime.

ii) Consider  $A = \{0, 1, \dots, n\}$ ,  $B = \{1, \dots, n\}$ . Consider a graph  $G$  with white vertices as  $r + 1$  element subsets of  $A$  and black vertices with  $r$  element subsets of  $B$ .

We join a black vertex  $V$  and a white vertex  $V'$  iff by removing the lowest element from the subset represented by  $V$  we get the elements of  $V'$ . For e.g if  $V = \{1, 3, 4, 5\}$  and  $V' = \{3, 4, 5\}$  we join them.

clearly degree(number of lines originating from a vertex) of each white vertex  $V'$  is its lowest element, e.g.  $V' = \{2, 3\}$  implies it is joined to  $\{0, 2, 3\}$  and  $\{1, 2, 3\}$  so degree of  $V' = 2$  which is its smallest element.

So sum of smallest elements

is the total edges(number of line segments) of  $G$  which is  $\binom{n+1}{r+1}$ .

again sum of smallest elements =  $\binom{n}{r} F(n, r)$

equating  $\binom{n+1}{r+1} = \binom{n}{r} F(n, r)$  implying

$$\boxed{F(n, r) = \frac{n - 1}{r - 1}}$$

61. Find all possible real numbers  $a, b, c, d, e$  which satisfy the following set of equations:

$$\begin{cases} 3a = (b+c+d)^3 \\ 3b = (c+d+e)^3 \\ 3c = (d+e+a)^3 \\ 3d = (e+a+b)^3 \\ 3e = (a+b+c)^3. \end{cases}$$

Ans: Without loss of generality, we may assume that  $a$  is the greatest among the five given numbers. Then  $d+e+a \geq c+d+e$ , and hence  $c = (d+e+a)^3/3 \geq (c+d+e)^3/3 = b$ .

Also,  $e = (a+b+c)^3/3 \geq (b+c+d)^3/3 = a$ , and since  $a \geq e$ , we get  $a = e$ . Hence also  $e \geq b$ , and  $b = (c+d+e)^3/3 \geq (b+c+d)^3/3 = a$  - giving  $a = b$ . Also, since  $c \geq b$ , we get  $a = c$ , and finally we also get  $a = d$ . So, we need to solve  $3a = (3a)^3$  which gives the following solutions:  $(1/3, \dots, 1/3)$ ,  $(0, \dots, 0)$  and  $(-1/3, \dots, -1/3)$ .

62. If  $a, b$  are positive real numbers such that  $a+b=1$  prove that,

$$(a + \frac{1}{a})^2 + (b + \frac{1}{b})^2 \geq \frac{25}{2}.$$

Ans: Let

$$P = (a + \frac{1}{a})^2 + (b + \frac{1}{b})^2 = a^2 + b^2 + \frac{1}{a^2} + \frac{1}{b^2} + 2 + 2 = (a-b)^2 + 2ab + \frac{2}{ab} + (\frac{1}{a} - \frac{1}{b})^2$$

$P$  is minimum iff both  $(a-b)^2$  and  $(\frac{1}{a} - \frac{1}{b})^2$  are 0 i.e.  $a = b$

But  $a+b=1$ . So  $a=b=1/2$

So minimum value of  $P$  is  $(1/2 + 2)^2 + (1/2 + 2)^2 = \frac{25}{2}$

So  $(a + \frac{1}{a})^2 + (b + \frac{1}{b})^2 \geq \frac{25}{2}.$

63. Suppose  $f$  is a function such that  $f(x) > 0$  and  $f'(x)$  is continuous for every real number  $x$ . If  $f'(t) \geq \sqrt{f(t)}$  for all  $t$ , then show that  $\sqrt{f(x)} \geq \sqrt{f(1)} + \frac{1}{2}(x - 1)$  for all  $x \geq 1$ .

Ans: We have,

$$f'(t) \geq \sqrt{f(t)} \implies \frac{f'(t)}{\sqrt{f(t)}} dt \geq dt$$

$$\implies \int_1^x \frac{f'(t)}{\sqrt{f(t)}} dt \geq \int_1^x dt = x - 1$$

$$\implies \left[ \frac{\sqrt{f(x)} - \sqrt{f(1)}}{\frac{1}{2}} \right] \geq x - 1$$

$$\implies \boxed{\sqrt{f(x)} \geq \sqrt{f(1)} + \frac{1}{2}(x - 1)}$$

64. Study the derivative of the function

$f(x) = \ln x - (x - 1)$  for  $x > 0$ ,  
and roughly sketch the graph of  $f(x)$  on plain paper.

Ans:

$$f(x) = \ln x - (x - 1), \quad f'(x) = \frac{1}{x} - 1, \quad f''(x) = \frac{-1}{x^2}$$

Clearly  $f'(x) = 0 \implies x = 1$  Clearly  $f'(x) \geq 0$  for  $0 < x \leq 1$  and  $f'(x) < 0$  for  $x > 1$

So  $f$  is maximum at  $x = 1$ . It increases at left of  $x = 1$  and decreases at right of  $x = 1$

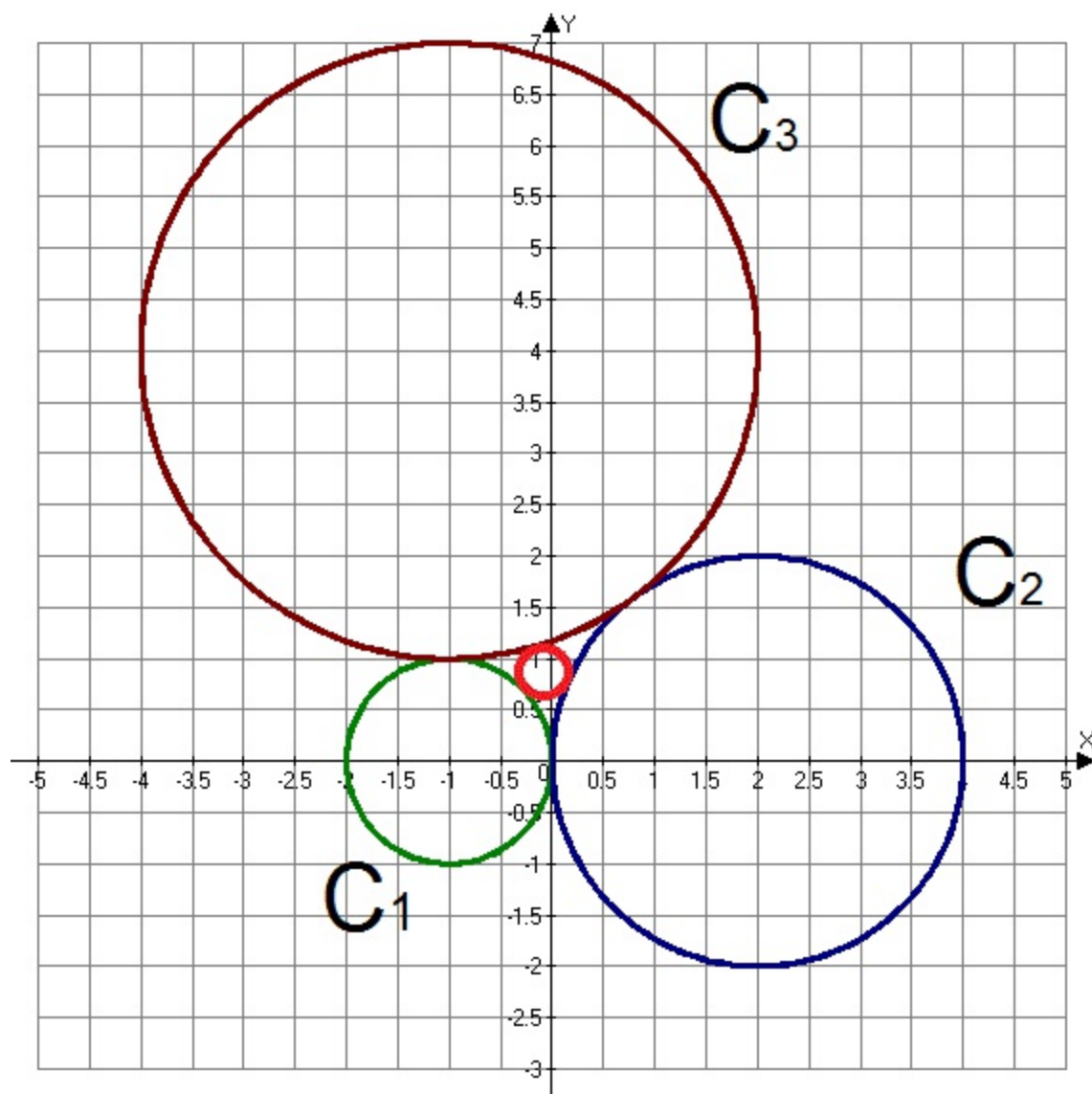
Maximum of  $f(x) = f(1) = \ln 1 - (1 - 1) = 0$

Clearly  $f''(x) < 0$  for all  $x > 0$ . Hence it is concave down.

**Using above information we trace the curve.**

65. The circles  $C_1, C_2, C_3$  with radii 1, 2, 3 respectively touch each other externally. Centres of  $C_1, C_2$  lie on  $x$ -axis.  $C_3$  touches them from top. Find ordinate of centre of circle that lies in region enclosed by  $C_1, C_2, C_3$  and touches all of them.

Ans:



WLOG, assume  $C_1, C_2$  pass through origin since only y-coordinate is needed.  
So centres of  $C_1, C_2$  are  $A(-1, 0)$  and  $B(2, 0)$  respectively.

Let centres of  $C_3$  be  $C(\alpha, \beta)$  and centre of required circle  $P$  be  $D(a, b)$  with radius  $r$ .

$C_3$  touch  $C_1, C_2$  hence,

$$(\alpha + 1)^2 + (\beta)^2 = (3 + 1)^2 = 16$$

$$(\alpha - 2)^2 + (\beta)^2 = (3 + 2)^2 = 25$$

Solving, centre of  $C_3$  is  $C(-1, 4)$

$P$  touches  $C_1, C_2, C_3 \implies$

$$(a+1)^2 + b^2 = (r+1)^2, (a-2)^2 + b^2 = (r+2)^2, (a+1)^2 + b^2 = (r+3)^2$$

Solving we get  $b = 2/3$  [Since  $C_3$  touches from top, we take positive value of  $b$ ]

**So required ordinate = 2/3.**

66. If  $a, b$  and  $c$  are the lengths of the sides of a triangle  $ABC$  and if  $p_1, p_2$  and  $p_3$  are the lengths of the perpendiculars drawn from the circumcentre onto the sides  $BC, CA$  and  $AB$  respectively, then show that

$$\frac{a}{p_1} + \frac{b}{p_2} + \frac{c}{p_3} = \frac{abc}{4p_1p_2p_3}.$$

Ans: From geometry, we have,

$a = 2R \sin A$  and  $p_1 = R \cos A$  where  $R$  is the circumradius of  $\triangle ABC$  and  $A$  is  $\angle BAC$ .

Hence we have,

$$\frac{a}{p_1} = \frac{2R \sin A}{R \cos A} = 2 \tan A.$$

Similarly we have,

$$\frac{b}{p_2} = 2 \tan B \quad \text{and} \quad \frac{c}{p_3} = 2 \tan C.$$

For the angles  $A, B$  and  $C$  of a triangle we have,

$$\tan A + \tan B + \tan C = \tan A \tan B \tan C.$$

$$\therefore \frac{a}{p_1} + \frac{b}{p_2} + \frac{c}{p_3} = 2 \tan A + 2 \tan B + 2 \tan C = 2(\tan A + \tan B + \tan C) = 2 \tan A \tan B \tan C = \frac{1}{4} (2 \tan A)(2 \tan B)(2 \tan C) =$$

$$\frac{1}{4} \cdot \frac{a}{p_1} \cdot \frac{b}{p_2} \cdot \frac{c}{p_3} = \frac{abc}{4p_1p_2p_3}.$$

67. Show that there is exactly one value of  $x$  which satisfies the equation

$$2\cos^2(x^3 + x) = 2^x + 2^{-x}.$$

Ans: we know that  $2\cos^2(x^3 + x) \leq 2$

Also  $2^x + 2^{-x} \geq 2$  [By **AM-GM**]

So  $2^x + 2^{-x} = 2\cos^2(x^3 + x)$  iff  $2^x + 2^{-x} = 2$  and  $2\cos^2(x^3 + x) = 2$  holds.

Now  $2^x + 2^{-x} = 2$  iff  $2^x = 2^{-x}$  or  $2^{2x} = 1 = 2^0$  so  $x = 0$

Also  $x = 0$  satisfies  $2\cos^2(x^3 + x) = 2$  so exactly one value of  $x$  satisfies

$$2\cos^2(x^3 + x) = 2^x + 2^{-x}$$

which is  $x = 0$ .

68. If  $A = \int_0^\pi \frac{\cos(x)}{(x+2)^2} dx$ , then show that,

$$\int_0^{\frac{\pi}{2}} \frac{\sin(x)\cos(x)}{x+1} dx = \frac{1}{2} \left( \frac{1}{2} + \frac{1}{\pi+2} - A \right).$$

Ans:

$$A = \int_0^\pi \frac{\cos x \cdot dx}{(x+2)^2}$$

put  $x = 2t$

$$\frac{A}{2} = \int_0^{\frac{\pi}{2}} \frac{\cos 2t \cdot dt}{(2t+2)^2}$$

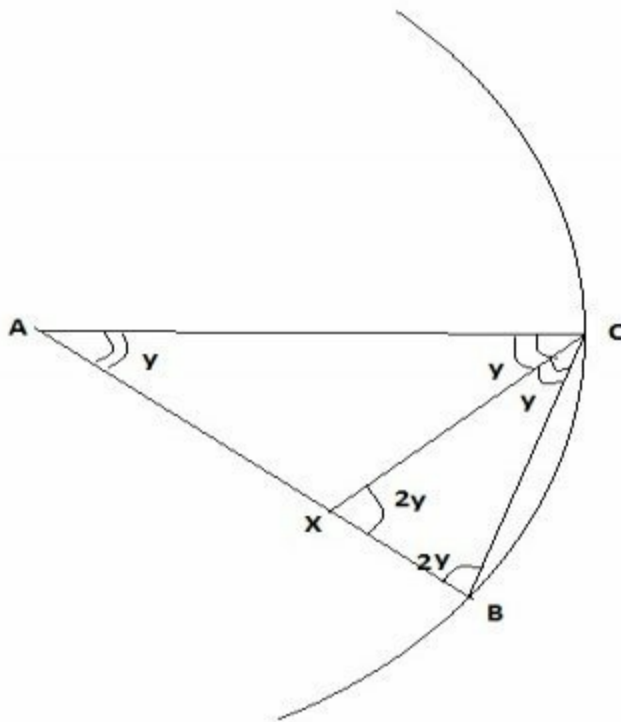
$$\frac{A}{2} = \frac{-\cos 2t}{2(2t+2)} \Big|_0^\pi - \int_0^{\frac{\pi}{2}} \frac{\sin 2t \cdot dt}{2t+2} \quad [\text{integration by parts}]$$

by evaluating the limits and  $\sin 2t = 2 \sin t \cdot \cos t$  we get,

$$\int_0^{\frac{\pi}{2}} \frac{\sin x \cdot \cos x \cdot dx}{x+1} = \frac{1}{2} \left( \frac{1}{2} + \frac{1}{\pi+2} - A \right)$$

69. Let  $X$  be a point on a straight line segment  $AB$  such that  $AB \cdot BX = AX^2$ . Let  $C$  be a point on the circle with centre at  $A$  and radius  $AB$  such that  $BC = AX$ . (see figure). Show that  $\angle BAC = 36^\circ$ .

Ans:



Ans: Refer to the *figure attached above*

We are given following information:

- 1.)  $AB \cdot BX = AX \cdot AX$
- 2.)  $BC = AX$

From (1) we easily infer  $AB/AX = AX/BX$ ;  $\Rightarrow AC/AX = AX/BX$ ; [Since  $AB = AC = \text{radius of the circle we r working with}$ ]

$\Rightarrow AC/BC = AX/BX$ ; [ $BC = AX$ ]

$\Rightarrow CX$  is the internal bisector of  $\angle ACB$ ; let  $\angle ACX = \angle BCX = y$

And we also observe readily that  $\triangle ACB$  is isosceles.

$\Rightarrow \angle ABC = \angle ACB = 2y$

Again using the given fact

$AB \cdot BX = AX \cdot AX \Rightarrow AB \cdot BX = BC \cdot BC; \Rightarrow AB/BC = BC/BX;$

So, we are having  $\triangle ABC$  and  $\triangle XBC$  in which

i)  $AB/BC = BC/BX$

and

ii)  $\angle ABC = \angle XBC = 2y$

$\Rightarrow \triangle ABC \sim \triangle XBC$

so now corresponding angles of the similar  $\triangle s$  will be equal.

$\Rightarrow$  the opposite  $\angle$  of  $AB$  in  $\triangle ABC =$  the opposite  $\angle$  of  $BC$  in  $\triangle XBC$

$\Rightarrow \angle BXC = \angle ACB = 2y; \Rightarrow$  from  $\triangle XBC, 2y + 2y + y = 180^\circ; \Rightarrow 5y = 180^\circ; \Rightarrow y = 36^\circ;$

Hence readily we get  $\angle BAC = y = 36^\circ$ .

70. Evaluate:

$$\lim_{n \rightarrow \infty} \left( \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n} \right).$$

Ans:

$$\begin{aligned} & \lim_{n \rightarrow \infty} \left( \frac{1}{n+1} + \frac{1}{n+2} + \dots + \frac{1}{n+n} \right) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \frac{1}{n+i} = \lim_{n \rightarrow \infty} \frac{1}{n} \cdot \sum_{i=1}^n \frac{1}{1+\frac{i}{n}} = \int_0^1 \frac{1}{1+x} = \ln(1+x) \Big|_0^1 = \ln 2 \end{aligned}$$

71. Evaluate:

$$\lim_{n \rightarrow \infty} \left[ \left(1 + \frac{1}{2n}\right) \left(1 + \frac{3}{2n}\right) \left(1 + \frac{5}{2n}\right) \dots \left(1 + \frac{2n-1}{2n}\right) \right]^{\frac{1}{2n}}.$$

Ans: ,

$$\begin{aligned}
I &= \lim_{n \rightarrow \infty} \left\{ \prod_{r=1}^n \left( 1 + \frac{2r-1}{2n} \right) \right\}^{\frac{1}{2n}} \\
\Rightarrow \ln I &= \lim_{n \rightarrow \infty} \frac{1}{2n} \sum_{r=1}^n \ln \left( 1 + \frac{2r-1}{2n} \right) \\
&= \lim_{n \rightarrow \infty} \left\{ \frac{1}{2n} \sum_{r=1}^{2n} \ln \left( 1 + \frac{r}{2n} \right) - \frac{1}{2n} \sum_{r=1}^n \ln \left( 1 + \frac{2r}{2n} \right) \right\} \\
&= \lim_{2n \rightarrow \infty} \frac{1}{2n} \sum_{r=1}^{2n} \ln \left( 1 + \frac{r}{2n} \right) - \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n} \sum_{r=1}^n \ln \left( 1 + \frac{r}{n} \right) \\
&= \int_0^1 \ln(1+x) dx - \frac{1}{2} \int_0^1 \ln(1+x) dx \\
&= \frac{1}{2} \int_0^1 \ln(1+x) dx \\
&= \frac{1}{2} \int_1^2 \ln x dx \\
&= \frac{1}{2} [x \ln x - x]_1^2 \\
&= \frac{1}{2} \{2 \ln 2 - 1\} \\
&= \ln 2 - \frac{1}{2} \\
\Rightarrow I &= \frac{2}{\sqrt{e}}
\end{aligned}$$

72. Show that for all real  $x$ , the expression  $ax^2 + bx + c$  (where  $a, b, c$  are real constants with  $a > 0$ ) has minimum value  $\frac{4ac - b^2}{4a}$ . Also find the value of  $x$  for which the minimum value is attained.

Ans: Let  $f(x) = ax^2 + bx + c \Rightarrow f'(x) = 2ax + b \Rightarrow f''(x) = 2a$

Now equating  $f'(x) = 0$  we have  $x = -b/2a$ . Also  $f''(x)$  is always positive [ $a > 0$  given]

So  $f(x)$  is minimum at  $x = -b/2a$ .

The minimum value is  $a\left(\frac{-b}{2a}\right)^2 + b\left(\frac{-b}{2a}\right) + c = \frac{(4ac - b^2)}{4a}$ .

73. If  $\frac{\sin^4(x)}{a} + \frac{\cos^4(x)}{b} = \frac{1}{a+b}$ , then show that,  

$$\frac{\sin^6(x)}{a^2} + \frac{\cos^6(x)}{b^2} = \frac{1}{(a+b)^2}.$$

Ans: We have

$$\begin{aligned} \frac{\sin^4 x}{a} + \frac{\cos^4 x}{b} &= \frac{1}{a+b} \\ \Rightarrow \frac{\tan^4 x}{a} + \frac{1}{b} &= \frac{\sec^4 x}{a+b} \\ \Rightarrow \frac{z^2}{a} + \frac{1}{b} &= \frac{(1+z)^2}{a+b} \quad \text{where } z = \tan^2 x \end{aligned}$$

which simplifies off to

$$\begin{aligned} (bz - a)^2 &= 0 \\ \Rightarrow \tan^2 x &= \frac{\sin^2 x}{\cos^2 x} = \frac{a}{b} \\ \text{Thus } \frac{\sin^2 x}{a} &= \frac{\cos^2 x}{b} = k \quad \text{where } k = \frac{1}{a+b} \end{aligned}$$

Therefore required expression 
$$= \frac{a^3 k^3}{a^2} + \frac{b^3 k^3}{b^2} = \frac{1}{(a+b)^2}.$$

74. Consider the circle of radius 1 with its centre at the point (0, 1). From this initial position, the circle is rolled along the positive  $x$ -axis without slipping. Find the locus of point  $P$  on the circumference of the circle which is at the origin at initial position of circle.

Ans: Let the linear speed of the circle be  $v$ , angular speed be  $\omega$ , time  $t$  and radius  $r$  (I know radius is given, but still...). Further let  $\theta$  be the angle subtended at the center by the point at time  $t$ . Since the point is originally at the origin, hence initial phase  $\theta_0 = 0$ .  $x$  and  $y$  denotes the co-ordinates of the

point  $P$  at any time  $t$

Now, due to the motion of the  $x$  co-ordinate of the point, we have  
 $x = vt + r \sin \theta$ . Since the circle rolls without slipping, we have  $v = r\omega$  and  
hence  $x = r\omega t + r \sin \theta$

Now, coming to the  $y$ - coordinate, see that the  $y$  co-ordinate can be  
represented as a function of  $\theta$ , viz,  $y = r(1 - \cos \theta)$ . (Draw the diagram at any  
time  $t$  and try to see why this equation holds ....)

Now,  $\theta = \theta_0 + \omega t \implies \theta = \omega t$

Thus we have  $x = r\theta + r \sin \theta$

Now here  $r = 1$ , so we have  $y = 1 - \cos \theta$  and  $x = \theta + \sin \theta$  which is the  
required locus of the point.

75. Let  $a_0$  and  $B_0$  be two positive integers. Define  $a_n, b_n$  for  $n \geq 1$  using the  
relations  $a_n = a_{n-1} + 2b_{n-1}$ ,  $b_n = a_{n-1} + b_{n-1}$  and let  $c_n = \frac{a_n}{b_n}$  for  $n = 0, 1, 2, \dots$

a) Write  $(\sqrt{2} - c_{n+1})$  in terms of  $(\sqrt{2} - c_n)$

b) Show that  $|\sqrt{2} - c_{n+1}| < \frac{1}{1 + \sqrt{2}} |\sqrt{2} - c_n|$

c) Show that  $\lim_{n \rightarrow \infty} c_n = \sqrt{2}$ .

Ans: a) 
$$\sqrt{2} - c_{n+1} = \sqrt{2} - \frac{a_{n+1}}{b_{n+1}} = \sqrt{2} - \frac{a_n + 2b_n}{a_n + b_n}$$

$$= \frac{(a_n + \sqrt{2}b_n)(\sqrt{2} - 1)}{a_n + b_n} = (\sqrt{2} - 1) \frac{(\frac{a_n}{b_n} - \sqrt{2})}{\frac{a_n}{b_n} + 1}$$

$$= (1 - \sqrt{2}) \frac{\sqrt{2} - c_n}{c_n + 1}$$

b) 
$$\sqrt{2} - c_{n+1} = (1 - \sqrt{2}) \frac{\sqrt{2} - c_n}{c_n + 1} = \frac{1}{1 + \sqrt{2}} \frac{c_n - \sqrt{2}}{c_n + 1}$$

$$\implies |\sqrt{2} - c_{n+1}| = \frac{1}{1 + \sqrt{2}} \left| \frac{c_n - \sqrt{2}}{c_n + 1} \right|$$

We have,  $\frac{1}{c_n + 1} < 1$

$$\Rightarrow \frac{1}{1 + \sqrt{2}} \left| \frac{c_n - \sqrt{2}}{c_n + 1} \right| < \left| \frac{\sqrt{2} - c_n}{1 + \sqrt{2}} \right|$$

$$\Rightarrow |\sqrt{2} - c_{n+1}| < \frac{1}{1 + \sqrt{2}} |\sqrt{2} - c_n|$$

c) Let  $\lim_{n \rightarrow \infty} c_n = L$

Now,  $c_{n+1} = \frac{a_{n+1}}{b_{n+1}} = \frac{a_n + 2b_n}{a_n + b_n} = \frac{c_n + 2}{c_n + 1}$

So  $L = \frac{L + 2}{L + 1} \Rightarrow L^2 = 2 \Rightarrow L = \sqrt{2}$

So  $\boxed{\lim_{n \rightarrow \infty} c_n = \sqrt{2}}$ .

76. Let  $S$  be the set of all sequences  $(a_1, a_2, \dots)$  of non-negative integers such that

(i)  $a_1 \geq a_2 \geq \dots$ ; and

(ii) there exists a positive integer  $N$  such that  $a_n = 0$  for all  $n \geq N$ .

Define the dual of the sequence  $(a_1, a_2, \dots)$  belonging to  $S$  to be the sequence  $(b_1, b_2, \dots)$ , where, for  $m \geq 1$  and  $b_m$  is the number of  $a_n$ 's which are greater than or equal to  $m$ .

(i) Show that the dual of a sequence in  $S$  belongs to  $S$ .

(ii) Show that the dual of the dual of a sequence in  $S$  is the original sequence itself.

(iii) Show that the duals of the distinct sequences in  $S$  are distinct.

Ans: 1) Since number of  $a_n$ 's which greater than  $i$  is  $\geq$  number which greater than  $i + 1$

So  $b_i \geq b_{i+1}$  for all  $i \in \mathbb{Z}^+$

And since number of  $a_n$ 's which  $\geq a_1 + 1$  is 0

This give us  $(b_n)$  belongs to  $S$

2) Since number of  $a_n$  which  $\geq a_i$  is equal to  $i$

So number of  $b_n$  which  $\geq i$  is equal to  $a_i$

So dual of dual of sequence is original sequence

3) Let  $f : S \rightarrow S$  is function that  $f(x) =$  dual of  $x$

We have  $f(f(x)) = x$  so  $f$  is injective

Hence the duals of the distinct sequences in  $S$  are distinct.

77. Find the value of

$$\int_2^{11} \frac{dx}{1-x}.$$

Ans:

$$\int_2^{11} \frac{dx}{1-x} = - \int_2^{11} \frac{dx}{x-1}$$

$$= -[\ln(x-1)]_2^{11} = -[\ln(11-1) - \ln(2-1)]$$

$$= -[\ln 10 - \ln 1] = -[\ln 10 - 0] = -\ln 10.$$

78. If  $f(x)$  is a real-valued function of a real variable  $x$ , such that  $2f(x) + 3f(-x) = 15 - 4x$  for all  $x$ , find the function  $f(x)$ .

Ans: We have,

$$2f(x) + 3f(-x) = 15 - 4x, \text{ for all real } x.$$

Substituting  $x$  by  $-x$  we get

$2f(-x) + 3f(x) = 15 + 4x$  for all real  $x$ .

Solving these two equations, we get,

$$\boxed{f(x) = 4x + 3} \text{ and also } f(-x) = -4x + 3.$$

79. Solve

$$6x^2 - 25x + 12 + \frac{25}{x} + \frac{6}{x^2} = 0.$$

Ans:

$$\begin{aligned} 6x^2 - 25x + 12 + \frac{25}{x} + \frac{6}{x^2} &= 0 \\ \Rightarrow 6 \left( x^2 + \frac{1}{x^2} \right) - 25 \left( x - \frac{1}{x} \right) + 12 &= 0 \\ \Rightarrow 6 \left[ \left( x - \frac{1}{x} \right)^2 + 2 \right] - 25 \left( x - \frac{1}{x} \right) + 12 &= 0 \\ \Rightarrow 6 \left( x - \frac{1}{x} \right)^2 - 25 \left( x - \frac{1}{x} \right) + 24 &= 0. \\ \left( x - \frac{1}{x} \right) &= t, \end{aligned}$$

Putting  $t = x - \frac{1}{x}$ , we get,

$$\begin{aligned} 6t^2 - 25t + 24 &= 0 \\ \Rightarrow (3t - 8)(2t - 3) &= 0 \\ \Rightarrow t &= \frac{8}{3} \text{ or } \frac{3}{2}. \end{aligned}$$

Now,

$$\begin{aligned} t = \frac{8}{3} &\Rightarrow x - \frac{1}{x} = \frac{8}{3} \\ \Rightarrow 3x^2 - 8x - 3 &= 0 \\ \Rightarrow (3x + 1)(x - 3) &= 0 \\ \Rightarrow x &= -\frac{1}{3} \text{ or } 3. \end{aligned}$$

Similarly,

$$\begin{aligned} t = \frac{3}{2} &\Rightarrow x - \frac{1}{x} = \frac{3}{2} \\ \Rightarrow 2x^2 - 3x - 2 &= 0 \\ \Rightarrow (2x + 1)(x - 2) &= 0 \\ \Rightarrow x &= -\frac{1}{2} \text{ or } 2. \end{aligned}$$

Hence solutions are

$$\boxed{x = -\frac{1}{2}, -\frac{1}{3}, 2, 3.}$$

80. If  $k$  is an odd positive integer, prove that for any  $n \geq 1$ ,  $1^k + 2^k + \dots + n^k$  is divisible by  $\frac{n(n+1)}{2}$ .

Ans: Let  $S = 1^k + 2^k + \dots + n^k$  again  $S = n^k + (n-1)^k + \dots + 1^k$

Adding we get,  $2S = (1^k + n^k) + (2^k + (n-1)^k) + \dots + (n^k + 1^k)$

Now, for odd  $k$ ,

$1+n|1^k + n^k$ ,  $2+(n-1)|2^k + (n-1)^k$ , .....,  $n+1|n^k + 1^k$  that is,  $n+1$  divides each term of  $2S$ .

So  $n+1|2S$

Again for odd  $k$

$n|2n^k$ ,  $2+n-2|2^k + (n-2)^k$ ,  $3+n-3|3^k + (n-3)^k$ , .....,  $n-2+2|(n-2)^k + 2^k$ ,  $n-1+1|(n-1)^k + 1^k$

So  $n|2S$

But  $n, n+1$  are coprime. So  $n(n+1)|2S$

Surely  $n(n+1)$  is even and  $2S$  is also so.

Hence  $\boxed{\frac{n(n+1)}{2} | S}$ .

81. Find the number of rational numbers  $\frac{m}{n}$ , where  $m, n$  are relatively prime positive integers satisfying  $m < n$  and  $mn = 25!$ .

Ans: The prime factors of  $25!$  are 2, 3, 5, 7, 11, 13, 17, 19, 23. So there are total 9 prime divisors of  $25!$

So number of ways  $25!$  can be resolved in two relatively prime factors  
 $= 2^{9-1} = 2^8$

Among the integer one is smaller, other bigger. Hence they can be expressed in form  $\frac{m}{n}$ ,  $m < n$  and  $mn = 25!$

So  $Answer = 2^8$ .

82. Let  $\{x_n\}$  be a sequence  $x_1 = 2, x_2 = 1$   
 $2x_n - 3x_{n-1} + x_{n-2} = 0$

for  $n > 2$ . Find an expression for  $x_n$ .

Ans: Putting  $x = 3$ , we will get  $x_3 = \frac{1}{2}$

Putting  $x = 4$ , we will get  $x_4 = \frac{1}{4}$

Putting  $x = 5$ , we will get  $x_5 = \frac{1}{8}$

So clearly we can see, that  $x_1 = 2, x_2 = 1, x_3 = \frac{1}{2}, x_4 = \frac{1}{4}, x_5 = \frac{1}{8}$ .

So let us claim that  $x_n = 2^{2-n}$ .

then we will use this to prove the given equation.

$$2 \cdot 2^{2-n} - 3 \cdot 2^{2-n+1} + 2^{2-n+2} = 2 \cdot 2^{2-n} - 6 \cdot 2^{2-n} + 4 \cdot 2^{2-n}$$

$$2x_n - 6x_n + 4x_n = 0$$

83. Show that,

$$2(\sqrt{251} - 1) < \sum_{k=1}^{250} \frac{1}{\sqrt{k}} < 2(\sqrt{250})$$

Ans: for  $k > 1$

$$\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k} + \sqrt{k}} < \frac{2}{\sqrt{k} + \sqrt{k-1}} = 2(\sqrt{k} - \sqrt{k-1})$$

$$\frac{1}{\sqrt{k}} = \frac{2}{\sqrt{k} + \sqrt{k}} > \frac{2}{\sqrt{k+1} + \sqrt{k}} = 2(\sqrt{k+1} - \sqrt{k})$$

Now,

Using 1<sup>st</sup> relation,

$$\frac{1}{\sqrt{1}} < 2(1 - 0)$$

$$\frac{1}{\sqrt{2}} < 2(\sqrt{2} - 1)$$

$$\frac{1}{\sqrt{250}} < 2(\sqrt{250} - \sqrt{249})$$

Adding we get,

$$\sum_{k=1}^{250} \frac{1}{\sqrt{k}} < 2\sqrt{250}$$

Similarly using second relation we can achieve,

$$2(\sqrt{251} - 1) < \sum_{k=1}^{250} \frac{1}{\sqrt{k}}$$

Combining the results we get,

$$2(\sqrt{251} - 1) < \sum_{k=1}^{250} \frac{1}{\sqrt{k}} < 2(\sqrt{250})$$

84. Let  $x = (x_1, x_2, \dots, x_n)$ ,  $y = (y_1, y_2, \dots, y_n)$ , where  $x_1, x_2, \dots, x_n, y_1, y_2, \dots, y_n$  are real numbers. We write  $x > y$ , if for some  $k$ ,  $1 \leq k \leq n-1$ ,  $x_1 = y_1, x_2 = y_2, \dots, x_k = y_k$  but  $x_{k+1} > y_{k+1}$ . Show that for  $u = (u_1, \dots, u_n)$ ,  $v = (v_1, \dots, v_n)$ ,  $w = (w_1, \dots, w_n)$  and  $z = (z_1, \dots, z_n)$ , if  $u > v$  and  $w > z$ , then  $(u + w) > (v + z)$ .

Ans:  $(u_1 + w_1 + u_2 + w_2 + \dots + u_k + w_k, \dots, u_m + w_m, \dots, u_n + w_n)$  and  $v + z = (v_1 + z_1 + v_2 + z_2 + \dots + v_k + z_k, \dots, v_m + z_m, \dots, v_n + z_n)$

Since  $u > v$  and  $w > z$ , we have  $u_i + w_i = v_i + z_i$  for all  $i \leq k$  and for  $i > k$   $u_i + w_i > v_i + z_i$

Hence we have,  $(u + w) > (v + z)$ .

85. Show that if  $n$  is any odd integer greater than 1, then  $n^5 - n$  is divisible by 80.

Ans: For odd  $n$  we have,

$$n^4 \equiv 1 \pmod{16} \text{ [See Tomato subjective 35(a).]}$$

$$\implies 16 | n^4 - 1.$$

If  $5 | n$ , then we have  $5 \cdot 16 | n(n^4 - 1)$ . [Since  $\gcd(5, 16) = 1$ .]

$$\implies 80 | n^5 - n.$$

Now if  $5 \nmid n$ , then,

$$n \equiv \pm 1, \pm 2 \pmod{5}$$

$$\implies n^2 \equiv \pm 1 \pmod{5}$$

$$\implies n^4 \equiv 1 \pmod{5}$$

$$\implies 5 | n^4 - 1.$$

$$\text{Hence } 5 \cdot 16 | n^4 - 1 \implies 80 | n^5 - n.$$

86. Let  $S = \{1, 2, 3, \dots, n\}$ . Find the number of unordered pairs  $\{A, B\}$  of subsets of  $S$  such that  $A$  and  $B$  are disjoint, where  $A$  or  $B$  or both may be empty.

Ans: You can simply pick, for each element, if it goes in  $A$ , in  $B$ , or in neither, for  $3^n$  ordered pairs of subsets.

Each ordered pair except for  $(\emptyset, \emptyset)$  is counted twice so the number of

unordered pairs is  $\frac{3^n + 1}{2}$ .

87. Consider the set  $SSS$  of all integers between and including  $\$100$  and  $\$999999$ . Call two integers  $\$x$  and  $\$y$  in  $SSS$  to be in same equivalence class if the digits appearing in  $\$x$  and  $\$y$  are the same. For example, if  $\$x = 1010$ ,  $\$y = 1000$  and  $\$z = 1201$ , then  $\$x$  and  $\$y$  are in same equivalence class, but  $\$y$  and  $\$z$  are not. Find the number of distinct equivalence classes that can be formed out of  $SSS$ .

Ans: For one digit classes, there are 9 classes, each for all numbers formed by each digit excluding 0.

For two digit classes, there are 45 classes for the same reason.

For three digit classes, there are  $\binom{10}{3} = 120$  classes;

For four digit classes, there are  $\binom{10}{4} = 210$  classes, and four five digit

classes, there are  $\binom{10}{5} = 252$  classes. So, in total, there are  $9 + 45 + 120 + 210 + 252 = \boxed{636}$  classes.

88. Suppose that one moves along the points  $(m, n)$  in the plane where  $m$  and  $n$  are integers in such a way that each move in a diagonal step, that is, consists of one unit to the right or left followed by one unit either up or down.

(a) What points  $(p, q)$  can be reached from the origin?

(b) What is the maximum number of moves needed to reach such a point  $(p, q)$ ?

Ans: The invariant is  $p + q \pmod{2}$ . So, from the origin, we will reach all points  $(2m + k, 2n + k)$ , where  $m, n \in \mathbb{Z}$ , and either  $k = 0$  or  $k = 1$ . That is coordinates with same parity can be reached.

For the maximum, assume that the point is  $(p, q)$ , where  $p \geq q$ . (Otherwise, the point symmetric to the line  $y = x$  will have the same number of moves required.) We can reach the y-coordinate  $q$  in  $q$  moves; all these moves should be to the same direction as the line  $y = \pm x$  in the quadrant goes.

This way, we need another  $p - q$  moves to reach the x-coordinate  $p$ . These moves should be in a zigzag pattern to ensure that we stay in the same y-coordinate. So in total, we need  $q + p - q = p$  moves. Since we assumed that

$p$  is the larger of the two values, our answer would be  $\boxed{\max(|p|, |q|)}$ .

89. Let  $P(z) = az^2 + bz + c$ , where  $a, b, c$  are complex numbers.

(a) If  $P(z)$  is real for all real numbers  $z$ , show that  $a, b, c$  are real numbers.

(b) In addition to (a) above, assume that  $P(z)$  is not real whenever  $z$  is not real. Show that  $a = 0$ .

Ans:

(a) If  $P(z)$  is real for all real numbers  $z$ , show that  $a, b, c$  are real numbers.

[/quote]

$P(0)$  is real  $\implies c$  is real.  $P(1) + P(-1) = 2a + 2c$  is real  $\implies a$  is real  
 $\implies P(1) - a = b$  is real and hence  $a, b, c$  are real.

(b) Suppose assume  $a \neq 0$ . Then, consider  $Q(z) = z^2 + \frac{b}{a}z + \frac{c}{a}$ . Let it have roots  $\alpha, \beta$ . If  $\alpha, \beta$  are imaginary, then  $P(\alpha) = 0$  must be imaginary which is false. So,  $\alpha, \beta$  are real. Now, using the fact that  $\alpha + \beta = -\frac{b}{a}$ , we have  
 coefficient of  $i$  in  $P\left(\frac{\alpha + \beta}{2} + i\right) = (\alpha + \beta) + \frac{b}{a} = 0$  and so,  $P\left(\frac{\alpha + \beta}{2} + i\right)$  is real while  $\left(\frac{\alpha + \beta}{2} + i\right)$  is imaginary which contradicts our assumption that  $a \neq 0 \implies a = 0$ .

90. A partition of a set  $S$  is formed by disjoint nonempty subsets of  $S$  whose union is  $S$ . For example  $\{\{1, 3, 5\}, \{2\}, \{4, 6\}\}$  is a partition of the set  $T = \{1, 2, 3, 4, 5, 6\}$  consisting of subsets  $\{1, 3, 5\}, \{2\}$  and  $\{4, 6\}$ . However  $\{\{1, 2, 3, 5\}, \{3, 4, 6\}\}$  is not a partition of  $T$ .

If there are  $k$  nonempty subsets in a partition, then it is called a partition into  $k$  classes. Let  $S_k^n$  stand for the number of different partitions of a set with  $n$  elements into  $k$  classes.

(i) find  $S_2^n$

(ii) Show that  $S_k^{n+1} = S_{k-1}^n + kS_k^n$ .

Ans: 2)

see partitioning  $n + 1$  elements in  $k + 1$  can happen in two ways.

let the elements are  $a_1, a_2, \dots, a_{n+1}$

then either case 1:  $a_1, a_2, \dots, a_n$  form  $k$  partitions which can be done in  $S_k^n$  ways and then the remaining element choose one of these  $k$ -partitions (so,  $k$

ways) and go to it.

so under case 1  $kS_k^n$  different possibilities.

case 2 :  $a_1, a_2, \dots, a_n$  form  $k$  partitions (which can be done in  $S_{k-1}^n$  ways) and the remaining element  $a_{n+1}$  forms a new partition which consists of  $a_{n+1}$  only.

so under case 2  $S_{k-1}^n$  different possibilities.

$$\text{so } S_k^{n+1} = S_{k-1}^n + kS_k^n$$

$$1) \text{ using 2) we get } S_2^n = 1 + 2 + 2^2 + \dots + 2^{n-2} = 2^{n-1} - 1$$

91. For a finite set  $A$ , let  $|A|$  denote the number of elements in the set  $A$ .

(a) Let  $F$  be the set of all functions

$$f: \{1, 2, \dots, n\} \rightarrow \{1, 2, \dots, k\}, (n \geq 3, k \geq 2)$$

satisfying  $f(i) \neq f(i+1)$  for every  $i, 1 \leq i \leq n-1$

Show that  $|F| = k(k-1)^{n-1}$

(b) Let  $c(n, k)$  denote the number of functions in  $F$  satisfying  $f(n) \neq f(1)$

For  $n \geq 4$  prove that

$$c(n, k) = k(k-1)^{n-1} - c(n-1, k).$$

(c) Using (b) prove that for  $n \geq 3$

$$c(n, k) = k(k-1)^n + (-1)^n(k-1).$$

Ans: a) no of options in first position is  $k$  and for 2nd position  $k-1$ , 3rd position  $k-1$ , so on.

so total no of functions are  $|F| = k(k-1)^{n-1}$ .

(b) for any function in  $F$  either case 1  $f(n) \neq f(1)$  or case 2  $f(n) = f(1)$

case 1 : no of functions under this case  $c(n, k)$

case 2: here  $f(1) = f(n) \neq f(n-1)$

so,  $f(n-1) \neq f(1)$

so here the total no of functions are  $c(n-1, k)$

so,  $c(n, k-1) + c(n, k) = |F| = k(k-1)^{n-1}$

(c) using (b)

$$c(n, k) = k(k-1)^{n-1} - k(k-1)^{n-2} + k(k-1)^{n-3} + \dots = k(k-1)^n + (-1)^n(k-1)$$

92. Let  $N = \{1, 2, \dots, n\}$  be a set of elements called voters. Let  $C = \{S : S \text{ is a proper subset of } N\}$  be the set of all subsets of  $N$ . members of  $C$  are called coalitions. Let  $f$  be a function from  $C$  to  $(0, 1)$ . A coalition  $S$  being an improper subset of  $N$  is said to be winning if  $f(S) = 1$ ; it is said to be losing coalition if  $f(S) = 0$ . Such a function  $f$  is called a voting game if the following conditions hold:

(a)  $N$  is a winning coalition.

(b) the empty set  $\phi$  is a losing coalition.

(c) if  $S$  is a winning coalition and  $S \subseteq S'$ , then  $S'$  is also winning.

(d) if both  $S$  and  $S'$  are winning coalitions, then  $S$  and  $S'$  have a common voter.

Show that the maximum number of winning coalitions of a voting game is  $2^{n-1}$ . Also find a voting game for which the number of winning coalitions is  $2^{n-1}$ .

Ans: We have, every subset  $S$  of  $N$  has a complement  $S^c$  such that  $S \cap S^c = \phi$

Clearly, if  $S$  belongs to a winning coalition,  $S^c$  belongs to a losing coalition. But the converse is not necessarily true. So to maximise winning coalitions,

we always put  $S$  in winning side and  $S^c$  in losing side so that there is a bijection between these two partitioned sets.

Since the number of elements in winning and losing side is same(=  $p$  say) this way, and number of subsets of  $N = 2^n$ , we have,

$$2p = 2^n \implies p = 2^{n-1}$$

**To construct a voting game with winning coalition  $2^{n-1}$  we do the following:**

\*Put  $\{1\}$  in winning side.

\*By condition (c),  $\{1, 2\}, \{1, 3\}, \dots, \{1, n\}$  are the two element winning subsets of  $N$

$\{1, 2, 3\}, \{1, 2, 4\}, \dots, \{1, 2, n\}, \{1, 3, 4\}, \dots, \{1, 3, n\}, \dots$ , are the three element subsets,.....,  $N$  is the  $n$  subset of winning side.

\*All the complements of above are losing.

We observe that, the total number of winning coalitions is the sum of 0-combinations, 1-combinations, 2-combinations, .....,  $(n-1)$  combinations of the elements in  $N$  excepting 1

So, number of winning coalitions=

$$\binom{n-1}{0} + \binom{n-1}{1} + \binom{n-1}{2} + \dots + \binom{n-1}{n-1} = 2^{n-1}.$$

93. If  $c$  is a real number with  $0 \leq c \leq 1$ , then show that the values taken by the

function  $y = \frac{x^2 + 2x + c}{x^2 + 4x + 3c}$ , as  $x$  varies over real numbers, range over all real numbers.

Ans: Cross multiplying and simplifying the given expression, we get,

$x^2(y-1) + 2x(2y-1) + (3y-1)c = 0$  Now since  $x$  is real, discriminant of above expression  $D \geq 0$

$$\implies 4(2y-1)^2 - 4(y-1)(3y-1)c \geq 0 \implies y^2(4-3c-4y(1-c)) + (1-c) \geq 0$$

The above is satisfied iff this quadratic in  $y$  has discriminant  $D' \leq 0$

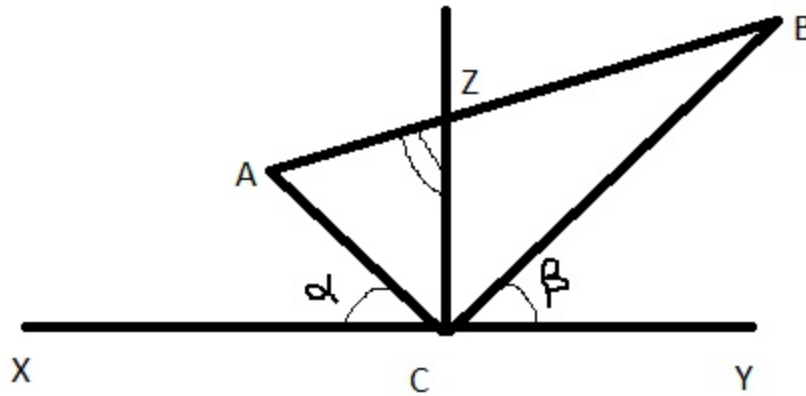
$$\implies 16(1-c)^2 - 4(1-c)(4-3c) \leq 0 \implies c(c-1) \leq 0 \text{ implying, } 0 < c < 1$$

which is the given condition,

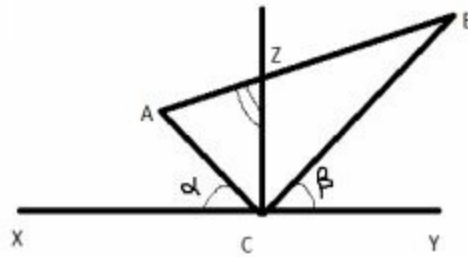
hence  $y$  will always be real.

94. In the adjoining figure,  $CZ$  is perpendicular to  $XY$ , and the ratio of lengths  $AZ$  to  $ZB$  is  $1 : 2$ . The angle  $ACX$  is  $\alpha$  and angle  $BCY$  is  $\beta$ . Find expression of angle  $AZC$  in terms of  $A$  and  $B$ .

Attachments:



In the adjoining figure,  $CZ$  is perpendicular to  $XY$ , and the ratio of lengths  $AZ$  to  $ZB$  is  $1 : 2$ . The angle  $ACX$  is  $\alpha$  and angle  $BCY$  is  $\beta$ . Find expression of angle  $AZC$  in terms of  $A$  and  $B$ .

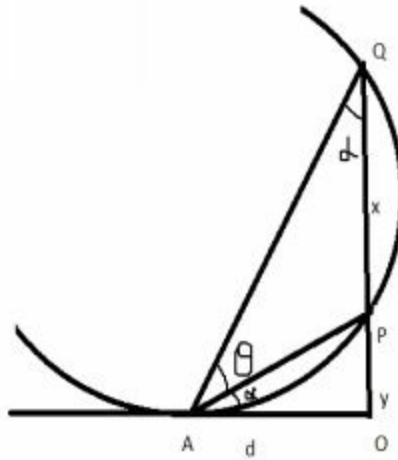


Ans: Applying **mn theorem** (Some also call **projection theorem**) of trigonometry,

$$(2+1) \cot(\angle AZC) = 2 \cot(\angle ZCB) - \cot(\angle ZCA) = 2 \cot\left(\frac{\pi}{2} - \beta\right) - \cot \pi/2 - \alpha$$

So,  $\boxed{\angle AZC = \cot^{-1}\left(\frac{2\tan\beta - \tan\alpha}{3}\right)}$

95. A man walking towards a building, on which a flagstaff is fixed vertically, observes the angle subtended by the flagstaff to be greatest when he is at a distance  $d$  from the building, If  $\theta$  be the observed greatest angle, show that the length of the flagstaff is  $2d \tan \theta$ .



Ans: Let  $PQ$  be the flagstaff(see above figure). Then maximum angle subtended will be at  $A$  where circle through  $P$  and  $Q$  has  $AO$  as tangent.

Let  $\angle PQA = \alpha$  then  $\angle PAO = \alpha$  [angle in alternate segment]

$$\begin{aligned}
 \text{So, } PQ &= OQ - OP = d[\tan(\alpha + \theta) - \tan \alpha] \\
 &= d\left[\tan\left(\frac{\pi}{2} - \alpha\right) - \tan(\alpha)\right] \text{ [from tr. } QOA, 2\alpha + \theta = \frac{\pi}{2}] \\
 &= d[\cot \alpha - \tan \alpha] = d\left[\frac{1 - \tan^2 \alpha}{\tan \alpha}\right] = 2d\left[\frac{\cos^2 \alpha - \sin^2 \alpha}{2 \cos \alpha \sin \alpha}\right] = 2d \cot 2\alpha = 2d \cot\left(\frac{\pi}{2} - \theta\right) \\
 &= 2d \tan \theta \\
 \text{So, } \boxed{\text{length of flagstaff} = 2d \tan \theta}
 \end{aligned}$$

96. Let  $ABCD$  be a cyclic quadrilateral with lengths  $AB = p, BC = q, CD = r$

and  $DA = s$ . Show that  $\frac{AC}{BD} = \frac{ps + qr}{pq + rs}$ .

Ans:  $(AB, BC, CD, DA) = (p, q, r, s)$  and  $(AC, BD) = (x, y)$

$$x^2 = p^2 + q^2 - 2pq \cos \angle B = r^2 + s^2 + 2rs \cos \angle B \implies x^2 = \frac{(pr + qs)(ps + qr)}{pq + rs}$$

$$y^2 = \frac{(pr + qs)(pq + rs)}{ps + qr}$$

Likewise (by 2nd Cosine rule),

Dividing and taking square root,  $\frac{x}{y} = \frac{ps + qr}{pq + rs}$  or

$$\boxed{\frac{AC}{BD} = \frac{ps + qr}{pq + rs}}$$

97. Prove by induction or otherwise that

$$\int_0^{\pi/2} \frac{\sin(2n+1)x}{\sin x} dx = \frac{\pi}{2} \text{ for every integer } n \geq 0.$$

Ans: Let us assume,

$$P(n) : \int_0^{\pi/2} \frac{\sin(2n+1)x}{\sin x} dx = \frac{\pi}{2}. \text{ Clearly, for } n=0, P(n) \text{ is true.}$$

Now we assume that  $P(n)$  is true for  $n = m - 1$ .

Hence,

$$\int_0^{\pi/2} \frac{\sin(2m-1)x}{\sin x} dx = \frac{\pi}{2}. \text{ Also we have,}$$

$$\sin(2m+1)x - \sin(2m-1)x = 2 \sin x \cos 2mx.$$

$$\Rightarrow \frac{\sin(2m+1)x}{\sin x} - \frac{\sin(2m-1)x}{\sin x} = 2 \cos 2mx.$$

$$\Rightarrow \int_0^{\pi/2} \frac{\sin(2m+1)x}{\sin x} dx - \int_0^{\pi/2} \frac{\sin(2m-1)x}{\sin x} dx = \int_0^{\pi/2} 2 \cos 2mx = 2 \left[ \frac{\sin 2mx}{2m} \right]_0^{\pi/2} = 0.$$

$$\therefore \int_0^{\pi/2} \frac{\sin(2m+1)x}{\sin x} dx = \int_0^{\pi/2} \frac{\sin(2m-1)x}{\sin x} dx = \frac{\pi}{2}.$$

Hence  $P(m)$  is also true. Thus by induction hypothesis, we can say that  $P(n)$  is true for all integer  $n \geq 0$ .

98. Find the maximum among  $1, 2^{1/2}, 3^{1/3}, 4^{1/4}, \dots$

Ans: Suppose,  $f(x) = x^{1/x}$ .

Hence,

$$f'(x) = x^{1/x} \cdot \frac{1}{x^2} (1 - \ln x).$$

Thus after  $x = e$ , the function  $f(x)$  is decreasing in nature. Hence  $3^{1/3} > n^{1/n} \forall n > 3$ .

Now we have to determine the greatest among  $1, 2^{1/2}, 3^{1/3}$ .

Clearly,  $1 < \sqrt{2} = 2^{1/2}$ .

Also,  $8 < 9 \implies 8^{1/6} < 9^{1/6} \implies 2^{1/2} < 3^{1/3}$ .

Hence  $3^{1/3}$  is the greatest among all the listed real numbers.

99. A pair of complex numbers  $z_1, z_2$  is said to have the property  $P$  if for every complex number  $z$  we find real numbers  $r$  and  $s$  such that  $z = rz_1 + sz_2$ . Show that a pair of complex numbers has property  $P$  if and only if the points  $z_1, z_2$  and  $0$  on complex plane are not collinear.

Ans: Let us assume the contrary. So  $z_1, z_2, 0$  are collinear and  $z_1, z_2$  has same argument  $\theta$ .

So  $z_1 = |z_1|(\cos\theta + i\sin\theta), z_2 = |z_2|(\cos\theta + i\sin\theta)$

So  $z = rz_1 + sz_2 = (r|z_1| + s|z_2|)(\cos\theta + i\sin\theta)$

So  $z$  is collinear to  $0, z_1, z_2$ .

But this way, the complex numbers not collinear to them cannot be represented. So contradiction arises.

Hence, a pair of complex numbers has property  $P$  if and only if the points  $z_1, z_2$  and  $0$  on complex plane are not collinear.

100. Let  $f(x, y) = x^2 + y^2$ . Consider the region, including the boundary, enclosed by  $y = \frac{x}{2}$ ,  $y = \frac{-x}{2}$ , and  $x = y^2 + 1$ . Find the maximum value of  $f(x, y)$  in this region.

Ans: To maximise,

$f(x, y) = x^2 + y^2$  in the region  $y = \frac{x}{2}$ ,  $y = \frac{-x}{2}$ ,  $x = y^2 + 1$

Taking equations,  $y = x/2$ ,  $x = y^2 + 1$  and solving we get  $x = 2$ ,  $y = 1$

Taking equations,  $y = -x/2$ ,  $x = y^2 + 1$  and solving we get  $x = 2$ ,  $y = -1$

Taking equations  $y = x/2$ ,  $y = -x/2$ , we get  $x = y = 0$

So corner points are  $A(2, 1)$ ,  $B(2, -1)$ ,  $C(0, 0)$

At  $A$ ,  $f = 2^2 + 1^2 = 5$

At  $B$ ,  $f = 2^2 + (-1)^2 = 5$

At  $C$ ,  $f = 0^2 + 0^2 = 0$

At both  $A, B$  value of  $f(x, y)$  is more than that the value of  $f(x, y)$  at  $C$ .

So maximum value of  $f(x, y) = x^2 + y^2$  in the given region is 5.

101. Let  $A$  and  $B$  be two fixed points 3cm apart.

a) Let  $P$  be any point non-collinear with  $A$  and  $B$ , such that  $\angle APB = 120^\circ$ . The tangent at  $P$  to the circle passing through the points  $P, A$  and  $B$  meets the extended line  $AB$  at point  $K$ . Find the length of the segments  $KB$  and  $AK$ .

b) Hence or otherwise, prove that the locus of all points in the plane such that  $PA = 2PB$  is a circle.

Ans: Let  $\angle KPB = \theta$  So angle  $\angle BAP = \theta$  [angle in alternate segment]

In triangle  $PKB$ ,

$$\frac{PB}{\sin \angle PKB} = \frac{KB}{\sin \theta} \dots \dots \dots i$$

In triangle  $PKA$

$$\frac{2x}{\sin \angle PKA} = \frac{KP}{\sin \theta} \dots \dots \dots \frac{ii}{i} \Rightarrow$$

$$2 = \frac{KP}{KB} \Rightarrow KP^2 = 4KB^2 \quad \text{Now } KA.KB = KP^2 [\text{power of point}]$$

$$\Rightarrow KB(KB + 3) = KP^2 = 4KB^2 \Rightarrow KB = 1cm$$

So  $KP = 2KB = 2cm$

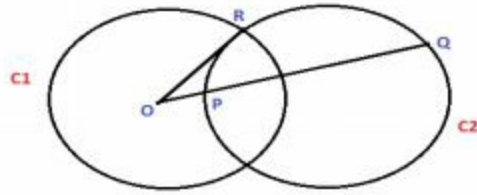
b) Let us assume  $P(h, k)$  be required point and for convenience that coordinates of  $A, B$  are  $(0, 0), (3, 0)$  respectively. By condition,  
 $PA = 2PB \Rightarrow PA^2 = 4PB^2 \Rightarrow h^2 + k^2 = 4[(h - 3)^2 + k^2]$   
 $\Rightarrow \boxed{h^2 + k^2 - 8h + 12 = 0}$   
 Above is the locus of a circle. So locus of  $P$  is circle.

102. Let  $a$  be a non zero complex number such that  $|a| \neq 1$ . Let  $P$  be the point  $a$  in the complex plane and let  $Q$  be the point  $1/\bar{a}$ . Let  $C_1$  be the circle  $\{z : |z| = 1\}$  and let  $C_2$  be the circle passing through  $P$  and  $Q$ . Show that  $C_1$  and  $C_2$  intersect orthogonally.

Ans: Let the circles intersect at  $R$ . We see that,  $O, P$  and  $Q$  are on same straight line since argument of  $P, Q$  are same. Let  $P = |a|(\cos \theta + i \sin \theta)$ , then  
 $Q = 1/\bar{a} = \frac{1}{|a|(\cos \theta - i \sin \theta)} = \frac{1}{|a|}(\cos \theta + i \sin \theta)$

So  $OP.OQ = |a| \times \frac{1}{|a|} = 1 = OR^2$

So by power of a point,  $OR$  is tangent to  $C_2$ . Also it is radius of  $C_1$ , So  $C_1$  and  $C_2$  intersect orthogonally.



103. In a certain game, 30 balls of  $k$  different colours are kept inside a sealed box .You are told only the value of  $k$  but not the number of balls of each colour . Based on this, you have to guess whether it is possible to split the balls into 10 groups of each, such that in each group the three balls are of different colours. Your answer is to be a simple *YES* or *NO*. You win or loss a point according as your guess correct or not. For what values of  $k$ , you can say *NO* and be sure of winning? For what values of  $k$ , you can say *YES* and be sure of winning? Justify your solution.

Ans: **Solution:** No & sure of winning :-  $k = 1, 2$ .

**Justification:** If colours are there the 3 balls cannot be of different colours & if  $k \geq 3$  then there exists a possibility of splitting into two groups of 3 each in each group 3 balls are of different colours.

For  $k = 3$  take 10 ball of each colour & the splitting is possible.

**Solution:** Yes & sure of winning  $k = 21, \dots, 30$

**Justification:** By we can say that number of ball of a certain colour is at most two. So we can keep them in different boxes. For  $k \leq 20$  it is not possible .

For example take  $(k - 1)$  balls of different colours &  $(30 - k + 1)$  ball of different colour. Now by  $(30 - k + 1)$  ball of same colour are placed in two

groups. So at least one group contains more than one ball of the same colour.

104. Let  $f(x)$  be a polynomial with integer coefficients. Suppose that there exist distinct integers  $a_1, a_2, a_3, a_4$ , such that  $f(a_1) = f(a_2) = f(a_3) = f(a_4) = 3$ . Show that there does not exist any integer  $b$  with  $f(b) = 14$ .

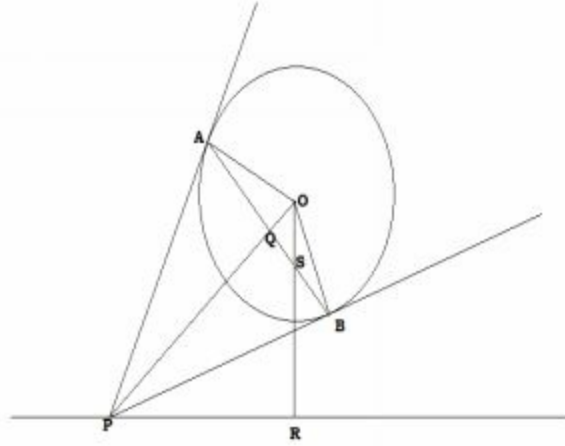
Ans: Consider the auxiliary polynomial  $g(x) = f(x) - 3$ . Clearly, according to the problem,  $g(x)$  has four distinct integer roots  $a_1, a_2, a_3, a_4$ . Hence we may write

$g(x) = (x - a_1)(x - a_2)(x - a_3)(x - a_4)Q(x)$  where  $Q(x)$  is an integer coefficient polynomial. (Since by factor theorem if ' $t$ ' is a root of a polynomial  $f(x)$  then  $(x - t)$  is its factor).

Suppose there exists an integer  $b$  such that  $f(b) = 14$ ; then  $g(b) = 14 - 3 = 11$ . Hence  $g(b) = (b - a_1)(b - a_2)(b - a_3)(b - a_4)Q(b)$ . Since  $a_i$ 's are distinct, so are  $(b - a_i)$ 's. Therefore the equation indicates that 11 can be written as the product of at least 4 different integers which is impossible (as 11 is a prime). Indeed even if we consider  $-1, 1$  and  $11$ , we have only three integers whose product is 11.

Thus we have a contradiction and proving that there does not exist an integer  $b$  such that  $f(b) = 14$ .

105. Tangents are drawn to a given circle from a point on a given straight line, which does not meet the given circle. Prove that the locus of mid-point of chord joining the two points of contact of the tangents with the circle is a circle.



Ans: In the adjoining figure, we shall try to show that if the point  $P$  varies in the straight line, the position of  $S$  remains unchanged.

From similarities of  $\triangle OQB$  and  $\triangle PQB$ , we get,

$$\frac{QB}{PB} = \frac{OQ}{OB} \Rightarrow \frac{QB^2}{PB^2} = \frac{OQ^2}{OB^2} \Rightarrow \frac{OB^2 - OQ^2}{PB^2} = \frac{OQ^2}{OB^2} \Rightarrow OB^4 = OQ^2(PB^2 + OB^2) \Rightarrow OB^4 = OQ^2 \cdot OP^2,$$

Now from similarities of  $\triangle OQS$  and  $\triangle OPR$  we get,

$$\frac{OQ}{OS} = \frac{OR}{OP} \Rightarrow \frac{OQ^2}{OS^2} = \frac{OR^2}{OP^2} \Rightarrow OS^2 \cdot OR^2 = OQ^2 \cdot OP^2 = OB^4 \Rightarrow OS \cdot OR = OB^2 \Rightarrow OS = \frac{OB^2}{OR}.$$

Clearly  $OB$  is the radius of the circle and  $OR$  is the distance of the line from the center of the circle. Hence  $OB$  and  $OR$  are fixed. Thus the point  $S$  is fixed. Now as the point  $P$  moves in the straight line, the point  $Q$  moves maintaining  $\angle OQS = \frac{\pi}{2}$  where  $O$  and  $S$  are fixed. Hence the locus of  $Q$  is a circle with fixed diameter  $OS$ .

106. How many 6 – letter words can be formed using the letters  $A, B$  and  $C$

so that each letter appears at least once in the word?

Ans: Number of ways 6 – letter words can be formed using letters  $A, B, C$  so that each letter appears at least once in the word

= Number of ways 6 – letter word formed using  $A, B, C$  – Number of ways 6 letter word formed using any two of  $A, B, C$  + Number of 6 – letter words formed by using  $A, B, C$ . [ using inclusion exclusion principle. ]

$$= 3^6 - \binom{3}{2} 2^6 + \binom{3}{1} 1^6$$

$$= \boxed{555}$$

107. Consider the system of equations  $x + y = 2, ax + y = b$ . Find conditions on  $a$  and  $b$  under which

- (i) the system has exactly one solution;
- (ii) the system has no solution;
- (iii) the system has more than one solution.

Ans:

$$x + y = 2, ax + y = b$$

(i) the system has exactly one solution if two lines intersect each other, that is  $a \neq 1$

(ii) no solution if the lines are parallel, that is  $a = 1$  &  $b \neq 2$

(iii) the system has more than one solution if the two lines are same, that is  $a = 1$  &  $b = 2$ .

108. If any one pair among the straight lines

$ax + by = a + b, bx - (a + b)y = -a, (a + b)x - ay = b$  intersect, then show that the three straight lines are concurrent.

Ans: Three lines are concurrent if each of them is linear combination of other two & they are not parallel.

Now given one pair intersect that is they are not parallel.

Now

$$\begin{aligned} ax + by &= a + b \dots (i) \\ bx - (a + b)y &= -a \dots (ii) \\ (a + b)x - ay &= b \dots (iii) \\ (i) + (ii) - (iii) &= 0 \end{aligned}$$

So they are concurrent.

109. Find the vertices of the two right angles triangles, each having area 18 and such that the point  $(2, 4)$  lies on the hypotenuse, and the other two sides are formed by the  $x$  and  $y$  axes.

Ans: Suppose the vertices are  $(a, 0)$  and  $(0, b)$ . Clearly  $\frac{1}{2}ab = 18$  or  $ab = 36$ .

Also the equation of the line through  $(0, a)$ ,  $(b, 0)$  is  $\frac{x}{a} + \frac{y}{b} = 1$ . Since we know that  $(2, 4)$  is on that line, there fore  $\frac{2}{a} + \frac{4}{b} = 1$ .

In this equation, lets replace  $a$  by  $\frac{36}{b}$ . Hence we get  $\frac{2}{\frac{36}{b}} + \frac{4}{b} = 1$ .

.

Therefore we get a quadratic in  $b$ .  $(b - 12)(b - 6) = 0$

Thus  $b = 12$  or  $6$  implying  $a = 3$  or  $6$

.

The other two vertices are:  $(0, 12)$  and  $(3, 0)$  OR  $(0, 6)$  and  $(6, 0)$ .

110. Let  $f : \mathcal{N} \rightarrow \mathcal{N}$  be the function defined by  $f(0) = 0$ ,  $f(1) = 1$ , and  $f(n) = f(n-1) + f(n-2)$ , for  $n \geq 2$ , where  $\mathcal{N}$  is the set of all non negative integers. Prove the following results:

i)  $f(n) < f(n+1)$  for all  $n \geq 2$

ii) There exists precisely four non-negative integers  $n$  for which  $f(f(n)) = f(n)$ .

iii)  $f(5n)$  is divisible by 5, for all  $n$ .

Ans: Clearly  $f(2) = 1$ , and  $f(3) = f(2) + f(1) = 2$ . Hence  $f(3)$  is positive. We use strong form of induction and assume that up to  $n$ , all  $f(n)$  is positive.

Then  $f(n+1) = f(n) + f(n-1)$ .

Since  $(n-1)$  is positive. Hence  $f(n+1) > f(n)$ . This solves first part.

For second part we observe

$$f(f(0)) = f(0)$$

$$f(f(1)) = f(1) \text{ (as } f(1) = 1)$$

As  $f(2) = f(0) + f(1) = 1$ , we have  $f(f(2)) = f(1) = 1 = f(2)$ .

Next observe that

$$f(3) = f(2) + f(1) = 1 + 1 = 2$$

$$f(4) = f(3) + f(2) = 2 + 1 = 3$$

$$f(5) = f(4) + f(3) = 3 + 2 = 5$$

So  $f(f(5)) = f(5)$  as  $f(5) = 5$

So we got four numbers 0, 1, 2, 5, for which  $f(f(n)) = f(n)$  by observation.

We wish to show that this does not happen for any value of  $n$  greater than 5.

Clearly, if  $f(f(n)) = f(n)$ , then  $f(n) = n$ . We will show that  $f(n) > n$  for all  $n$  greater than 5. (If we can show that then it will automatically show that  $f(n)$  is not equal  $n$  for any value greater than 5, implying  $f(f(n))$  not equal to  $f(n)$  for any other value apart from the ones that we have found earlier by trial and error).

Note that  $f(6) = f(5) + f(4) = 5 + 3 = 8$ . That is  $f(6) > 6$ . Similarly  $f(7) = f(6) + f(5) = 8 + 5 = 13$  implying  $f(7) > 7$ .

We will again use strong for of induction. Suppose  $f(k) > k$  for all values of  $k$  from 6 to  $n$ . Using this assumption we will show that  $f(n+1) > n+1$ .

$f(n+1) = f(n) + f(n-1)$ . By induction  $f(n) > n$  and  $f(n-1) > n-1$ . This implies  $f(n+1) > n + n - 1 = 2n - 1$ .

But  $2n-1 > n+1$  (as  $n > 2$ ). Hence it follows that  $f(n+1) > n+1$ . This solves second part.

Finally we wish to show that 5 divides  $f(5n)$ .

We again use induction. First note that  $f(5) = 5$  hence divisible by 5. Assume that the claim is true for  $n-1$ . That is 5 divides  $f(5(n-1))$ . Using this assumption we wish to show that 5 divides  $f(5n)$ . Now note that:

$$\begin{aligned}
 f(5n) &= f(5n-1) + f(5n-2) \\
 &= f(5n-2) + f(5n-3) + f(5n-3) + f(5n-4) \\
 &= f(5n-3) + f(5n-4) + f(5n-4) + f(5n-5) + f(5n-4) + f(5n-5) + f(5n-4) \\
 &= f(5n-4) + f(5n-5) + f(5n-4) + f(5n-4) + f(5n-5) + f(5n-4) + f(5n-5) + f(5n-4) \\
 &= 5f(5n-4) + 3f(5n-5)
 \end{aligned}$$

Clearly  $5f(5n-4)$  is divisible by 5. Also  $3f(5n-5)$  is divisible by 5 as by inductive assumption  $f(5n-5)$  is divisible by 5. Hence their sum  $5f(5n-4) + 3f(5n-5)$  is divisible by 5. This shows 5 divides  $f(5n)$ .

This completes the proof of third part.

111. All the permutation of the letters  $a, b, c, d, e$  are written down and arranged in alphabetical order as in dictionary. Thus the arrangement  $abcde$  is in first position and  $abced$  is in second position. What is the position of the word  $debac$ ?

Ans: Number of arrangements with:

\* $a$  at first= $4!$

\* $b$  at first= $4!$

\* $c$  at first= $4!$

Now, Number of arrangements with:

\* $d$  first,  $a$  second= $3!$

\* $d$  first,  $b$  second= $3!$

\* $d$  first,  $c$  second= $3!$

Now, number of words before  $debac$  with  $d$  first,  $e$  second is two, namely  $deabc$  and  $deacb$ .

So total words before  $debac = 3 \times 4! + 3 \times 3! + 2 = 92$

So position of  $debac = \boxed{93^{rd}}$

112. Show that a necessary and sufficient condition for the line  $ax + by + c = 0$  where  $a, b, c$  are non-zero real numbers, to pass through  $1^{st}$  quadrant is either  $ac < 0$  or  $bc < 0$ .

Ans:

$$ax + by + c = 0 \implies \frac{x}{\frac{-c}{a}} + \frac{y}{\frac{-c}{b}} = 1$$

Now the given line pass though the origin iff either of the intercepts are

positive, i.e.,

either  $\frac{-c}{a} > 0$  or  $\frac{-c}{b} > 0$

i.e., either  $ac > 0$  or  $bc > 0$  [proved]

113. Consider the function,

$$f(x) = \lim_{n \rightarrow \infty} \frac{\ln(2+x) - x^{2n} \sin x}{1 + x^{2n}}, x > 0$$

i) Is  $f(x)$  continuous at  $x = 1$ ? Justify.

ii) Show that  $f(x)$  does not vanish in the interval  $0 \leq x \leq \frac{\pi}{2}$ , and indicate the points where  $f(x)$  changes its sign.

Ans: we have  $x^{2n} \rightarrow \infty$  when  $x > 1$ , and  $x^{2n} \rightarrow 0$  when  $x < 1$ , if  $n \rightarrow \infty$   
so,

$$\text{LHL} = \lim_{x \rightarrow 1^-} \lim_{n \rightarrow \infty} \frac{\ln(2+x) - x^{2n} \sin x}{1 + x^{2n}} = \ln 3,$$

$$\text{RHL} = \lim_{x \rightarrow 1^+} \lim_{n \rightarrow \infty} \frac{\ln(2+x) - x^{2n} \sin x}{1 + x^{2n}}$$

$$= \lim_{x \rightarrow 1^+} \lim_{n \rightarrow \infty} \frac{\frac{\ln(2+x)}{x^{2n}} - \sin x}{(1 + 1/x^{2n})} = -\sin 1 \quad \text{hence discontinuous.}$$

for second part, intermediate value theorem is applicable iff function is continuous. So  $f(x)$  does not vanish in the interval  $0 \leq x \leq \frac{\pi}{2}$  though it changes sign at neighbourhood of  $x = 1$ .

114. A function  $f : \mathcal{R} \rightarrow \mathcal{R}$  is called periodic if for some constant  $a > 0$ ,

$f(x+a) = f(x)$  for every real number  $x$ . Show that the function  $f(x) = \cos x + \cos\left(\frac{\sqrt{3}}{2}x\right)$  is not periodic.

Ans: Let us assume

$$f(x+a) = f(x), a \neq 0 \implies \cos(x+a) + \cos\frac{\sqrt{3}}{2}(x+a) = \cos x + \cos\left(\frac{\sqrt{3}}{2}x\right)$$

In a particular case, when  $x = 0$ ,

$$\cos(a) + \cos\frac{\sqrt{3}}{2}a = 1 + 1 = 2$$

Which is possible iff  $\cos a = 1, \cos\frac{\sqrt{3}}{2}a = 1$

or,  $a = 2\pi n, \frac{\sqrt{3}}{2}a = 2\pi k$  for some  $k, n \in \{0, 1, 2, \dots\}$

$$\text{So } \frac{a}{\frac{\sqrt{3}}{2}a} = \frac{n}{k} \implies \frac{2}{\sqrt{3}} = \frac{n}{k}$$

Which is contradictory since **RHS** of above is rational whereas **LHS** is irrational.

So claim of question is proved.

115. Show that for all positive integer  $n$ , 7 divides  $3^{2n+1} + 2^{n+2}$

Ans: Let  $P_n = 3^{2n+1} + 2^{n+2}$

Clearly  $7|P_1 = 35$

Let 7 divide  $P_n$  for  $n = m$ . Then  $P_m = 7p$  for some positive integer  $p$

$$\begin{aligned} \text{So } P_{m+1} &= 3^{2m+3} + 2^{m+3} = 9 \cdot 3^{2m+1} + 2 \cdot 2^{m+2} \\ &= 2 \cdot 7p + 7 \cdot 3^{2m+1} \end{aligned}$$

Clearly above is divisible by 7.

Hence  $7|3^{2n+1} + 2^{n+2}$  by induction.

116. Prove that  $2^{2n} - 3n - 1$  is divisible by 9 for all  $n \geq 1$

Ans:

$$\begin{aligned} 2^{2n} - 3n - 1 &= 4^n - 3n - 1 = (1 + 3)^n - 3n - 1 \\ &= 1 + \binom{n}{1}3 + \binom{n}{2}3^2 + \dots + 3^n - 3n - 1 = 9\left(\binom{n}{2} + \binom{n}{3}3 + \dots + 3^{n-2}\right) \end{aligned}$$

Clearly, above is divisible by 9.

117. How many numbers less than  $10^8$  are there, sum of whose digits equals 7?

Ans: Required solution = coefficient of  $x^7$  in  $(1 + x + x^2 + \dots + x^9)^8 = (1 - x^{10})^8(1 - x)^{-8}$

which is  $\boxed{\binom{14}{7}}$

118. i) Find the number of all possible ordered  $k$ -tuples of non-negative

integers  $(n_1, n_2, \dots, n_k)$  such that  $\sum_{i=1}^k n_i = 100$ .

ii) Show that the number of possible ordered 4-tuples of non-negative integers  $(n_1, n_2, n_3, n_4)$  such that

$$\sum_{i=1}^4 n_i \leq 100 \quad \text{is} \quad \binom{104}{4}$$

Ans: i) Required number = number of non-negative integral solution of

$$(n_1 + n_2 + \dots + n_k = 100)$$

$$= \binom{k + 100 - 1}{100} = \binom{k + 99}{100}$$

ii) Required number = Number of non-negative integral solutions of

$$\left(\sum_{i=1}^4 n_i = 0\right)$$

+ Number of non-negative integral solutions of  $\left(\sum_{i=1}^4 n_i = 1\right)$  + Number of non-

negative integral solutions of  $\left(\sum_{i=1}^4 n_i = 2\right)$  + ..... + Number of non-

negative integral solutions of  $\left(\sum_{i=1}^4 n_i = 100\right)$

$$= \binom{4 + 0 - 1}{0} + \binom{4 + 1 - 1}{1} + \dots + \binom{4 + 100 - 1}{100}$$

$$= \binom{3}{0} + \binom{4}{1} + \binom{5}{2} + \dots + \binom{103}{100}$$

$$= \binom{5}{1} + \binom{5}{2} + \binom{6}{3} + \dots + \binom{103}{100}$$

$$= \binom{6}{2} + \binom{6}{3} + \dots + \binom{103}{100}$$

.....

.....

$$= \binom{103}{99} + \binom{103}{100} = \binom{104}{100} = \binom{104}{4}$$

119. For a real number  $x$ , let  $[x]$  denote the largest integer less than or equal to  $x$ , and  $\{x\}$  denote  $x - [x]$ . Find all solutions of the equation  $13[x] + 25\{x\} = 271$

Ans: Let  $x = a + n$ , where  $[x] = a$ , i.e, the integral part and  $n$  is the fractional part,  $0 \leq n < 1$ .

$$13[x] + 25\{x\} = 271 \implies 13a + 25(a + n - a) = 271 \implies 25n = 271 - 13a$$

Since 271 and  $13a$  are integers so  $25n$  is surely integer.

Let  $n = b/25$ ,  $0 \leq b < 25$  and  $b$  is integer

$$\text{So } a = \frac{271 - 25n}{13} = \frac{271 - b}{13}$$

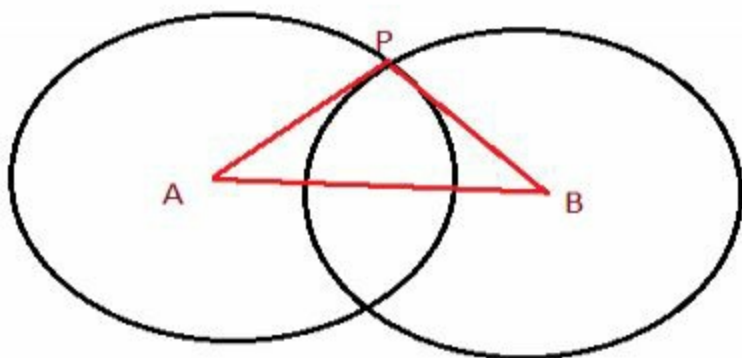
The only integers  $b$  such that  $0 \leq b < 25$  and  $a$  is integer, are,  $b = 11, 24$

So required solutions are  $n = 11/25, a = 20$  and  $n = 24/25, a = 19$

$$\text{So } \boxed{x = \frac{511}{25}, \frac{499}{25}}.$$

120. Let the circles

$x^2 + y^2 - 2cy - a^2 = 0$  and  $x^2 + y^2 - 2bx + a^2 = 0$  with centres at  $A$  and  $B$  intersect at  $P$  and  $Q$ . Show that the points  $A, B, P, Q$  and  $O(0, 0)$  lie on a circle.



Ans: Let  $P(h, k)$  be one of the intersecting point.

Centre of first circle is  $A(0, c)$ , centre of second circle, is  $B(b, 0)$

We know, angle between radii of circles at point of intersection  $= \frac{\pi}{2}$

So  $PA^2 + PB^2 = AB^2 \implies (h^2 + (k - c)^2) + ((h - b)^2 + k^2) = b^2 + c^2$   
 $\implies h^2 + k^2 - kc - bh = 0$  Above is the locus of circles  $x^2 + y^2 - bx - cy = 0$ .

It obviously passes through  $Q$  since  $Q$  is another intersecting point. Also it passes through  $O(0, 0)$

It is easily checked that,  $A(0, c)$  and  $B(b, 0)$  satisfies the above locus.

Hence, the points  $A, B, P, Q$  and  $O(0, 0)$  lie on a circle.

121. i) In the identity,

$$\frac{n!}{x(x+1)(x+2)\dots(x+n)} = \sum_{k=0}^n \frac{A_k}{x+k}$$

$$A_k = (-1)^k \binom{n}{k}$$

Prove that

ii) Deduce that

$$\binom{n}{0} \frac{1}{1.2} - \binom{n}{1} \frac{1}{2.3} + \binom{n}{2} \frac{1}{3.4} - \dots + (-1)^n \binom{n}{n} \frac{1}{(n+1)(n+2)} = \frac{1}{n+2}$$

Ans: i) We have,

$$\frac{n!}{x(x-1)(x-2)\dots(x-n)} = \frac{A_0}{x} + \frac{A_1}{x+1} + \dots + \frac{A_n}{x+n}$$

$$\Rightarrow n! = A_0(x+1)(x+2)\dots(x+n) + A_1x(x+2)\dots(x+n) + \dots + A_nx(x+1)(x+2)\dots(x+n-1)$$

Putting,  $x = 0, -1, -2, \dots, -n$  we get

$$n! = A_0 n! \Rightarrow A_0 = \frac{n!}{n!0!} = (-1)^0 \binom{n}{0}$$

$$n! = A_1(-1)1.2\dots(n-1) \Rightarrow A_1 = (-1)^1 \frac{n!}{(n-1)!1!} = (-1)^1 \binom{n}{1}$$

$$n! = A_2(-1)(-2)1.2\dots(n-2) \Rightarrow A_2 = (-1)^2 \binom{n}{2} \dots\dots\dots$$

.....

$$A_k = (-1)^k \binom{n}{k}$$

This way,

ii) Putting  $x = 1$ ,

$$\frac{n!}{1.2.3\dots(1+n)} = \sum_{k=0}^n \frac{A_k}{1+k}$$

$$\Rightarrow \frac{1}{n+1} = \binom{n}{0} \frac{1}{1} - \binom{n}{1} \frac{1}{2} + \dots + (-1)^n \binom{n}{n} \frac{1}{n+1} \dots\dots\dots a$$

Putting  $x = 2$

$$\frac{n!}{2.3.4\dots(2+n)} = \sum_{k=0}^n \frac{A_k}{2+k}$$

$$\Rightarrow \frac{1}{(n+1)(n+2)} = \binom{n}{0} \frac{1}{2} - \binom{n}{1} \frac{1}{3} + \dots + (-1)^n \binom{n}{n} \frac{1}{n+2} \dots\dots\dots b$$

$a - b$  gives,

$$\frac{1}{n+1} \left[ 1 - \frac{1}{n+2} \right] = \binom{n}{0} \left[ \frac{1}{1} - \frac{1}{2} \right] - \binom{n}{1} \left[ \frac{1}{2} - \frac{1}{3} \right] + \dots + (-1)^n \binom{n}{n} \left[ \frac{1}{n+1} - \frac{1}{n+2} \right]$$

$$\Rightarrow \binom{n}{0} \frac{1}{1.2} - \binom{n}{1} \frac{1}{2.3} + \binom{n}{2} \frac{1}{3.4} - \dots + (-1)^n \binom{n}{n} \frac{1}{(n+1)(n+2)} = \frac{1}{n+2}$$

122. Let  $D = a^2 + b^2 + c^2$ , where  $a$  and  $b$  are successive positive integers and  $c = ab$ . Prove that  $\sqrt{D}$  is an odd positive integer.

Ans:

$$\begin{aligned} b &= a + 1; \quad c = ab = a^2 + a. \\ D &= a^2 + b^2 + c^2 = a^2 + (a + 1)^2 + (a^2 + a)^2 \\ &= a^2 + a^2 + 2a + 1 + a^4 + 2a^3 + a^2 \\ &= a^4 + 2a^3 + 3a^2 + 2a + 1 \\ &= (a^2 + a + 1)^2 \end{aligned}$$

Hence,  $\sqrt{D} = a^2 + a + 1 = a(a + 1) + 1$ .

Clearly,  $a(a + 1)$  is even. Hence  $\sqrt{D}$  is odd

123. Show that

$$\begin{aligned} &\frac{3}{1.2.4} + \frac{4}{2.3.5} + \frac{5}{3.4.6} + \dots + \frac{n+2}{n(n+1)(n+3)} \\ &= \frac{1}{6} \left[ \frac{29}{6} - \frac{4}{n+1} - \frac{1}{n+2} - \frac{1}{n+3} \right] \end{aligned}$$

Ans: The identity is true for  $n = 1$

Let it be true for  $n = m$

$$\begin{aligned} &\text{Adding } \frac{m+2+1}{(m+1)(m+1+1)(m+3+1)} \text{ to both sides we get,} \\ &\frac{3}{1.2.4} + \frac{4}{2.3.5} + \frac{5}{3.4.6} + \dots + \frac{m+3}{(m+1)(m+2)(m+4)} \\ &= \frac{1}{6} \left[ \frac{29}{6} - \frac{4}{m+1} - \frac{1}{m+2} - \frac{1}{m+3} \right] + \frac{m+3}{(m+1)(m+2)(m+4)} \\ &= \frac{1}{6} \left[ \frac{29}{6} - \frac{4}{m+1} - \frac{1}{m+2} - \frac{1}{m+3} \right] + \frac{2}{3(m+1)} - \frac{1}{2(m+2)} - \frac{1}{6(m+4)} \\ &= \frac{1}{6} \left[ \frac{29}{6} - \frac{4}{m+1} - \frac{1}{m+2} - \frac{1}{m+3} \right] + \frac{4}{m+1} - \frac{3}{m+2} - \frac{1}{m+4} \\ &= \frac{1}{6} \left[ \frac{29}{6} - \frac{4}{m+2} - \frac{1}{m+3} - \frac{1}{m+4} \right] \end{aligned}$$

So identity holds for  $n = m + 1$

Hence by induction the identity is proved.

124. Let  $u_n = (3 + \sqrt{5})^n + (3 - \sqrt{5})^n$  for  $n = 1, 2, 3, \dots$

(i) Show that for each  $n$ ,  $u_n$  is an integer.

(ii) Show that  $u_{n+1} = 6u_n - 4u_{n-1}$  for all  $n \geq 2$ .

(iii) Use (ii) above, to show that  $u_n$  is divisible by  $2^n$ .

Ans: For simplicity, we will first prove (ii).

$$\begin{aligned} & 6u_n - 4u_{n-1} \\ &= 6(3 + \sqrt{5})^n + 6(3 - \sqrt{5})^n - 4(3 + \sqrt{5})^{n-1} - 4(3 - \sqrt{5})^{n-1} \\ &= (3 + \sqrt{5})^{n-1}[6(3 + \sqrt{5}) - 4] + (3 - \sqrt{5})^{n-1}[6(3 - \sqrt{5}) - 4] \\ &= (3 + \sqrt{5})^{n-1}[14 + 6\sqrt{5}] + (3 - \sqrt{5})^{n-1}[14 - 6\sqrt{5}] \\ &= (3 + \sqrt{5})^{n-1}(3 + \sqrt{5})^2 + (3 - \sqrt{5})^{n-1}(3 - \sqrt{5})^2 \\ &= (3 + \sqrt{5})^{n+1} + (3 - \sqrt{5})^{n+1} \\ &= u_{n+1}. \end{aligned}$$

From the expression of  $u_n$ , we see that  $u_1$  and  $u_2$  are integers. Hence by (ii) that we have proved, we can say that  $u_n$  is an integer for all  $n$ .

Lastly we use induction to prove (iii).

We assume that  $2^n | u_n$  and  $2^{n-1} | u_{n-1}$ .

Hence  $2^{n+1} | 6u_n$  and  $2^{n+1} | 4u_{n-1}$ .

Then,  $2^{n+1} | 6u_n - 4u_{n-1} \implies 2^{n+1} | u_{n+1}$ .

Thus by induction, we can say that  $u_n$  is divisible by  $2^n$ .

125. Each pair in a group of 20 persons is classified by the existence of kinship relation and friendship relation between them. The following table of data is obtained from such a classification.

Determine (with justifications) whether each of the following statements is

supported by the following data:

- (i) Most of the friends are kin.
- (ii) Most of the kin are friends.

|                                                                                                                                                                                                      |     |     |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|-----|
| <p>Friendship </p> <p>Kinship </p> | Yes | No  |
| Yes                                                                                                                                                                                                  | 27  | 31  |
| No                                                                                                                                                                                                   | 3   | 129 |

Ans: i) From above, 27 pairs of people are both friends and kin, but only three pairs of people are friends but not kin.

So statement (i), i.e., most friends are kin, holds.

ii) 27 pairs of people are both kin and friends but 31 pairs are kin but not friends.

So statement (ii), i.e., most kin are friends, is false.

126. Find the different number of ways 5 different gifts can be presented to 3 children so that each child receives at least one gift.

Ans: i) If the gift distribution is (3, 1, 1)

Then total ways of distributing =  $\binom{3}{1} \frac{5!}{3!} = 60$

ii) If the gift distribution is (2, 2, 1)

The two children getting two gifts can be selected in  $\binom{3}{2} = 3$  ways.

So total ways of distributing gifts =  $3 \times \frac{5!}{2!2!}$

So total ways of distribution =  $90 + 60 = \boxed{150}$

127. We say that a sequence  $\{a_n\}$  has a property  $P$ , if there exists a positive integer  $m$  such that  $a_n \leq 1$  for every  $n \geq m$ . For each of the following sequences, determine whether it has the property  $P$  or not. [Do not use any result on limits]

$$\text{i) } a_n = \begin{cases} 0.9 + \frac{200}{n} & \text{if } n \text{ is even} \\ = \frac{1}{n} & \text{if } n \text{ is odd} \end{cases} \quad \text{ii) } a_n = \begin{cases} 1 + \frac{\cos \frac{n\pi}{2}}{n} & \text{if } n \text{ is even} \\ = \frac{1}{n} & \text{if } n \text{ is odd} \end{cases}$$

Ans: i) It has property  $P$

Clearly  $200/2000 = 0.1$  and for all  $n \geq 2001$

a)  $\frac{1}{2001}, \frac{1}{2003}, \frac{1}{2005}, \dots$  are all lesser than 1.

b)  $0.9 + \frac{200}{n}$  are lesser than 1 where  $n$  are the even integers  $\geq 2000$

ii) It does not have property  $P$

For every  $n = 4r$  [a positive integer]

$$\cos \frac{4r\pi}{2} = \cos 2r\pi = 1$$

$$\text{Hence } 1 + \frac{\cos \frac{4r\pi}{2}}{4r} = 1 + \frac{1}{4r} > 1$$

Since there are infinitely many such  $r$ , so there is no such  $n$  that

$a_n \leq 1 \forall n \geq m$  i) It has property  $P$

Clearly  $200/2000 = 0.1$  and for all  $n \geq 2001$

a)  $\frac{1}{2001}, \frac{1}{2003}, \frac{1}{2005}, \dots$  are all lesser than 1.

b)  $0.9 + \frac{200}{n}$  are lesser than 1 where  $n$  are the even integers  $\geq 2000$

ii) It does not have property  $P$

For every  $n = 4r$  [a positive integer]

$$\cos \frac{4r\pi}{2} = \cos 2r\pi = 1$$

$$\text{Hence } 1 + \frac{\cos \frac{4r\pi}{2}}{4r} = 1 + \frac{1}{4r} > 1$$

Since there are infinitely many such  $r$ , so there is no such  $n$  that  $a_n \leq 1 \forall n \geq m$

128. For any integer  $n$ , let  $\langle n \rangle$  be the integer nearest to  $\sqrt{n}$ .

a) Given a positive integer  $k$ , describe all positive integers  $n$  such that  $\langle n \rangle = k$ .

b) Show that

$$\sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n} = 3$$

Ans: a) We have,

$$k \leq \sqrt{n} + \frac{1}{2} \leq k + 1 \implies k - 1/2 \leq \sqrt{n} \leq k + 1/2$$

$$\implies k^2 - k + 1/4 \leq n \leq k^2 + k + 1/4$$

But since  $k, n$  are integral, we have,

$$k^2 - k + 1 \leq n \leq k^2 + k$$

$$\begin{aligned} \text{b) } \sum_{n=1}^{\infty} \frac{2^{\langle n \rangle} + 2^{-\langle n \rangle}}{2^n} &= \sum_{k=1}^{\infty} (2^k + 2^{-k}) \sum_{n=k^2-k+1}^{k^2+k} \frac{1}{2^n} \\ &= \sum_{k=1}^{\infty} (2^k + 2^{-k}) \left( \frac{1}{2^{k^2-k+1}} + \dots + \frac{1}{2^{k^2+k}} \right) = \sum_{k=1}^{\infty} \frac{2^k + 2^{-k}}{2^{k^2-k+1}} (1 + \dots + \frac{1}{2^{2k-1}}) \\ &= \sum_{k=1}^{\infty} \frac{2^k + 2^{-k}}{2^{k^2-k+1}} \left( \frac{1 - \frac{1}{2^{2k}}}{\frac{1}{2}} \right) = 2 \sum_{k=1}^{\infty} \left( \frac{1}{2^{k^2-2k+1}} + \frac{1}{2^{k^2+1}} - \frac{1}{2^{k^2+1}} - \frac{1}{2^{k^2+2k+1}} \right) \\ &= 2 \sum_{k=1}^{\infty} \left( \frac{1}{2^{(k-1)^2}} - \frac{1}{2^{(k+1)^2}} \right) \\ &= 2 \left[ \left( \frac{1}{2^0} + \frac{1}{2^1} + \frac{1}{2^2} + \dots \right) - \left( \frac{1}{2^2} + \frac{1}{2^3} + \frac{1}{2^4} + \dots \right) \right] = 2(1 + 1/2) = 3 \end{aligned}$$

129. Let  $PQ$  be the line segment of fixed length  $L$  with its two ends sliding along the  $X$ -axis and  $Y$ -axis respectively. Complete the rectangle where  $O$  is the origin. Show that the locus of foot of perpendicular from  $R$  to  $PQ$  is

given by  
 $x^{2/3} + y^{2/3} = L^{2/3}$ .

Ans: Let coordinates of  $Q$  be  $(h, 0)$ , then  $P$  is  $(\sqrt{L^2 - h^2}, 0)$ ,  $R$  is  $(\sqrt{L^2 - h^2}, h)$

Equation of  $PQ$  is

$\frac{x}{\sqrt{L^2 - h^2}} + \frac{y}{h} = 1$ ..... $i$  Let  $T$  be foot of perpendicular from  $R$  to  $PQ$ .

Since it is perpendicular to  $i$  and passes through  $R$  its equation is,

$$(y - h) = \frac{\sqrt{L^2 - h^2}}{h}(x - \sqrt{L^2 - h^2})$$
..... $ii$

Solving  $i, ii$  we have  $x = \frac{(L^2 - h^2)^{3/2}}{L^2}, y = \frac{h^3}{L^2}$

Evaluating  $x^{2/3}, y^{2/3}$  and adding, we get,

$$\boxed{x^{2/3} + y^{2/3} = L^{2/3}}$$

Which is the locus of an asteroid.

130. Out of a circular sheet of paper of radius  $a$ , a sector with central angle  $\theta$  is cut out and folded in shape of a conical funnel. Show that the volume of

the funnel is maximum when  $\theta$  equals  $2\pi\sqrt{\frac{2}{3}}$

Ans: Arc length of sector =  $r\theta$

if  $a$  = radius of base then,  $2\pi a = r\theta$  or  $a = \frac{r\theta}{2\pi}$

Slant height of cone = radius of sector =  $r$

so height is given by  $h^2 = r^2 - a^2 = r^2(1 - \frac{\theta^2}{4\pi^2})$

Volume  $V = \frac{1}{3}\pi a^2 h = \frac{1}{3}\pi \frac{r^2 \theta^2}{4\pi^2} r(1 - \frac{\theta^2}{4\pi^2})^{1/2}$

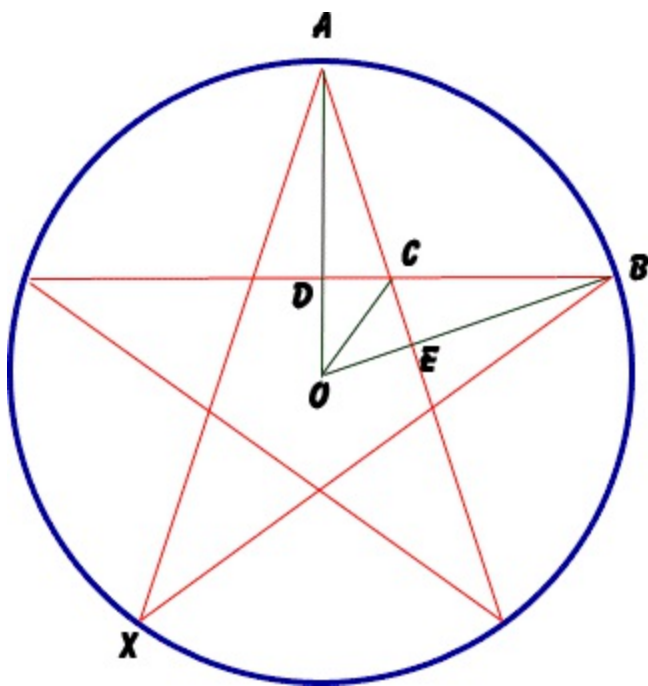
Let us maximise  $z = V^2 = \frac{1}{9} \frac{r^6 \theta^4}{16\pi^2} (1 - \frac{\theta^2}{4\pi^2}) = k(\theta^4 - \frac{\theta^6}{4\pi^2})$  where  $k = \frac{1}{9} \frac{r^6}{16\pi^2}$

So  $\frac{dz}{d\theta} = k(4\theta^3 - \frac{6\theta^5}{4\pi^2})$   
 $\frac{dz}{d\theta} = 0 \implies \theta = 2\pi\sqrt{\frac{2}{3}}$

It can be easily checked that  $\frac{d^2z}{d\theta^2}$  is negative at  $\theta = 2\pi\sqrt{\frac{2}{3}}$

Hence, the volume of the funnel is maximum when  $\theta$  equals  $2\pi\sqrt{\frac{2}{3}}$ .

131. A regular five-pointed star is inscribed in a circle of radius  $r$ . (See the figure). Show that the area of the region inside the star is  $\frac{10r^2 \tan \frac{\pi}{10}}{3 - \tan^2 \frac{\pi}{10}}$



Ans:

$AH = r$ , see diagram.

Rule of sines in the  $\triangle AHI : \frac{r}{\sin 126^\circ} = \frac{HI}{\sin 18^\circ}$ .

$LI \perp AH$ .

$\triangle HIL : LI = HI \cdot \sin 36^\circ$ .

$$LI = \frac{r \sin 18^\circ \sin 36^\circ}{\sin 126^\circ}.$$

$$\text{Area } \triangle AHI = \frac{r^2 \sin 18^\circ \sin 36^\circ}{2 \sin 126^\circ}.$$

Area of the region inside the star:

$$10 \cdot \frac{r^2 \sin 18^\circ \sin 36^\circ}{2 \sin 126^\circ}.$$

$$= 10 \cdot \frac{r^2 \sin 18^\circ 2 \sin 18^\circ \cos 18^\circ}{2 \sin 54^\circ}$$

$$= 10 \cdot \frac{r^2 \sin 18^\circ \sin 18^\circ \cos 18^\circ}{\sin 3(18^\circ)}$$

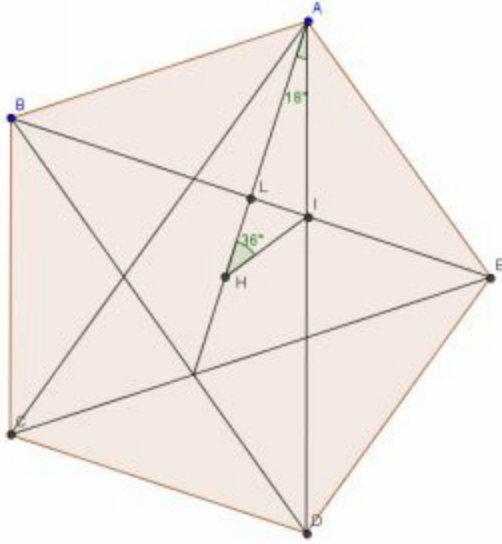
$$= 10 \cdot \frac{r^2 \sin 18^\circ \sin 18^\circ \cos 18^\circ}{3 \sin 18^\circ - 4 \sin^3 18^\circ}$$

$$= 10 \cdot \frac{r^2 \sin 18^\circ \cos 18^\circ}{3 - 4 \sin^2 18^\circ}$$

$$= 10 \cdot \frac{r^2 \tan 18^\circ}{\frac{3}{\cos^2 18^\circ} - 4 \tan^2 18^\circ}$$

$$= 10 \cdot \frac{r^2 \tan 18^\circ}{3(1 + \tan^2 18^\circ) - 4 \tan^2 18^\circ}$$

$$= 10 \cdot \frac{r^2 \tan 18^\circ}{3 - \tan^2 18^\circ}.$$



132. Using the identity  $\log(x) = \int_1^x \frac{dt}{t}$ ,  $x > 0$ , or otherwise, prove that

$$\frac{1}{n+1} \leq \log\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}.$$

Ans: Let  $f(x) = \log(x)$  and  $0 < a < b$

So  $f(x)$  is continuous and differentiable everywhere.

So by Lagrange's MVT, there exists at least one value of  $x$ , say  $x = c$ , such that,

$$f'(c) = \frac{f(b) - f(a)}{b - a} \implies \frac{1}{c} = \frac{f(b) - f(a)}{b - a}$$

$$\text{Now } a \leq c \leq b \implies \frac{1}{a} \geq \frac{1}{c} \geq \frac{1}{b}$$

$$\implies \frac{1}{b} \leq \frac{\log(b) - \log(a)}{b - a} \leq \frac{1}{a}$$

$$\implies \frac{1}{b} \leq \log\left(\frac{b}{a}\right) \leq \frac{1}{a}$$

Now putting  $b = n + 1, a = n$  [Since  $b > a$ ] we get,

$$\frac{1}{n+1} \leq \log\left(1 + \frac{1}{n}\right) \leq \frac{1}{n}.$$

133. a) Study the derivatives of the function

$f(x) = \frac{x+1}{(x-1)(x-7)}$  to make conclusions about the behaviour of the function as  $x$  ranges over all possible values for which above formula for  $f(x)$  is meaningful.

b) Use information obtained in (a) to draw a rough sketch of graph of  $f(x)$  on plain paper.

Ans:

$$f(x) = \frac{x+1}{(x-1)(x-7)} \quad f'(x) = -\frac{(x+5)(x-3)}{(x-1)^2(x-7)^2}$$

$f'(x) \geq 0$  in  $[-5, 3]$  and  $f'(x) < 0$  otherwise

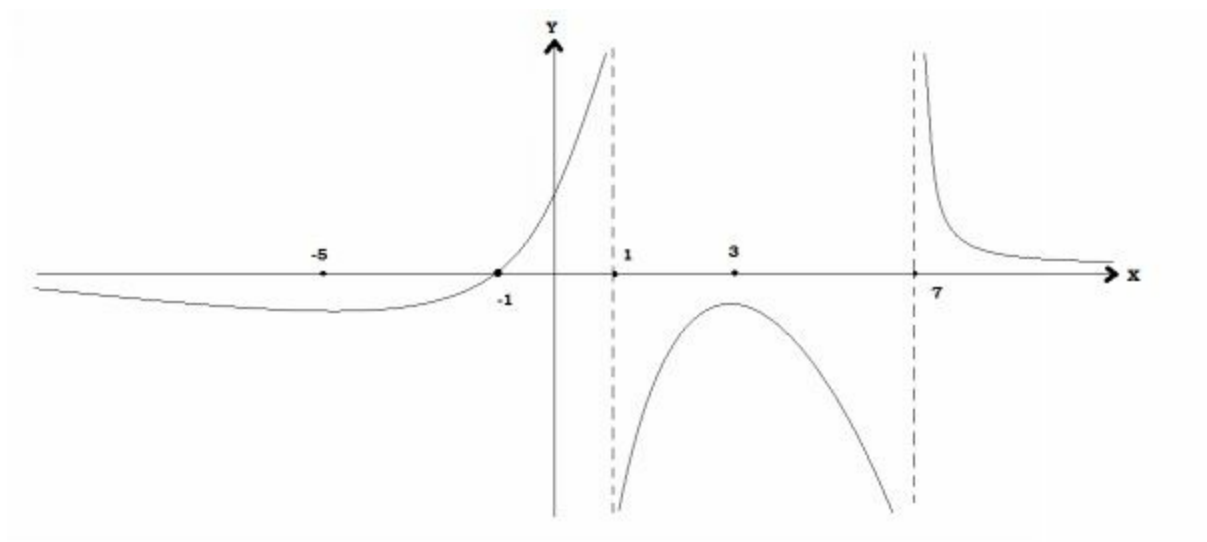
So  $f(x)$  increases in  $[-5, 3]$ , decreases in  $(3, \infty), (-\infty, -5)$

Hence  $-5$  is a point of minimum,  $3$  is point of maximum.

Maximum of  $f = f(3) = -1/2$

Minimum of  $f = f(-5) = -1/18$

Also  $f$  has infinite discontinuity at  $x = 1, 7$

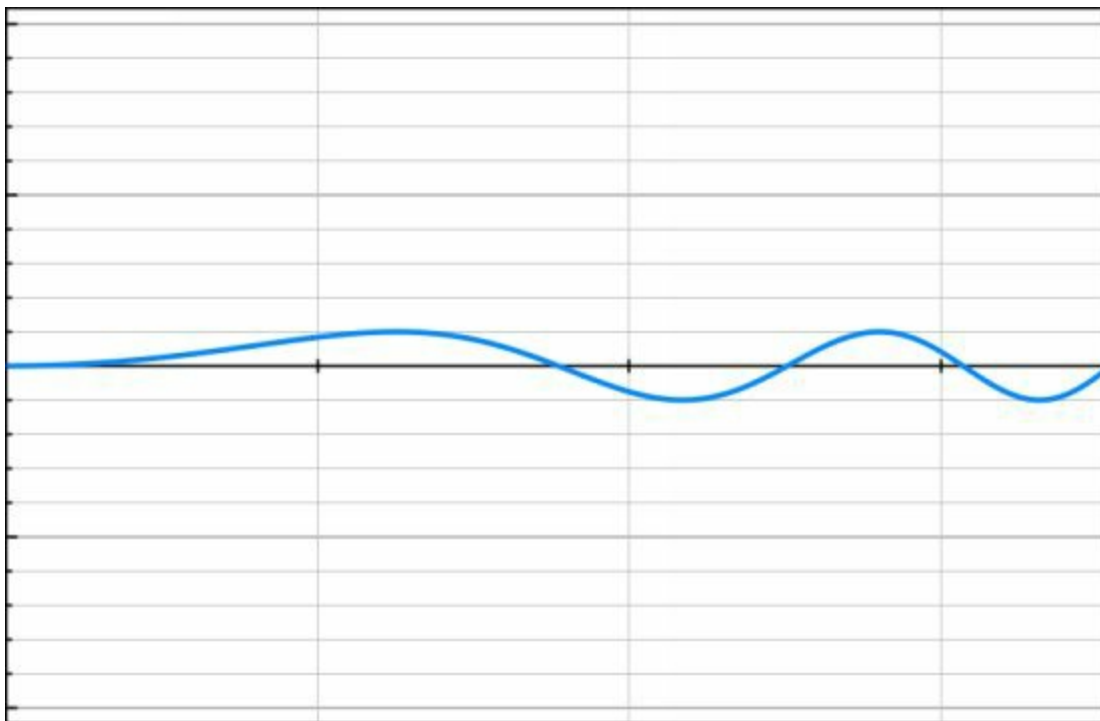


134. Sketch on plain paper, the graph of the function  $y = \sin(x^2)$  in the range  $0 \leq x \leq \sqrt{4\pi}$

Ans:

$y = \sin(x^2)$  is an even function, and it oscillates faster than  $y = \sin x$  as  $x^2$  increases more rapidly than  $x$

So the graph is:



135. Let  $k$  be a fixed odd positive integer. Find the minimum value of  $x^2 + y^2$ , where  $x, y$  are non negative integers and  $x + y = k$

Ans: Let  $f(x) = x^2 + y^2 = x^2 + (k - x)^2$  then  $f'(x) = 2x - 2(k - x)$ . Clearly  $f(x)$  decreases on left and increases on right of  $x = k/2$ . So the function is minimum at  $x = k/2$

But  $x$  is integer and  $k/2$  a fraction ( $k$  is odd). So we must check values of  $f(x)$  at integers nearest to  $x = k/2$  in its left and right which are  $x = (k - 1)/2$  and  $x = (k + 1)/2$  respectively.

Clearly  $f[(k - 1)/2] = (\frac{k - 1}{2})^2 + (\frac{k + 1}{2})^2 = f[(k + 1)/2]$

Hence required minimum value is  $(\frac{k - 1}{2})^2 + (\frac{k + 1}{2})^2 = \frac{k^2 + 1}{2}$

136. In a competition, six teams  $A, B, C, D, E, F$  play each other in the

preliminary round- called *round robin* tournament. Each game end in either a win or a loss. The winner is awarded two points while the loser is awarded zero points. After the round robin tournament, the three teams with the highest scores move to the final round. Based on the following information, find the score of each team at the end of the round robin tournament.

i) In a game between *E* and *F*, team *E* won.

ii) After each team has played four games, team *A* has 6 points, team *B* had 8 points and team *C* had

four points. The remaining matches yet to be played were

i) between *A* and *D*;

ii) between *B* and *E*; and

iii) between *C* and *F*

iii) The teams *D*, *E* and *F* had won their games against *A*, *B* and *C* respectively.

iv) Teams *A*, *B* and *D* had moved to the final round of the tournament.

Ans: i) Let us first consider *B*, till four rounds, it had got 8 points, hence won all his four matches against *A*, *C*, *D*, *F*. But it lost to *E* in its last game.

So final score of *B* is 8 points

ii) Now consider *A*, till four rounds it had 6 points, which means he lost one game till then. But in last round it lost to *D* and previously we saw that, *B* has beaten *A*.

So score of *A* is 6 points

iii) Till four rounds, *C* had 4 points, i.e, it won two, and according to previous observations, it lost to *A*, *B*, and won against *D*, *E*. Finally it lost to *F*.

So score of *C* is 4 points

iv) *D* had moved to final, but it had surely lost to *B*, *C* and won with *A*. But to make it to top three it must surpass 4 points of *C* and achieve at least 6 points. So it won against *E*, *F*.

So score of *D* is 6 points

v) *E* lost to *A*, *C*, *D* and beaten *B*, *F* from previous observations.

So score of *E* is 4 points

vi) We see that  $F$  won only with  $C$ .

Score of  $F$  is  $2 \text{ points}$

So the scores are

$A = 6 \text{ points}, B = 8 \text{ points}, C = 4 \text{ points}, D = 6 \text{ points}, E = 4 \text{ points}, F = 2 \text{ points}$

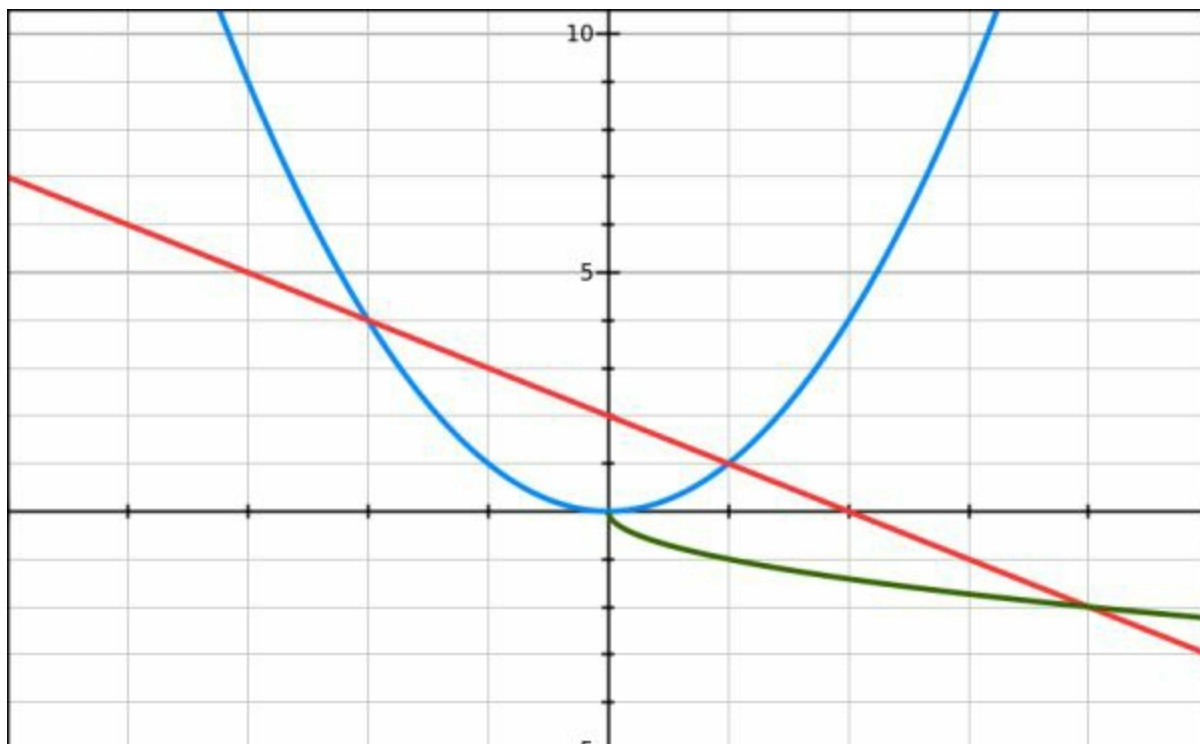
137. Find the area of the region in  $xy$ -plane, bounded by the graphs of  $y = x^2, x + y = 2$  and  $y = -\sqrt{x}$

Ans: Clearly the blue, red and green graphs depicts  $y = x^2, x + y = 2, y = -\sqrt{x}$  respectively.

$y = x^2$  and  $x + y = 2$  intersects at  $(1, 1)$  on first quadrant.  $y = -\sqrt{x}$  and  $y = x^2$  intersects at  $(0, 0)$ .  $x + y = 2$  and  $y = -\sqrt{x}$  intersects at  $(4, -2)$ .

So the corner points whose area are to be found are  $(0, 0), (1, 1), (4, -2)$ .

So the area is  $|\int_0^1 x^2 dx| + |\int_1^2 (2 - x) dx| + |\int_0^4 (-\sqrt{x}) dx| - |\int_2^4 (2 - x) dx|$   
 $= 25/6$  square units.



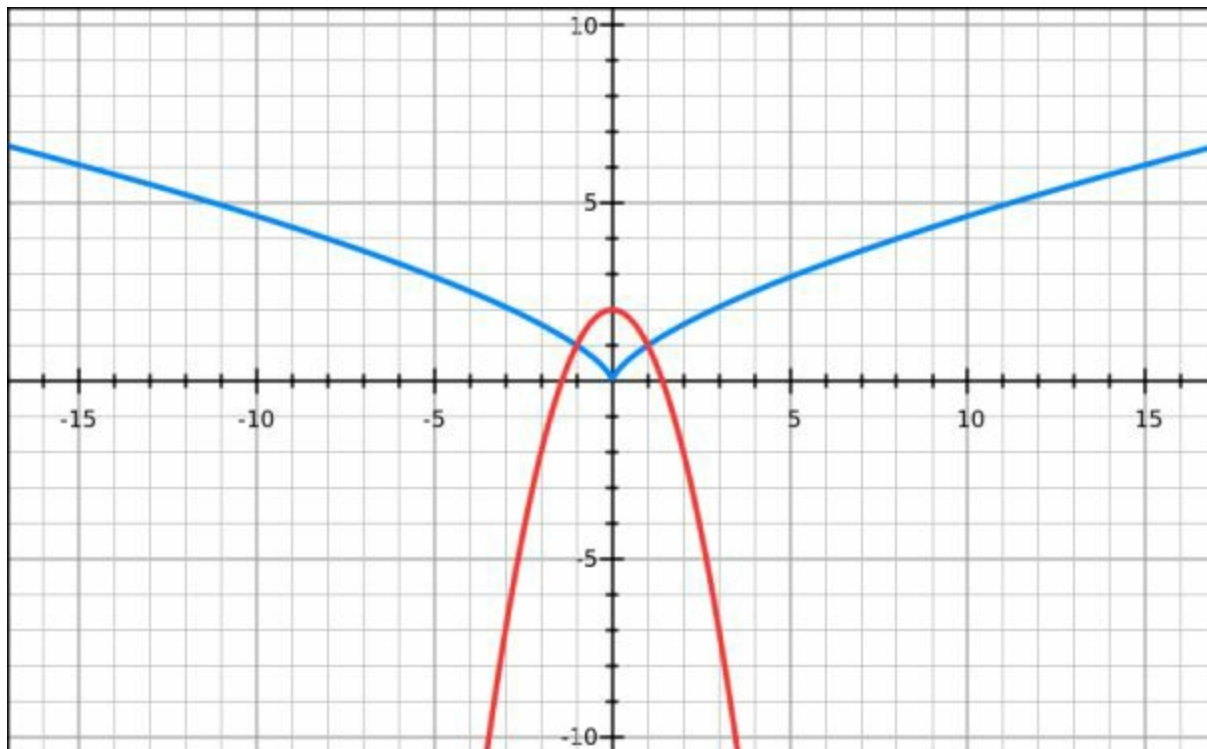
138. Show that the area of the bounded region enclosed between the curves  $y^3 = x^2$  and  $y = 2 - x^2$ , is  $2\frac{2}{15}$ .

Ans: clearly red and blue graphs are of  $y = 2 - x^2$  and  $y^3 = x^2$  respectively. Clearly since the graphs are symmetric about  $y$ -axis, **area enclosed by the graphs in first quadrant = area enclosed by them in second quadrant**

Clearly the graphs intersect at  $(1, 1)$  in first quadrant, and  $y = 2 - x^2$  cuts  $x$ -axis at  $(\sqrt{2}, 0)$

So required area 
$$= 2 \left[ \int_0^1 (2 - x^2) dx - \int_0^1 x^{2/3} dx \right] = 2\frac{2}{15}$$

Attachments:



139. a) Given  $m$  identical symbols, say  $H$ 's, show that, the number of boxes you can distribute them in  $k$  boxes marked  $1, 2, \dots, k$  so that no box goes empty

is  $\binom{m-1}{k-1}$ .

b) In an arrangement of  $m$   $H$ 's and  $n$   $T$ 's, an uninterrupted sequence of one kind of symbol is called a *run*. (For example, the arrangement  $HHHTHHTTTH$  of 6  $H$ 's and 4  $T$ 's opens with an  $H$  - run of length 3, followed successively with a  $T$  - run of length 1, an  $H$  - run of length 2, a  $T$  - run of length 3 and finally, a  $H$  - run of length 1. Find the arrangements of  $m$   $H$ 's and  $n$   $T$ 's in which there are exactly  $k$   $H$  - runs [You may use (a) above].

Ans: a) Let the boxes be denoted by variables  $x_1, x_2, \dots, x_k$

Then the required number is the number of positive integral solutions of

$x_1 + x_2 + \dots + x_k = m$  i.e., coefficient of  $x^m$  in  $(x^1 + x^2 + \dots + x^m)^k$ , i.e.,  

$$\binom{m-1}{m-k} = \binom{m-1}{k-1}$$

b) Clearly, by (a), the  $H's$  can be arranged in  $\binom{m-1}{k-1}$  ways.

But in between  $k$   $H$  - runs, there are  $k-1$  gaps where the  $T's$  can be placed. Also considering the extremities, there are two more places where the  $T's$  can be placed.

So there can be  $k-1, k$  or  $k+1$  runs of  $T$ .

So arrangement of

$$T's = \binom{n-1}{k-2} + \binom{n-1}{k-1} + \binom{n-1}{k} = \binom{n}{k-1} + \binom{n-1}{k}$$

$$= \left( \binom{m-1}{k-1} \right) \left( \binom{n}{k-1} + \binom{n-1}{k} \right)$$

So required total arrangements

140. Prove that:-

$$\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \dots + \binom{n}{3k} \leq (1/3)(2^n + 2)$$

where  $k$  is the greatest positive integer such that  $3k \leq n$ .

Ans: Let  $A$  be the required summation. This problem requires attention, let's

take the cube roots of unity,  $1, \omega, (\omega)^2$ . Now we have  $2^n = \sum_{k=0}^n \binom{n}{k}$ . Also

observe that  $(1 + \omega)^n = \sum_{k=0}^n \binom{n}{k} (\omega)^k$  and  $(1 + (\omega)^2)^n = \sum_{k=0}^n \binom{n}{k} (\omega)^{2k}$ . Now

add up these 3 equations and use the relation  $1 + \omega + \omega^2 = 0$  to get  
 $3A = 2^n + (1 + \omega)^n + (1 + (\omega)^2)^n = 2^n(-\omega^2)^n + (-\omega)^n$ .

Now  $-\omega = \frac{1 + i\sqrt{3}}{2} = \cos \frac{\pi}{3} + i \sin \frac{\pi}{3}$

$-\omega^2 = \frac{1 - i\sqrt{3}}{2} = \cos \frac{\pi}{3} - i \sin \frac{\pi}{3}$

So  $3A = 2^n + [\cos \frac{\pi}{3} + i \sin \frac{\pi}{3}]^n + [\cos \frac{\pi}{3} - i \sin \frac{\pi}{3}]^n$

Apply **De Moivre's theorem**,

$3A = 2^n + \cos \frac{n\pi}{3} + i \sin \frac{n\pi}{3} + \cos \frac{n\pi}{3} - i \sin \frac{n\pi}{3} = 2^n + 2\cos \frac{n\pi}{3} \leq 2^n + 2$

Hence  $A \leq (1/3)(2^n + 2)$

or  $\binom{n}{0} + \binom{n}{3} + \binom{n}{6} + \dots + \binom{n}{3k} \leq (1/3)(2^n + 2)$

Alternatively  $3A = 2^n + (-\omega^2)^n + (-\omega)^n$ . Taking modulus and using triangle inequality,  $|3A| = 3A \leq 2^n + |(-\omega^2)|^n + |(-\omega)|^n = 2^n + 2$ .

141. Show that the larger of the two areas into which the circle  $x^2 + y^2 = 64$  is divided by the curve  $y^2 = 12x$  is  $\frac{16}{3}(8\pi - \sqrt{3})$

Ans: Clearly the blue and red graphs are the upper half of  $x^2 + y^2 = 64$  and  $y^2 = 12x$  respectively.

Since both curves are symmetric about  $x$ -axis, it is sufficient to calculate the larger area of upper part and multiply by 2.

Clearly  $y^2 = 12x$  cuts  $x^2 + y^2 = 64$  at  $x = 4$  and  $x^2 + y^2 = 64$  cuts  $x$ -axis at  $x = 8$

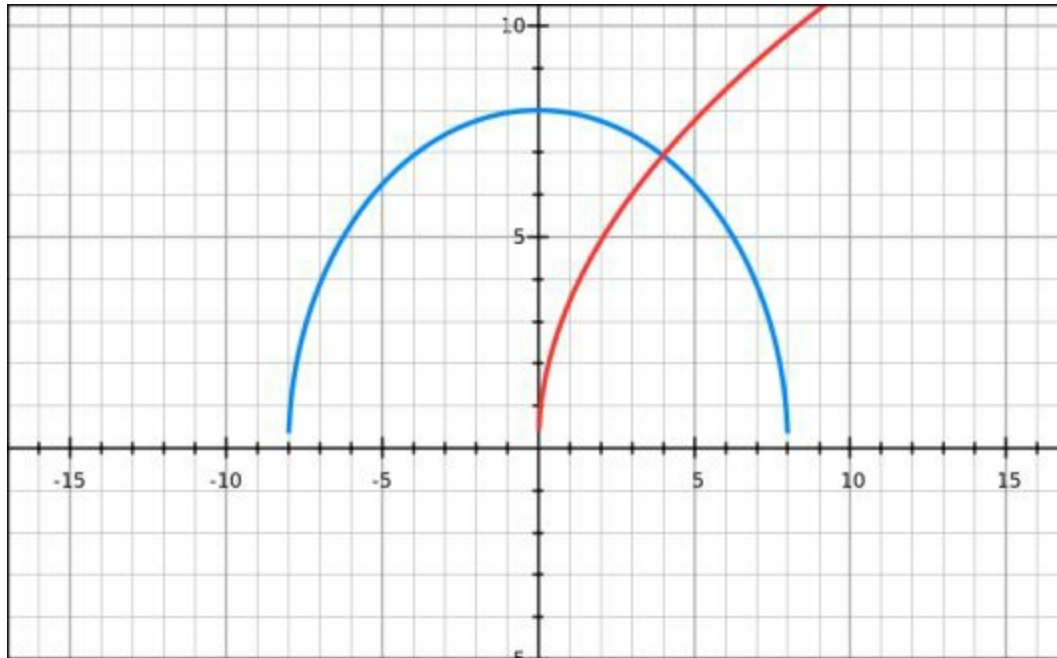
The smaller area between the curves

$= 2 \left[ \int_0^4 \sqrt{12x} dx + \int_4^8 \sqrt{64 - x^2} dx \right] = 2 \left[ \frac{4\sqrt{12}}{3} + \frac{32\pi}{3} \right]$

Area of the circle  $x^2 + y^2 = 64$  with radius 8 is  $64\pi$

$$\text{So required area} = 64\pi - 2 \cdot \left[ \frac{4\sqrt{12}}{3} + \frac{32\pi}{3} \right] = \frac{16}{3}(8\pi - \sqrt{3})$$

Attachments:



142. A cow is grazing with a rope around its neck and the other end of the rope is tied to a pole. The length of the rope is 10 metres. There are two boundary walls perpendicular to each other, one at a distance of 5 metres to the east of the pole and another at a distance of  $5\sqrt{2}$  metres to the north of the pole. Find the area the cow can graze on.

Ans: We have a circle with centre  $P$  and radius 10m. Let  $AC$  and  $AB$  be the boundary walls present. The striped region shows the area available for grazing. Now if we see carefully we can see that area of the shaded region is nothing but the sum of the areas of the rectangle  $AEPC$ , the two right angled triangles  $EPB$  and  $DPC$  and the sector  $BPC$ .

$$PB = PC = \text{radius of the circle} = 10m$$

$$PE = 5\sqrt{2}m$$

$$\text{AS } PE^2 + EB^2 = PB^2 \text{ (Pythagorean property of a right angled triangle)}$$

$$\text{From here, we get } EB = 5\sqrt{2}m$$

$$\text{Similarly, as } PD = 5m, PC = 10m$$

$$CD = 5\sqrt{2}m \text{ (pythagorean property)}$$

Therefore,

$$\text{Area of rectangle } AEPC = (5\sqrt{2} \times 5) = 25\sqrt{2}$$

$$\text{Area of triangle } PEB = \frac{1}{2} \times 5\sqrt{2} \times 5\sqrt{2} = 25$$

$$\text{Area of triangle } PDC = \frac{5 \times 5\sqrt{3}}{2} = \frac{25\sqrt{3}}{2}$$

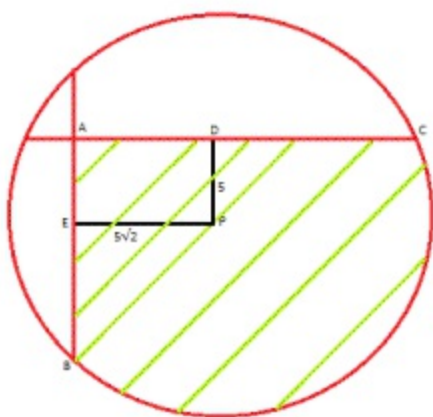
$$\text{Area of sector } BPC = 10 \times \pi^2 \frac{\angle BPC}{360}$$

$$\text{Now, } \angle BPC = 360 - (\angle BPE + \angle EPD + \angle EPD)$$

$$= 360 - [\arctan(BE/EP) + \arctan(CD/DP) + 90] = 360 - [\arctan(\frac{5\sqrt{2}}{5\sqrt{2}}) + \arctan(\frac{5\sqrt{3}}{5}) + 90] = 360 - (45 + 60 + 90) = 165$$

$$\text{Therefore area of sector } BPC = 100\pi \times \frac{165}{360} = \frac{275\pi}{6}$$

$$\text{Thus Total area of the Grazing Field} = 25\sqrt{2} + 25 + \frac{25\sqrt{3}}{2} + \frac{275\pi}{6}$$



143. Draw the graph(on plain paper) of  
 $f(x) = \min\{|x| - 1, |x - 1| - 1, |x - 2| - 1\}$

Ans: \*Graph of  $y = |x|$  is same as,  $y = x$ , where  $x$  is non negative. For negative values of  $x$ , it is the graph of  $y = x$  reflected about  $x$  - axis.

\*Now we apply transformations to find the remaining graphs.

\*Graph of  $y = |x| - 1$  can be found by lowering the graph of  $y = |x|$  by 1 unit along  $y$  - axis.

\*Graph of  $y = |x - 1| - 1$  can be found by first shift the graph of  $y = |x|$  along positive direction of  $x$  - axis by 1 unit, and then lowering it by 1 unit along  $y$  - axis.

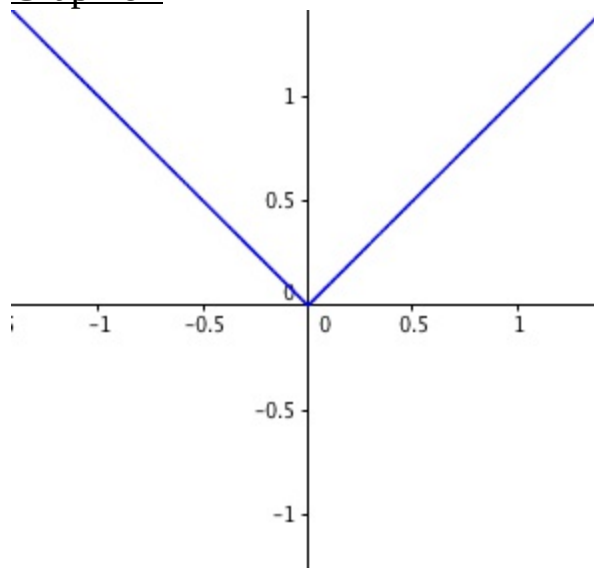
\*Graph of  $y = |x - 2| - 1$  can be found by first shift the graph of  $y = |x|$  along positive direction of  $x$  - axis by 2 unit, and then lowering it by 1 unit along  $y$  - axis.

\*Now we will plot all the graphs together and then consider the portion which are 'lowest'.

(all of them together)

(considering only the minimum portions)

Graph of  $|x|$



144. Sketch, on plain paper, the graph of  $y = \frac{x^2 + 1}{x^2 - 1}$

Ans: 1.**Domain:** The function is defined at all real numbers except  $x = 1$  and  $x = -1$

2.**Even/Odd:** Clearly  $f(x) = f(-x)$ . Hence it is sufficient to investigate the function for positive values of  $x$  and then reflect it about  $y - axis$ .

3.**Critical Points:** Critical Points are those values of  $x$  for which the first derivative of  $f'(x)$  is either 0 or undefined. here

$$f(x) = \frac{x^2 + 1}{x^2 - 1} \implies f'(x) = -\frac{4x}{(x^2 - 1)^2}$$

Hence critical points are  $x = 0, 1, -1$

4.**Monotonicity:** The first derivative is negative for all positive values of  $x$  (note that we are only investigating for positive  $x$  values, since we can then reflect the picture about  $y - axis$  as previously found). Hence the function is 'decreasing' for all positive value of  $x$ .

5.**Second Derivative:** We compute the second derivative to understand a

couple things:

convexity/concavity of the function

examine whether the critical points are maxima, minima, inflection points.

$$f''(x) = \frac{4(3x^2 + 1)}{(x^2 - 1)^3} \implies f''(0) = -4 < 0$$

So at  $x = 0$  we have local maxima. Since  $f(0) = -1$ , we have  $(0, -1)$  as a local maxima.

Also the second derivative is negative from  $x = 0$  to  $x = 1$  and positive after  $x = 1$ . Hence the curve is under-tangent (concave) from  $x = 0$  to  $x = 1$ , and above-tangent (convex) from  $x = 1$  onward.

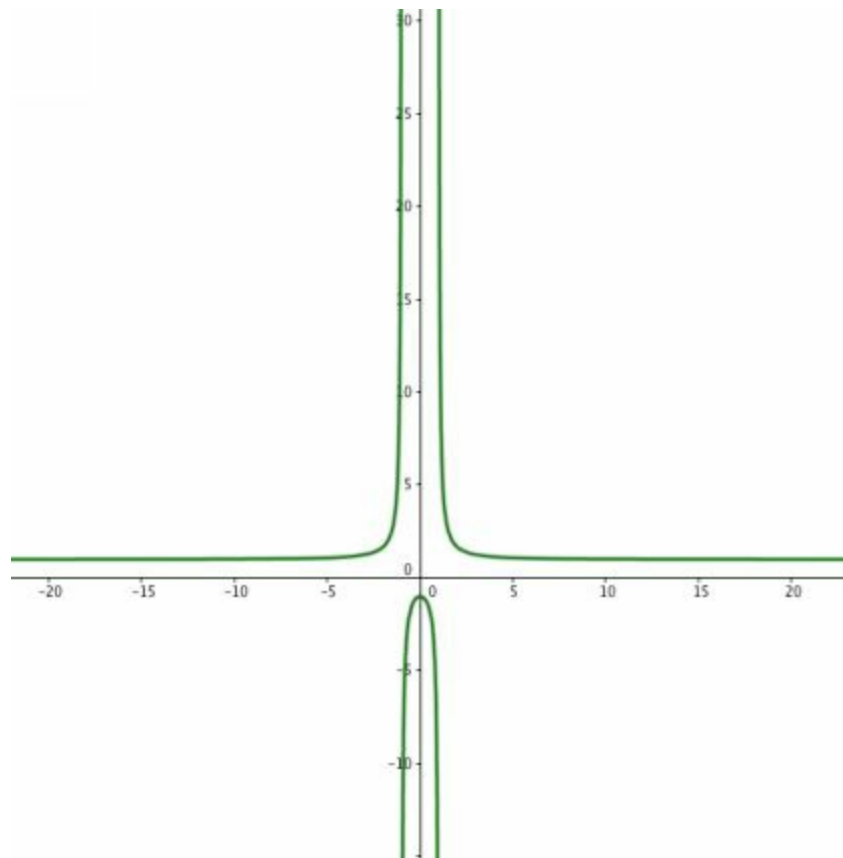
**6.Vertical Asymptote:** We next examine what happens near  $x = 1$ . We want to know what happens when we approach  $x = 1$  from left and from right. To that end we compute the following limits:

$$\lim_{x \rightarrow 1^+} \frac{x^2 + 1}{x^2 - 1} = -\infty, \quad \lim_{x \rightarrow 1^-} \frac{x^2 + 1}{x^2 - 1} = \infty$$

**7.Horizontal Asymptote:** Finally we examine what happens when  $x$  approaches  $\infty$ . To that end we compute the following:

$$\lim_{x \rightarrow \infty} \frac{x^2 + 1}{x^2 - 1} = 1$$

Using above informations we draw the following graph:



145. Find the number of positive integers  $\leq 6300$  such that none of them is divisible by 3, 5, 7.

Ans: Let  $A$  denote the occurrence of a multiple of 3,  $B$  denote that of 5 and  $C$  denote that of 7. Clearly the required number of cases  $= 6300 - |A + B + C| = 6300 - (|A| + |B| + |C| - |AB| - |BC| - |CA| + |ABC|)$ . Now  $|A| + |B| + |C| - |AB| - |BC| - |CA| + |ABC| =$  no. of multiples of 3 + no. of multiples of 5 + no. of multiples of 7 - no. of multiples of 15 - no. of multiples of 21 - no. of multiples of 35 + no. of multiples of 105.

146. Use the derivatives and left and right limits at the points of discontinuities, if any, of the function

$f(x) = \frac{x}{x+2} + \frac{x+2}{x}$  to make conclusions of the behaviour of the function as  $x$  ranges over all possible values. Using this, draw a rough sketch of the graph of function on plain paper.

Ans:

$$f(x) = \frac{x}{x+2} + \frac{x+2}{x} \implies f'(x) = \frac{2}{(x-2)^2} - \frac{2}{x^2}$$

$$f''(x) = -\frac{4}{(x-2)^3} + \frac{4}{x^3}$$

$$f'(x) = 0 \implies x = -1 \text{ and } f''(-1) = -8 < 0$$

So  $f(x)$  is maximum at  $x = -1$ .  $f(x)$  increases on left and decreases on right of  $x = -1$

Maximum value of  $f(x)$  is  $f(-1) = -2$

Clearly  $f(x)$  is undefined at  $x = 0, -2$ , hence discontinuous there.

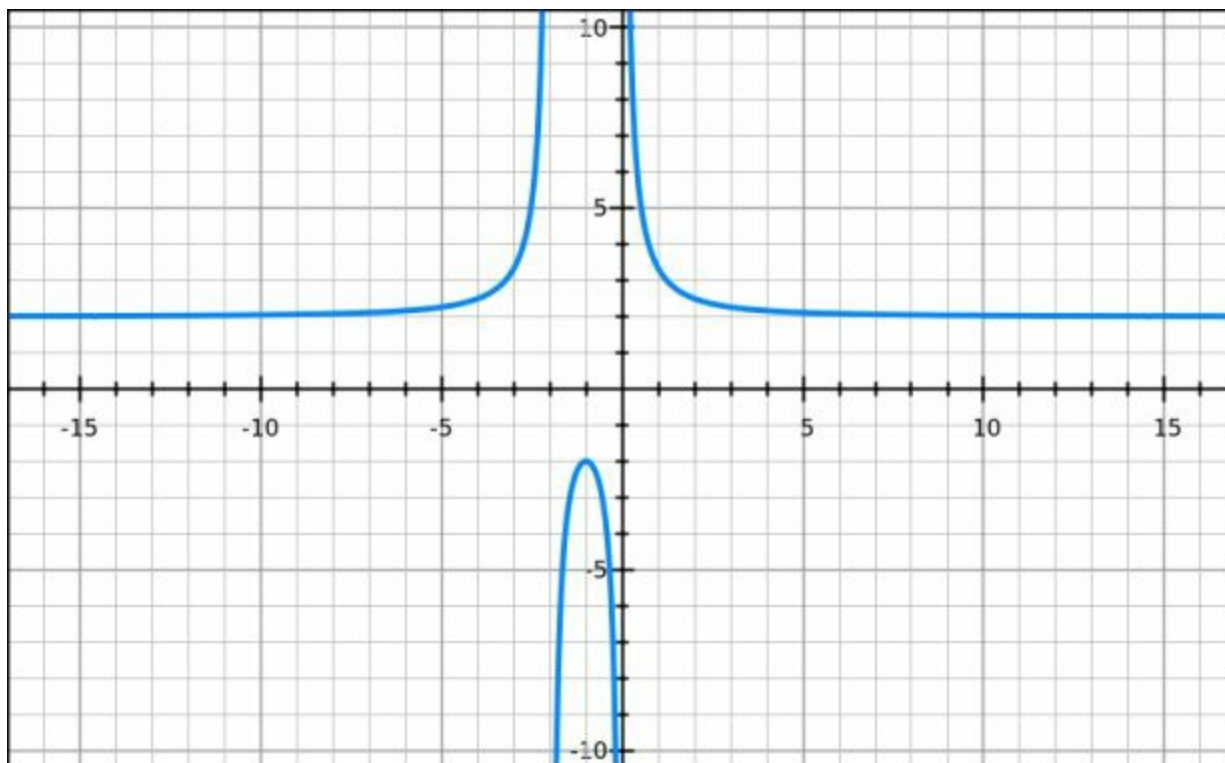
$$\lim_{x \rightarrow 0^+} f(x) = \infty, \lim_{x \rightarrow 0^-} f(x) = -\infty, \lim_{x \rightarrow -2^+} f(x) = -\infty, \lim_{x \rightarrow -2^-} f(x) = \infty$$

Let  $y = f(x) = \frac{x}{x+2} + \frac{x+2}{x} \implies x^2(y-2) + 2xy - 4x - 4 = 0$

Equating highest powers of  $x$  (i.e. coefficient of  $x^2$ , i.e.  $y-2$ ) and  $y$  (i.e. coefficient of  $y$ , i.e.,  $x$ ) to 0 we get  $y = 2, x = 0$

So asymptotes are  $x = 0, y = 2$ . Since equation is two degree, it has no more asymptotes.

Using above, we trace the curve:



147. For the following function  $f$  study its derivatives and use them to sketch its graph on plain on plain paper.

$$f(x) = \frac{x-1}{x+1} + \frac{x+1}{x-1} \text{ for } x \neq -1, 1$$

Ans:

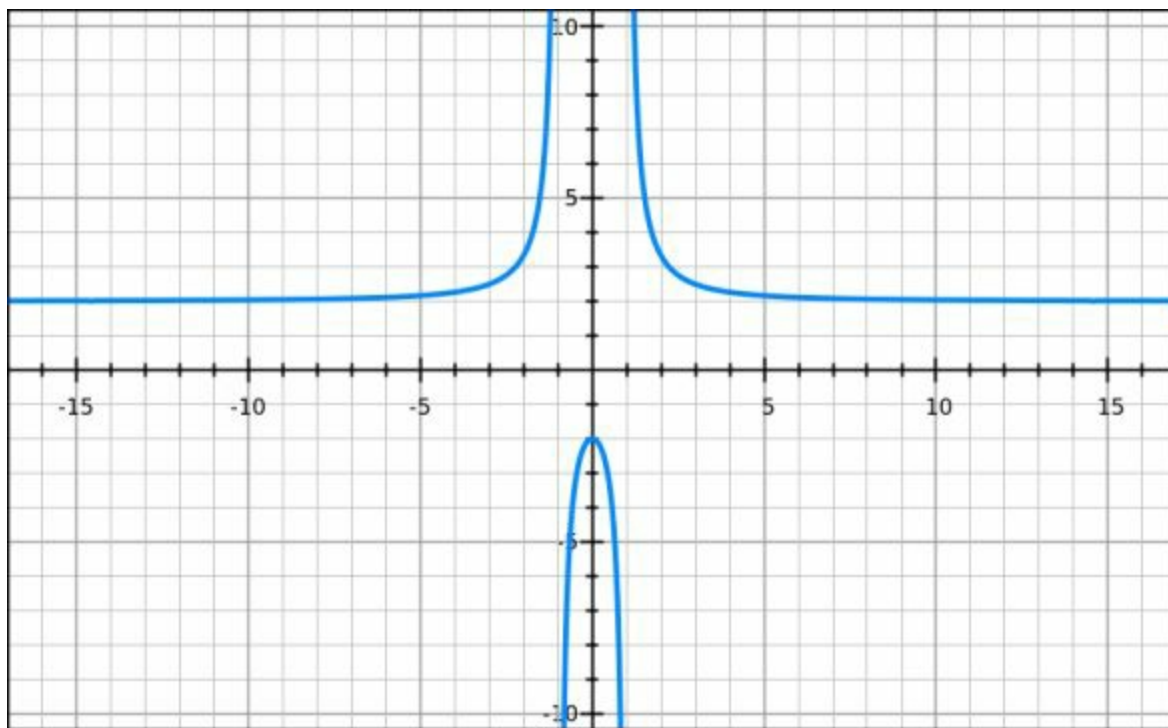
$$f(x) = \frac{x-1}{x+1} + \frac{x+1}{x-1} \implies f'(x) = \frac{2}{(x+1)^2} - \frac{2}{(x-1)^2}, f''(x) = \frac{-4}{(x+1)^3} + \frac{4}{(x-1)^3}$$

$f'(x) = 0 \implies x = 0$  and  $f''(0) = -8 < 0$  So 0 is a point of maximum.

$f(x)$  increases on left and decreases on right of  $x = 0$

Clearly  $f(x)$  is undefined, hence discontinuous at  $x = \pm 1$

Using above, we trace the curve:



148. Study the derivative of the function  $f(x) = x^3 - 3x^2 + 4$  and roughly sketch the graph of  $f(x)$  on plain paper.

Ans:

$$f(x) = x^3 - 3x^2 + 4 \implies f'(x) = 3x(x - 2)$$

Clearly  $f'(x) > 0$  when  $x < 0$  or  $x > 2$  and  $f'(x) \leq 0$  when  $0 \leq x \leq 2$

So  $f(x)$  decreases strictly in  $(0, 2)$  and increases strictly in  $(-\infty, 0)$  and  $(2, \infty)$

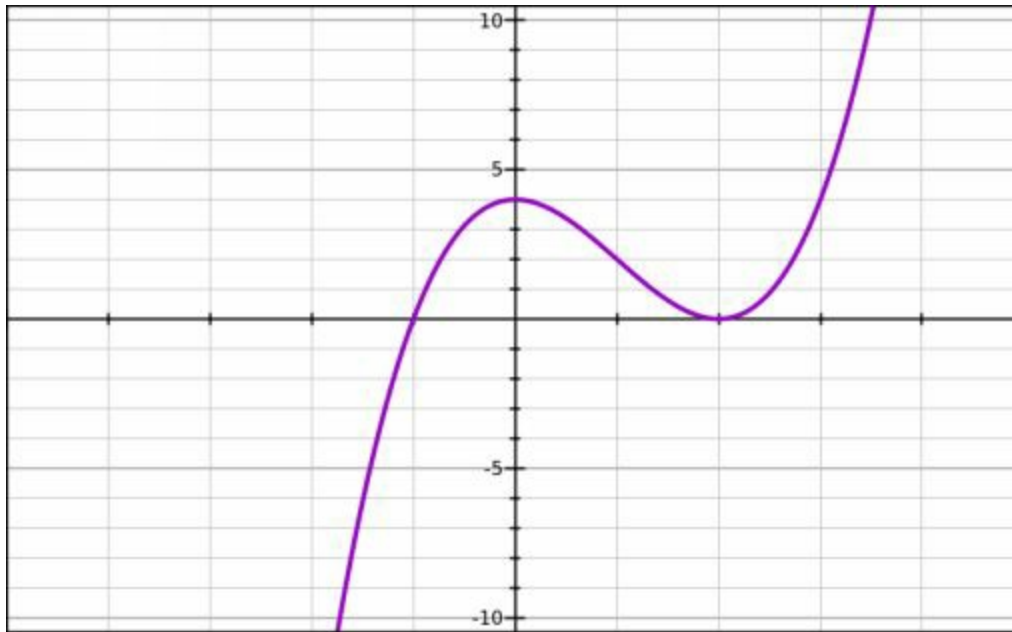
So  $f$  is maximum at  $x = 0$ , minimum at  $x = 2$ .

So maximum value of  $f$  is  $f(0) = 4$

Minimum of  $f$  is  $f(2) = 0$

So the graph is as follows:

Attachments:



149. Show that for every positive integer  $n$ ,  $\sqrt{n}$  is either an integer or an irrational number.

Ans: Let us assume  $\sqrt{n} = \frac{p}{q}$  where  $p, q \in \mathbb{Z}, q \neq 0, \gcd(p, q) = 1$

So  $n = \frac{p^2}{q^2}$  or  $q^2 n = p^2 \implies n | p^2 \implies n | p$ , putting  $p = rn$  for integral  $r$ , we get,

$$q^2 n = r^2 n^2 \implies q^2 = r^2 n \implies n | q^2 \implies n | q$$

But it is contradictory that  $n$  divides both  $p, q$  as  $\gcd(p, q) = 1$

So either  $q | p$  or  $n$  is not rational and expressible in  $p/q$  form.

I.e.  $\sqrt{n}$  is either an integer or an irrational number.

150. Suppose that the roots of  $x^2 + px + q = 0$  are rationals and  $p, q$  are

integers. Prove that the roots are integers.

Ans: Let the roots be  $\frac{p_1}{q_1}$  and  $\frac{p_2}{q_2}$ .

By rational root theorem we have  $q_1|1$  and  $q_2|1$ .

Hence  $q_1, q_2 \in \{-1, +1\}$  implying the roots to be integers.

151. Let  $X = \{0, 1, 2, 3, \dots, 99\}$ . For  $a, b$  in  $X$ , we define  $a * b$  to be the remainder obtained by dividing the product  $ab$  by 100. For example,  $9 * 18 = 62$  and  $7 * 5 = 35$ . Let  $x$  be an element in  $X$ . An element  $y$  in  $X$  is called the inverse in  $x$  if  $x * y = 1$ . Find which of the elements 1, 2, 3, 4, 5, 6, 7 have inverses and write down their inverses.

Ans: Clearly to find the inverse of a element  $a \in X$  we need to solve for  $x$  the equation  $a * x = 1$

i.e,  $a \times x \in S = (1, 101, 201, 301, \dots)$  where  $\times$  is the usual multiplication.

For  $a = 2, 4, 6$  it is impossible to have inverses as they are even and  $a \times x$  cannot have solution in  $S$ .

The smallest possible solution of  $1 \times x \in S$  is 1. For other elements in  $S$ , 1 has no inverse belonging to  $X$ . Hence inverse of 1 is unique.

In a similar manner, it can be shown that 67, 43 are inverses of 3, 7.

Finally 5 does not have inverse as no element in  $S$  is divisible by 5.

Hence 1, 3, 7 have inverses which are 1, 67, 43 respectively.

152. Sketch the curve  $y = 4x^3 - 3x + a$  on plain paper and show that the equation (in  $x$ )

$4x^3 - 3x + a$  (where the real constant  $a$  is such that  $0 < |a| < 1$ ) has three distinct real roots all of which has absolute values smaller than 1.

Ans:

$f(x) = 4x^3 - 3x + a \implies f(1) = 1 + a > 0, f(-1) = -1 + a, \forall 0 < |a| < 1$   
 $f'(x) = 12x^2 - 3$  Clearly  $f$  increases when  $x > 1/2$  or  $x < -1/2$  and decreases when  $-1/2 < x < 1/2$ . So  $f$  is maximum at  $x = -1/2$  minimum at  $x = 1/2$ .

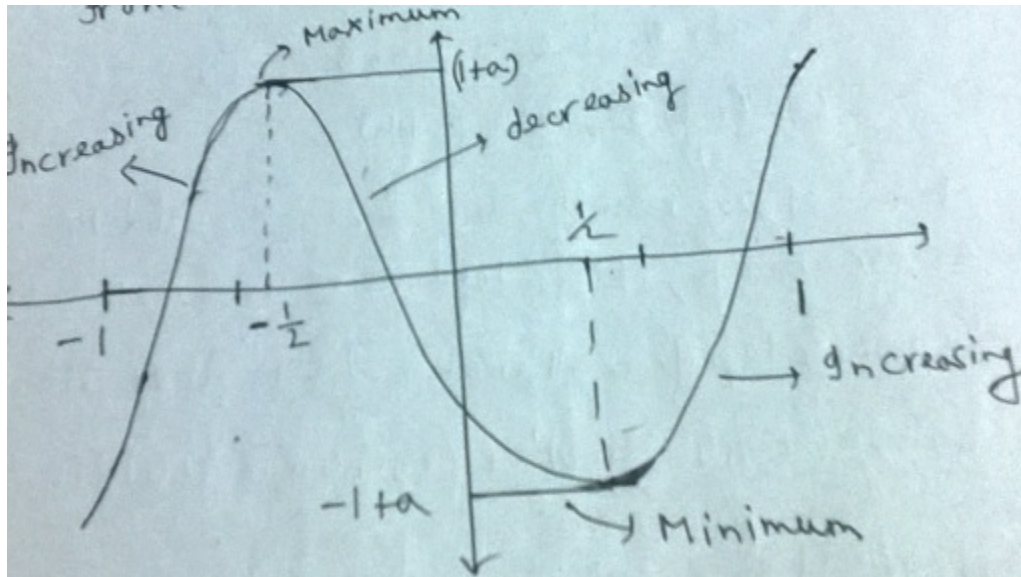
$$f(1/2) = -1 + a < 0, f(-1/2) = 1 + a > 0$$

Using above we trace the curve of  $f(x)$ .

$f(-1)$  is negative,  $f$  increases in  $(-1, -1/2)$  hence intersects  $x$ -axis once in that interval. Again, it assumes maximum value  $(1 + a) > 0$  at  $x = -1/2$  and thereafter decreases upto  $x = 1/2$  where it assumes minima  $(-1 + a) < 0$ . So it has a root at  $(-1/2, 1/2)$ . Lastly  $f(1) > 0$  and  $f$  increases strictly in  $(1/2, 1)$ . So again  $f$  has a root in  $(1/2, 1)$ .

So all three roots of  $f$  has absolute values less than 1.

**the required graph:**



153. Sketch the set  $A \cap B$  in the Argand plane, where  $A = \{z : |\frac{z+1}{z-1}| \leq 1\}$  and  $B = \{z : |z| - \Re z \leq 1\}$

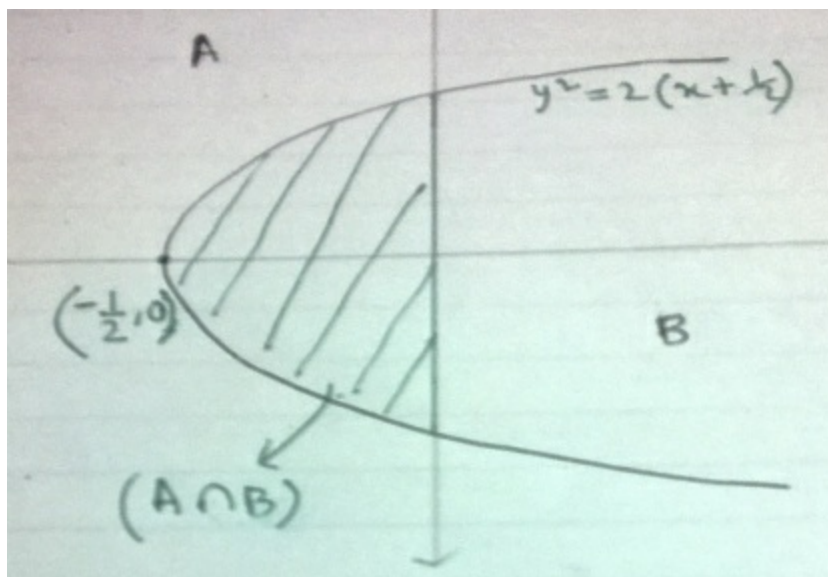
Ans:

$z = x + iy$  where  $x$  and  $y$  are real.

Then for set  $A$ , we have  $(x+1)^2 + y^2 \leq (x-1)^2 + y^2 \implies x \leq 0$

For set  $B$  we've,  $x^2 + y^2 \leq (1+x)^2 \implies y^2 \leq 2x+1$  which is the region within a parabola with its vertex at  $(-\frac{1}{2}, 0)$  and the real axis as its axis.

Hence  $A \cap B$  denotes the left half of the region within the aforesaid parabola, that is the region lying in the second and third quadrants.



154. Let  $[x]$  denote the largest integer less than or equal to  $x$ . For example,

$$\left[\frac{4}{1}\right] = 4; \left[\frac{4}{2}\right] = 2$$

. Draw a rough sketch of the graphs of the following on plain paper:

(i)  $f(x) = [x]$

(ii)  $g(x) = x - [x]$

(iii)  $h(x) = \frac{1}{[x]}$

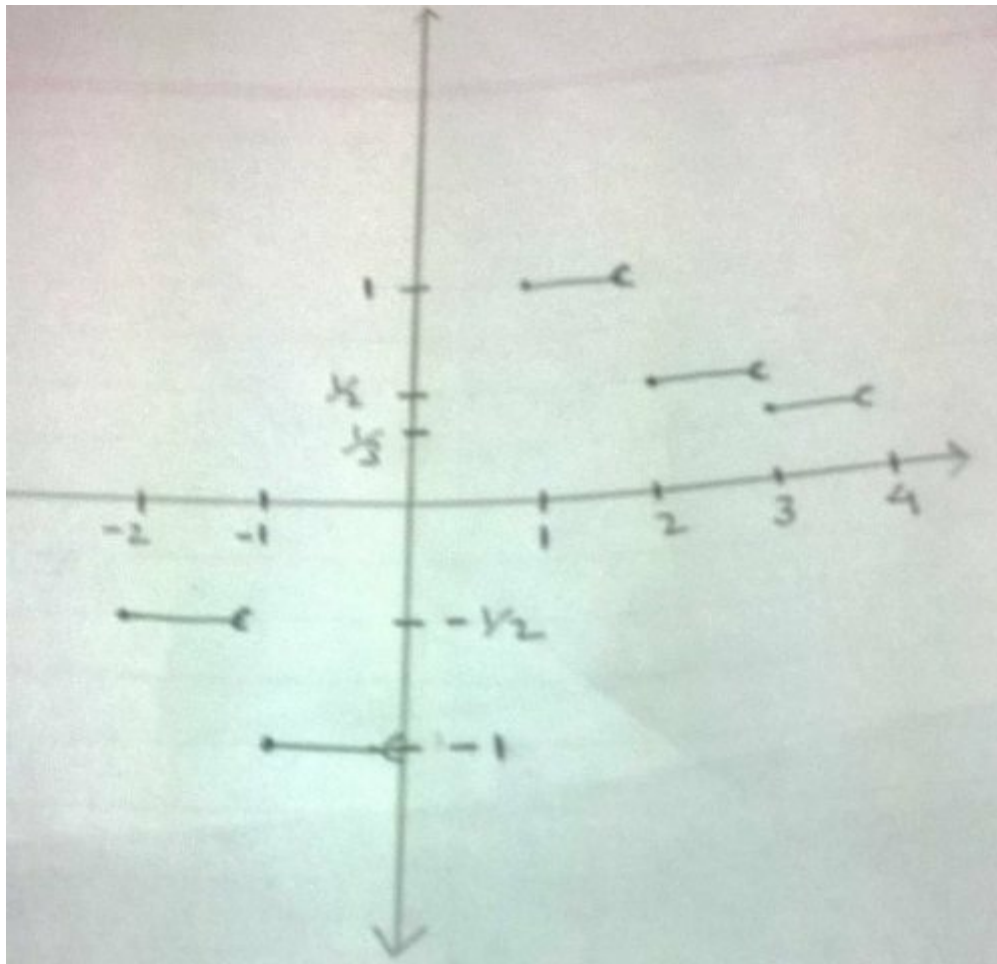
Ans: i) Clearly  $a \in \mathbb{I}$ ,  $f(x) = a - 1$  where  $a - 1 \leq x \leq a$ .

ii)  $g(x) = x - [x] = \{x\}$  where  $\{x\}$  = fractional part of  $x$ . Clearly  $0 \leq \{x\} \leq 1$

iii)  $\begin{cases} -2 \leq x \leq -1, [x] = -2, 1/[x] = -1/2 \\ -1 \leq x \leq 0, [x] = -1, 1/[x] = -1, \\ 1 \leq x \leq 2, [x] = 1, 1/[x] = 1, \\ 2 \leq x \leq 3, [x] = 2, 1/[x] = 1/2, \\ 3 \leq x \leq 4, [x] = 3, 1/[x] = 1/3 \end{cases}$

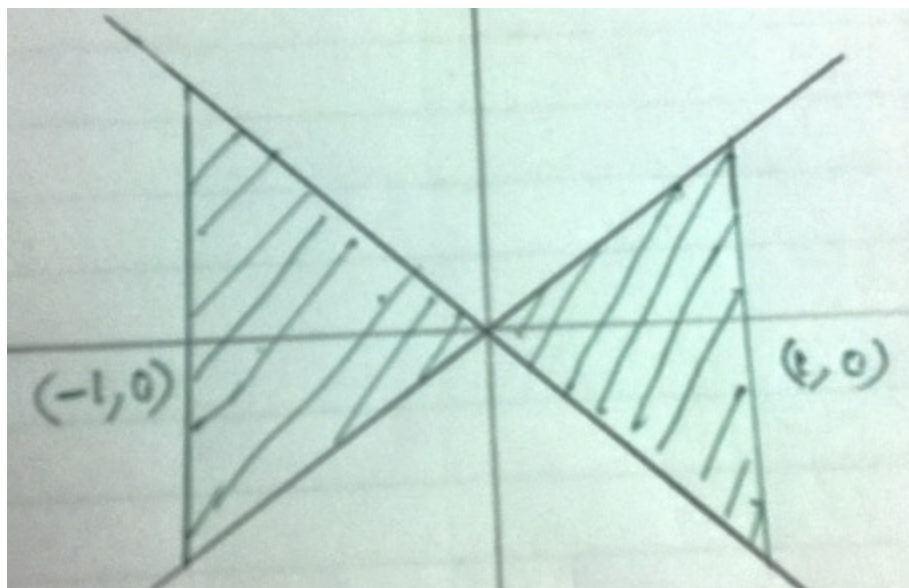
and so on.

So the graphs are, respectively:



155. Draw the region of points  $(x, y)$  in the plane, which satisfy  $|y| \leq |x| \leq 1$ .

Ans: For tracing the region, first we draw  $y = |x|$ , and reflect it about  $x$ -axis to get the graph of  $|y| = |x|$ , then we shade the region sandwiched between the two lines  $y = x$  and  $y = -x$  and containing  $x$ -axis, in the range  $[-1, 1]$  to get the required region.



156. Sketch, on plain paper, the regions represented on the plane by the following:

i)  $|y| = \sin x$ ;

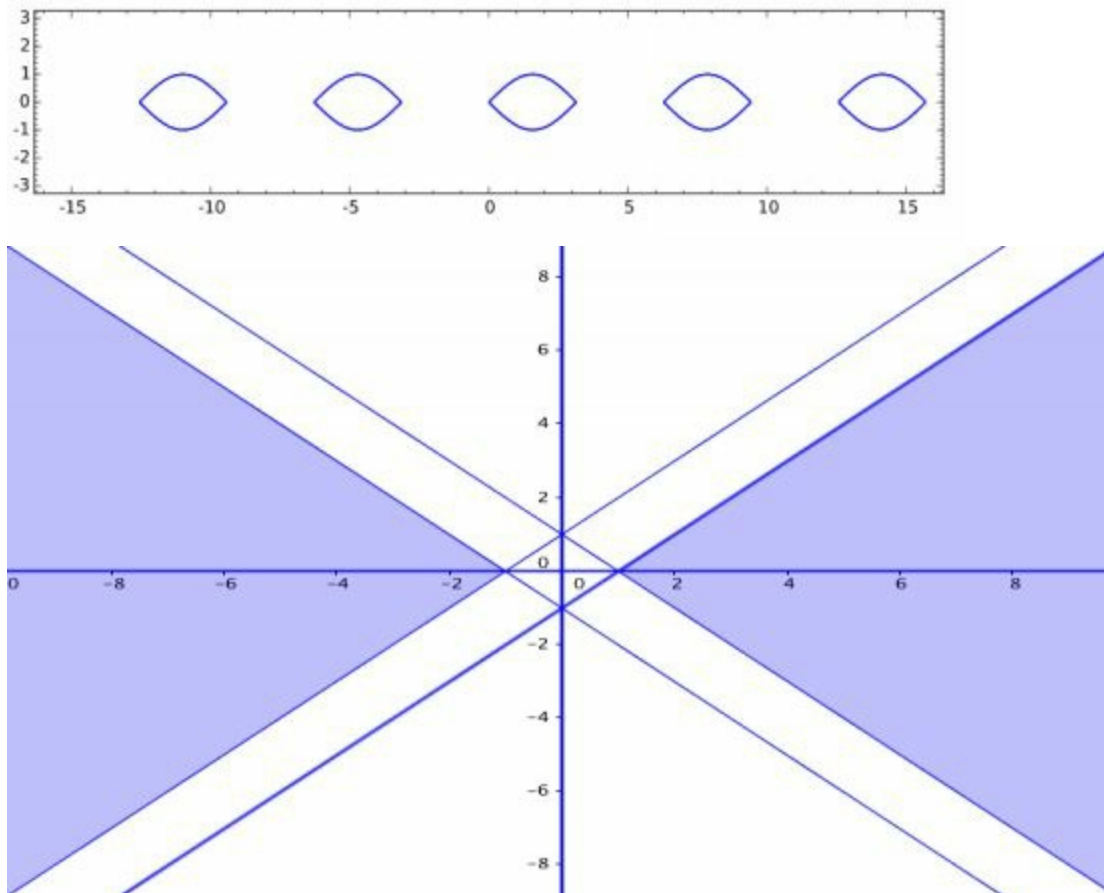
ii)  $|x| - |y| \geq 1$

Ans: i) For transforming  $y = \sin x$  to  $|y| = \sin x$  we omit the portions where sine curve is negative, and trace both the original sine curve and its reflection about  $x$ -axis where sine curve is positive.

ii) First we transform  $y = x - 1$  to  $y = |x| - 1$  by erasing the curve for negative values of  $x$  and reflecting the curve for positive values of  $x$  about  $y$ -axis. Then we transform  $y = |x| - 1$  to  $|y| = |x| - 1$  by omitting the

portions of  $y = |x| - 1$  where  $y$  is negative, and tracing both the original curve  $y = |x| - 1$  and its reflection about  $x$ -axis where  $y$  is positive. Lastly we see that,  $(0,0)$  doesn't lie in the region  $|x| - |y| \geq 1$  so we shade the regions, which does not contain the origin.

The graphs are as follows:



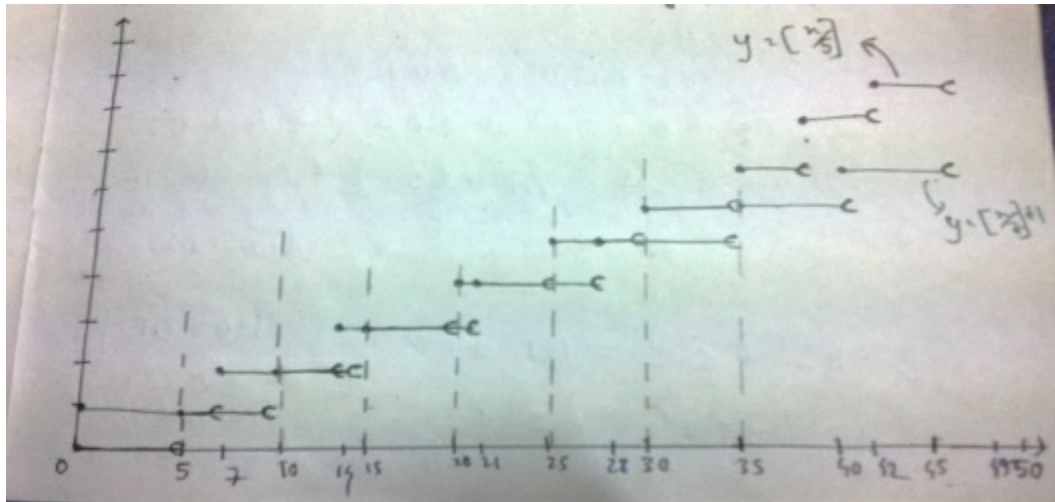
157. Find all positive integers  $x$  such that,  $[x/5] - [x/7] = 1$ , where, for any real number  $t$ ,  $[t]$  is the greatest integer less than or equal to  $t$ .

Ans: We trace the curves of  $y = [x/5]$  and  $y = [x/7] + 1$ .

Clearly integer values of  $x$  where the curves intersect are

5, 6, 10, 11, 12, 13, 15, 16, 17, 18, 19, 21, 22, 23, 24, 28, 29 hence they are required

integral solutions.



158. Using Calculus, sketch the graph on plain paper of the function  $f(x) = x^2 + x + 1/x + 1/x^2$  for  $x \neq 0$

Show that the function  $f$  defined as above for *positive real* numbers attains a unique minimum. What is the minimum value of function? What is the value of  $x$  at which minimum is attained?

Ans:

$$f(x) = x^2 + x + 1/x + 1/x^2 \implies f'(x) = 2x + 1 - 1/x^2 - 2/x^3$$

$$f'(x) = 0 \implies (x^2 - 1)(2x^2 + x + 2) = 0. \text{ Since } x \text{ is positive, only root of } f'(x) = 0 \text{ is } x = 1$$

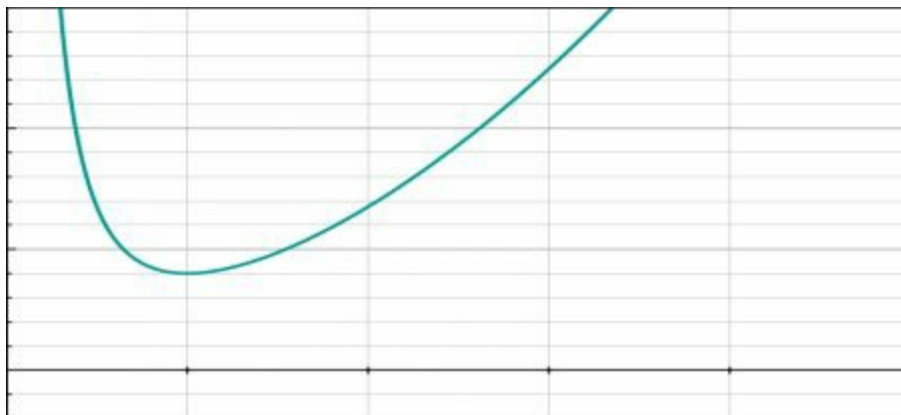
Clearly  $f'(x) < 0$  for all positive  $x$  in left of  $x = 1$  and  $f'(x) > 0$  for all  $x$  in right of  $x = 1$ .

Hence  $f$  decreases in left of  $x = 1$ , reaches minimum at  $x = 1$  and increases thereafter.

Since  $f'$  changes sign only at  $x = 1$  so  $f$  has unique minimum.

Minimum value of  $f(x)$  is  $f(1) = 4$

So required graph is:



159. Let  $[x]$  denote the largest integer (positive, negative or zero) less than or equal to  $x$ . Let  $y = f(x) = [x] + \sqrt{x - [x]}$  be defined for all real numbers  $x$ .

(i) Sketch on plain paper, the graph of the function  $f(x)$  in the range  $-5 \leq x \leq 5$

(ii) Show that, given any real number  $y_0$ , there is a real number  $x_0$  such that  $y_0 = f(x_0)$

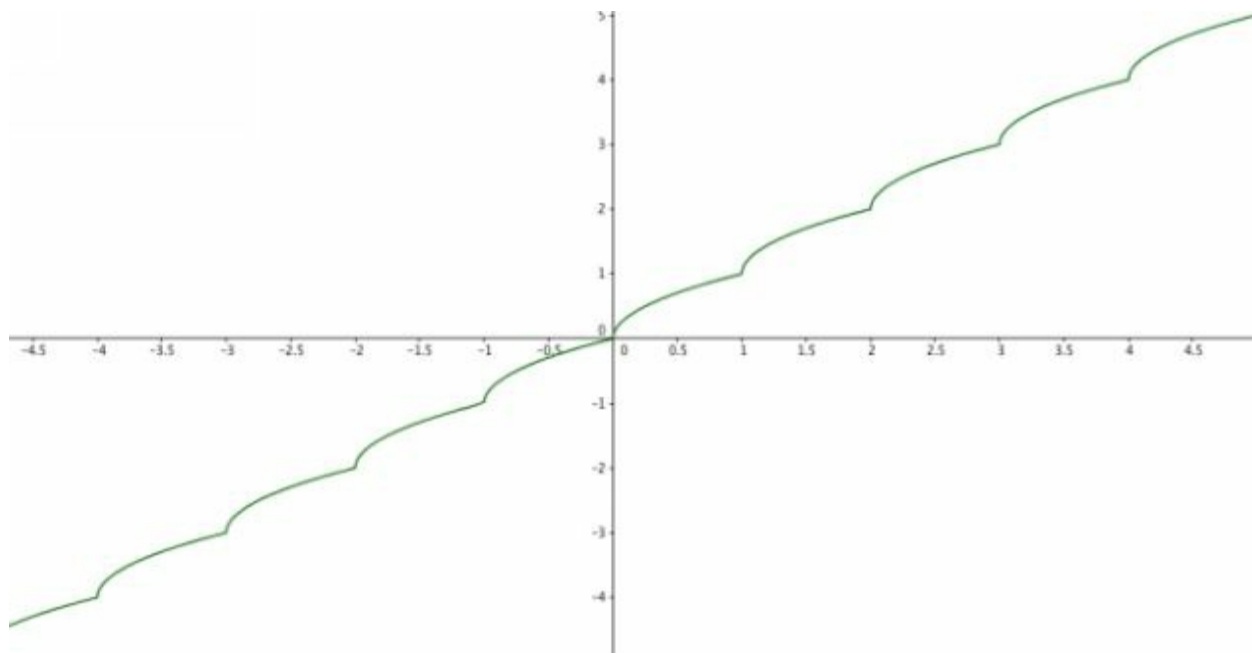
Ans: i) Note that graph of  $\sqrt{x - [x]}$  is same as of  $\sqrt{t}$ ,  $0 \leq t \leq 1$  which is the positive portion of the parabola  $y = t^2$  in the domain  $[0, 1]$ .

Clearly  $[x]$  part only increments (or decrements) it by integer quantity as  $[x]$  is constant between any two integers. That for any integer  $k$  for all  $x \in (k, k+1)$ ,  $f(x) = k + \sqrt{t}$ ,  $t \in (0, 1)$ .

Hence graph of  $f(x)$  is given below.

ii) Finally consider an arbitrary value  $y_0$ . We take  $x_0 = [y_0] + (y - [y_0])^2$ . Then  $f(x_0) = [x_0] + \sqrt{x_0 - [x_0]} = [y_0] + \sqrt{(y - [y_0])^2} = y_0$  (since  $0 \leq y - y_0 < 1 \implies 0 \leq (y - y_0)^2 < 1$ )

Alternatively, since it is clear from the graph that, every horizontal line in the plane cuts the graph of  $f(x)$  at least once, hence  $f(x)$  is invertible, i.e., given any real number  $y_0$ , there is a real number  $x_0$  such that  $y_0 = f(x_0)$



160. If  $\alpha, \beta, \gamma$  are roots of the equation  $x^3 + px^2 + qx + r = 0$ , find the equation whose roots are  $\alpha - \frac{1}{\beta\gamma}, \beta - \frac{1}{\gamma\alpha}, \gamma - \frac{1}{\alpha\beta}$ .

Ans: We have by Vieta's Theorem,  $\alpha\beta\gamma = -r$

So,  $\alpha - \frac{1}{\beta\gamma} = \alpha - \frac{\alpha}{\alpha\beta\gamma} = \alpha + \frac{\alpha}{r}$  Similar for,  $\beta - \frac{1}{\gamma\alpha}, \gamma - \frac{1}{\alpha\beta}$

Let  $y = x + x/r \implies x = \frac{ry}{r+1}$

Hence the required equation is obtained by putting  $\frac{ry}{r+1}$  in place of  $x$  in the given equation.

The transformed equation is  $\boxed{r^2y^3 + pr(r+1)y^2 + q(r+1)^2y + (r+1)^3 = 0}$

161. By considering the expression  $(1 + \sqrt{2})^n + (1 - \sqrt{2})^n$  where  $n$  is a positive integer, show that the integers  $[(1 + \sqrt{2})^n]$  are alternatively even and odd as  $n$  takes values  $1, 2, \dots$ . Here for any real number  $x$ ,  $[x]$  denotes the greatest integer less than or equal to  $x$ .

Ans: Let  $P_n = (1 + \sqrt{2})^n + (1 - \sqrt{2})^n$

$P_1 = (1 + \sqrt{2}) + (1 - \sqrt{2}) = 2$  which is even.

Let  $P_n$  and  $P_{n-1}$  be even integers. Let  $P_n = 2r, P_{n-1} = 2s$  where  $r, s \in \mathbb{Z}^+$

$$P_{n+1} = (1 + \sqrt{2})^{n+1} + (1 - \sqrt{2})^{n+1} = [(1 + \sqrt{2})^n + (1 - \sqrt{2})^n](1 + \sqrt{2} + 1 - \sqrt{2}) + (1 + \sqrt{2})(1 - \sqrt{2})[(1 + \sqrt{2})^{n-1} + (1 - \sqrt{2})^{n-1}] = 4r - 2s$$

Clearly above is even so by induction,  $(1 + \sqrt{2})^n + (1 - \sqrt{2})^n$  is always even.

Let  $(1 + \sqrt{2})^n + (1 - \sqrt{2})^n = 2H$

When  $n = 1, 3, 5, \dots, -1 < (1 - \sqrt{2})^n < 0$

So  $(1 + \sqrt{2})^n = 2H - (1 - \sqrt{2})^n$ . Clearly

$2H - 1 < -(1 - \sqrt{2})^n + 2H = (1 + \sqrt{2})^n < 2H$  So,

$[(1 + \sqrt{2})^n] = 2H - 1$  which is odd.

When  $n = 2, 4, 6, \dots, 0 < (1 - \sqrt{2})^n < 1$

$(1 + \sqrt{2})^n = 2H - (1 - \sqrt{2})^n$  Clearly

$2H < (1 + \sqrt{2})^n = 2H - (1 - \sqrt{2})^n < 2H + 1$ .

So  $[(1 + \sqrt{2})^n] = 2H$  which is even.

Hence, the integers  $[(1 + \sqrt{2})^n]$  are alternatively even and odd as  $n$  takes values  $1, 2, \dots$

162. Suppose that  $AB$  is an arc of a circle with a given radius and centre subtending an angle  $\theta$  ( $0 < \theta < \pi$  is fixed). Consider an arbitrary point  $P$  on

this arc and the product  $l(AP).l(PB)$ , where  $l(AP)$  and  $l(PB)$  denote the lengths of the straight lines  $AP$  and  $PB$ , respectively. Determine possible locations of  $P$  for this this product will be maximised. Justify your answer.

Ans: Let arc  $AP$  subtend  $\phi$  angle at centre. So arc  $PB$  subtend  $\theta - \phi$  angle at centre.

Let  $O$  be the centre of the circle.

In triangle  $APO \rightarrow \frac{l(AP)}{\sin \phi} = k$  [say] So  $l(AP) = k \cdot \sin \phi$

Similarly in triangle  $PBO$ ,  $l(PB) = c \cdot \sin(\theta - \phi)$  for some constant  $c$ .

Let  $g(\phi) = l(AP).l(PB) = kc \cdot \sin \phi \cdot \sin(\theta - \phi)$

$$g'(\phi) = kc[\cos \phi \sin(\theta - \phi) - \sin \phi \cos(\theta - \phi)] = 0 \implies \sin(\theta - 2\phi) = 0 \implies \phi = \theta/2$$

$$g''(\phi) = -2kc \cos(\theta - 2\phi), g''(\theta/2) = -2kc < 0 \text{ [since } k, c \text{ are positive as } \theta(0 < \theta < \pi)]$$

So  $g$  is maximum at  $\phi = \theta/2$ . that is  $AP, PB$  subtends exactly equal angles at centre and hence  $P$  is the mid point of arc  $AB$ .

163. i) If  $A + B + C = n\pi$  where  $n$  is a positive integer, show that  $\sin 2A + \sin 2B + \sin 2C = (-1)^{n-1} 4 \sin A \sin B \sin C$  ii) Let triangles  $ABC$  and  $DEF$  be inscribed in the same circle. If the triangles are of equal perimeter, then prove that  $\sin A + \sin B + \sin C = \sin D + \sin E + \sin F$  iii) State and prove the converse of (ii) above.

Ans:

i)

$$\sin 2A + \sin 2B + \sin 2C = 2 \sin(A+B) \sin(A-B) + 2 \sin C \cos C = 2 \sin(n\pi - C) \cos(A-B) + 2 \sin C \cos C = 2(-1)^{n-1} (\sin C \cos(A-B) + \sin C \cos C) \dots i$$

When  $n$  is odd,

$$\begin{aligned} i \implies 2 \sin C (\cos(A-B) + \cos C) &= 4 \sin C \cos\left(\frac{A-B+C}{2}\right) \cos\left(\frac{A-B-C}{2}\right) \\ &= 4 \sin C \cos\left(\frac{n\pi}{2} - B\right) \cos\left(\frac{n\pi}{2} - C\right) = 4 \sin C (-1)^{n-1} \sin B (-1)^{n-1} \sin C \text{ [as } n \text{ is} \end{aligned}$$

odd]

$$= 4\sin A \sin B \sin C$$

When  $n$  is even,

$$\begin{aligned} i \implies 2\sin C(\cos C - \cos(A - B)) &= 4\sin C \left[ \sin\left(\frac{C + A - B}{2}\right) \sin\left(\frac{A - B - C}{2}\right) \right] \\ &= 4\sin C \sin\left(\frac{n\pi - 2B}{2}\right) \sin\left(\frac{2A - n\pi}{2}\right) = -4\sin C (-1)^{n-1} \sin B (-1)^{n-1} \sin C \text{ [as } n \\ &\text{ is even]} \end{aligned}$$

$$= -4\sin A \sin B \sin C$$

Combining both cases,  $\sin 2A + \sin 2B + \sin 2C = (-1)^{n-1} 4\sin A \sin B \sin C$

ii) Let  $a, b, c$  and  $d, e, f$  be sides of  $ABC$  and  $DEF$  respectively, and  $R$  be the radius of circle.

Given

$$a + b + c = d + e + f \implies 2R(\sin A + \sin B + \sin C) = 2R(\sin C + \sin D + \sin F)$$

[by applying sine rule]

$$\implies (\sin A + \sin B + \sin C) = (\sin C + \sin D + \sin F)$$

iii) Conversely if

$$(\sin A + \sin B + \sin C) = 2R(\sin C + \sin D + \sin F) \implies \frac{a + b + c}{2R} = \frac{d + e + f}{2R} \implies a + b + c = d + e + f$$

i.e, perimeters are equal.

164. In a club of 80 members, 10 members play none of the games Tennis, Badminton and Cricket. 30 members play exactly one of the three games and 30 members play exactly two of the three games. 45 members play at least one of the games among Tennis and Badminton, whereas 18 members play both Tennis and Badminton. Determine the number of Cricket playing members.

Ans: Number of playing members =  $80 - 10 = 70$

Number of players playing exactly all three games =  $70 - (30 + 30) = 10$

18 play both tennis, badminton, among them 10 play all three games.

Hence,  $18 - 10 = 8$  play only tennis and badminton.

So  $30 - 8 = 22$  members play exactly two games with cricket between one of the games.

Now  $45 - 18 = 27$  play one of the combination of games:(cricket,tennis),  
(cricket,badminton),(tennis),(badminton)

So players playing only tennis or only badminton= $27 - 22 = 5$

30 players play only one game; so players playing only cricket= $30 - 5 = 25$

So 25 play only cricket, 10 play all games, 22 play exactly two games with one of them being cricket.

So total cricket playing members= $25 + 10 + 22 = \boxed{57}$

165. Find all  $(x, y)$  such that

$$\sin x + \sin y = \sin(x + y) \text{ and } |x| + |y| = 1$$

Ans:

$$\sin x + \sin y = \sin(x + y) \implies 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x-y}{2}\right) = 2\sin\left(\frac{x+y}{2}\right)\cos\left(\frac{x+y}{2}\right) \implies 2\sin\left(\frac{x+y}{2}\right)\sin x \sin y = 0$$

So the cases are

$$\text{i) } \sin(x + y) = 0 \implies x + y = n\pi \text{ but } |x| + |y| = 1 \geq |x + y| \geq x + y = n\pi \text{ so } n = 0$$

$$\text{So } x + y = 0 \implies x = -y \implies |x| = |y|$$

Plugging this in  $|x| + |y| = 1$  we get  $2|x| = 1 \implies x = \pm 1/2 \implies y = \mp 1/2$   
. So the solutions for this case are  $(1/2, -1/2), (-1/2, 1/2)$

$$\text{ii) } \sin x = 0 \implies x = n\pi \text{ So } |n\pi| + |y| = 1 \implies |y| = 1 - |n\pi| \implies n = 0$$

[since  $|y| \geq 0$ ]

So  $y = \pm 1$  Solutions for this case are  $(0, \pm 1)$ .

$$\text{iii) } \sin y = 0, \text{ solutions for this case are } (\pm 1, 0)$$

So the complete set of solutions is  $\{(1/2, -1/2), (-1/2, 1/2), (0, \pm 1), (\pm 1, 0)\}$

166. i) Determine  $m$  so that the equation

$x^4 - (3m + 2)x^2 + m^2$  has four real roots in arithmetic progression.

ii) Let  $a$  and  $b$  be two real numbers. If the roots of the equation

$x^2 - ax - b = 0$  have absolute values less than one, show that each of the following conditions holds:

(i)  $|b| \leq 1$  (ii)  $a + b < 1$  and (iii)  $b - a < 1$ .

Ans: i) Let the roots be  $(a - 2d), (a - d), a, (a + d)$

by Vieta's rule  $(a - 2d) + (a - d) + a + (a + d) = 0 \implies 2a = d$  i.e., roots are  $-3a, -a, a, 3a$

Product of roots taking two at a time

$$= (-3a)(-a) + (-3a)a + (-3a)3a + (-a)a + (-a)3a + a.3a = -3m + 2$$

$$\implies 10m^2 = 3m + 2 \dots \dots \dots i$$

$$\text{product of roots} = (-3a)(-a)3a.a = m^2 \dots \dots \dots ii$$

Solving  $i, ii$  we get  $m = 6, -6/19$

ii) Let  $p, q$  be the roots.

$$\text{so } p + q = a, pq = -b \text{ so } |b| = |pq| = |p||q| \text{ but } |p| < 1, |q| < 1 \implies |b| < 1$$

$$a + b = p + q - pq$$

Let us contradictorily assume

$$a + b > 1 \implies p + q - pq > 1 \implies (p - 1)(1 - q) > 0 \text{ But since } |p| < 1, |q| < 1$$

so  $(p - 1) < 0, 1 - q > 0, (p - 1)(1 - q) < 0$  hence a contradiction arises, i.e.,  $a + b < 1$

$$b - a = -p - q - pq$$

Let us contradictorily assume that

$$b - a = -p - q - pq < 1 \implies -(p + 1)(q + 1) > 0 \text{ but } p < 1, q < 1 \text{ so}$$

$p + 1 > 0, q + 1 > 0$  so  $-(p + 1)(q + 1) < 0$  hence contradiction arises, i.e.,  $b - a < 1$

So all the three given conditions hold.

167. A rectangle  $OACB$  with the two axes as two sides, the origin  $O$  as a vertex is drawn in which the length of  $OA$  is four times the width  $OB$ . A

circle is drawn passing through the points  $B$  and  $C$  and touching  $OA$  at its mid-point, thus dividing the rectangle into three parts. Find the ratio of the areas of these three parts.

Ans: Follow figure, the coordinates marked are

$O(0, 0)$ ,  $A(4n, 0)$ ,  $C(4n, n)$ ,  $D(0, n)$ ,  $D(2n, 0)$ ,  $M(2n, k)$  for some  $k$ .

Equation of the circular part with centre  $M$  is  $(x - 2n)^2 + (y - k)^2 = k^2$

It passes through  $(0, n) \implies (0 - 2n)^2 + (n - k)^2 = k^2 \implies k = \frac{5}{2}n$

Let join of  $M$  and  $D$  intersect  $CB$  at  $Q$  (not in figure.)

So  $MQ = MD - QD = \frac{5}{2}n - n = \frac{3}{2}n$

Let  $\angle QMC = \theta$

In triangle  $MQC$   $\cos\theta = MQ/MC = \frac{\frac{3}{2}n}{\frac{5}{2}n} = 3/5 \implies \theta = \arccos(3/5)$

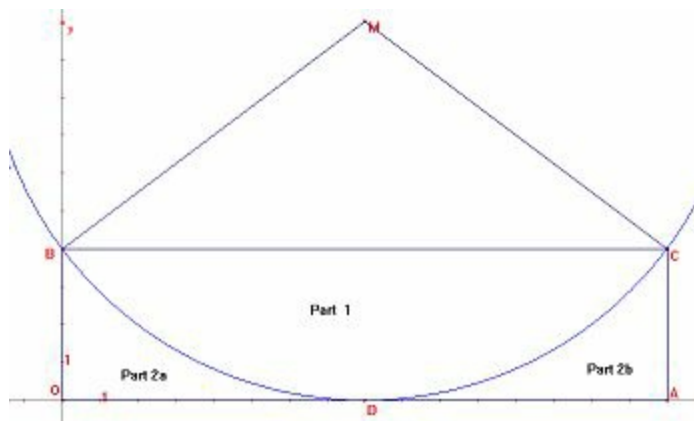
So  $ar.(BDCB) = ar.(BDCMB) - ar.(BMC) = \frac{\arccos(3/5)}{360} \pi k^2 - \frac{1}{2}(4n) \frac{3n}{2}$   
 $= [\frac{\arccos(3/5)}{360} \pi (5/2)^2 - 3] n^2$

$ar.(OBDO) = ar.(DCAO)$  Hence,

$ar.(OBDO) = \frac{1}{2} ar.(OBCA) - ar.(BDCB) = \frac{1}{2} [4 - (\frac{\arccos(3/5)}{360} \pi (5/2)^2 - 3)] n^2$

After calculations

$ar.(OBDO) : ar.(BDCB) : ar.(DCAO) = 1 : \frac{5\pi \arccos(3/5) - 864}{2016 - \arccos(3/5)5\pi} : 1$



168. An operation  $*$  on a set  $G$  is a mapping that associates with every pair of elements  $a$  and  $b$  of the set  $G$ , an unique element  $a * b$  of  $G$ .  $G$  is said to be a *group under operation  $*$* , if the following conditions hold:

- i)  $(a * b) * c = a * (b * c)$  for all elements  $a, b$  and  $c$  of  $G$ ;
- ii) there is an element  $e$  of  $G$  such that  $a * e = e * a = a$  for all elements  $a$  of  $G$  ;and
- iii) for each element  $a$  of  $G$ , there is an element  $a'$  of  $G$  such that  $a * a' = a' * a = e$ .

If  $G$  is the set whose elements are all subsets of a set  $X$ , and  $*$  is the operation of  $G$  defined as  $A * B = (A \cup B) \setminus (A \cap B)$ , show that  $G$  is a group under  $*$ .

(For any two subsets  $C$  and  $D$  of  $X$ ,  $C \setminus D$  denotes the set of all those elements which are in  $C$  but not in  $D$ ).

Ans: i)  $(a * b) * c = (A * B) * C$

Let  $A * B = (A \cup B) \setminus (A \cap B) = H$  Clearly as per question, Venn diagram of  $H$  is as in figure i.

$H * C = (A * B) * C$  Its Venn Diagram is as in figure ii.

Let  $B * C = M$ . Venn Diagram of  $M$  is as in figure iii.

$A * M = A * (B * C)$  Its Venn Diagram is as in figure iv.

Clearly shaded areas of *ii, iv* are identical. Hence,  $(A * B) * C = A * (B * C)$

ii) Let  $\phi$  be the null set. Then

$$A \cup \phi = A, A \cap \phi = \phi$$

So  $A * \phi = A \cup \phi \setminus A \cap \phi = A \setminus \phi = \phi \cup A \setminus \phi \cap A = \phi * A$ , i.e, elements that are not in  $\phi$  but are there in  $A$ , i.e.,  $A$

So there is  $\phi \in X$  such that  $A * \phi = \phi * A = A$

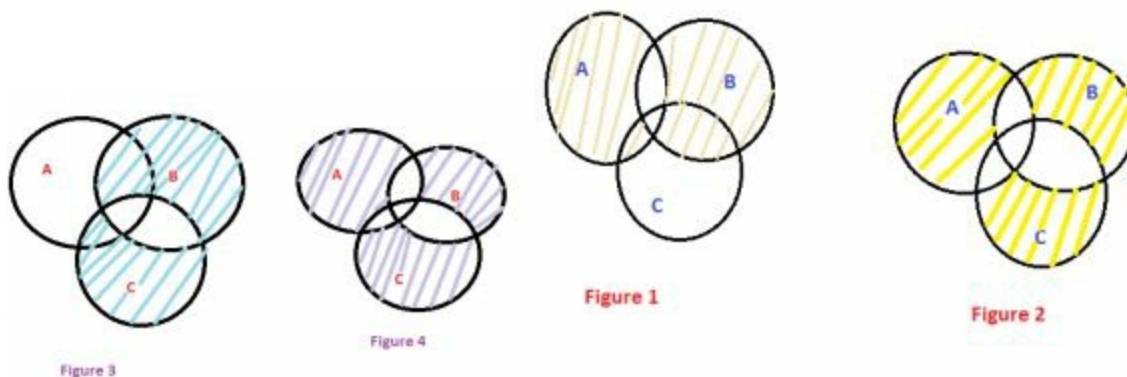
iii) We Have  $A \cup A = A, A \cap A = A$

So  $A * A = A \cup A \setminus A \cap A = A \setminus A$ , i.e. elements that are in  $A$  but does not belong to  $A$ , i.e.,  $\phi$

So for all  $A \in X$ , there is  $A \in X$  such that  $A * A = \phi$ .

So all conditions (i),(ii),(iii) are held.

Hence  $G$  is a group under  $*$ .



169. Show that if a prime number  $p$  is divided by 30, then the remainder is either prime or 1.

Ans: Let  $p = 30q + r$ , then we can see that if  $p < 30$  then remainder  $r = p$  if  $p > 30$ , then  $r \in 1, 2, \dots, 29$ , now clearly  $r \neq 2n, r \neq 3k, r \neq 5m$ , since prime factorisation of  $30 = 2 \times 3 \times 5$  so  $r$  cannot be a multiple of these numbers.

so remaining elements in the set of  $r$  are the prime no.s between 5 to 29 and 1. thus it is proved that if  $p$  is a **prime no.** divided by 30 then we get a remainder either 1 or a **prime no.** .

170. Consider the function

$f(x) = \begin{cases} \sin(\frac{\pi}{4}(x - [x])), & \text{if } [x] \text{ is odd, } x \geq 0 \\ \cos(\frac{\pi}{4}(1 - x + [x])), & \text{if } [x] \text{ is even, } x \geq 0 \end{cases}$  i) Sketch the graph of  $f$  on plain paper.

iii) Determine the point of discontinuities of  $f$  and the points where  $f$  is not differentiable.

Ans: The nature of function repeats after 2 units while starting from a fixed point. Hence it has period 2. Thus just analyzing  $f(x)$  in  $[0, 2]$  would complete our task.

Suppose, in  $[0, 1)$ ,  $f(x) = g(x)$ . We have

$$g(x) = \cos\left(\frac{\pi}{4}(1 - x)\right). \quad g(0) = \frac{1}{\sqrt{2}}. \quad \lim_{x \rightarrow 1^-} g(x) = 1.$$

$$g'(x) = \frac{\pi}{4} \sin\left(\frac{\pi}{4}(1 - x)\right). \quad \text{Hence in } [0, 1) \quad g'(x) > 0. \quad \lim_{x \rightarrow 1^-} g'(x) = 0.$$

$$g''(x) = -\frac{\pi^2}{16} \cos\left(\frac{\pi}{4}(1 - x)\right). \quad \text{Hence in } [0, 1) \quad g''(x) < 0 \text{ implying concavity in } [0, 1).$$

Suppose, in  $[1, 2]$ ,  $f(x) = h(x)$ . We have

$$h(x) = \sin\left(\frac{\pi}{4}(x - 1)\right). \quad h(1) = 0. \quad h(2) = \frac{1}{\sqrt{2}}.$$

$$h'(x) = \frac{\pi}{4} \cos\left(\frac{\pi}{4}(x - 1)\right). \quad \text{Hence in } [1, 2] \quad h'(x) > 0.$$

$$h''(x) = -\frac{\pi^2}{16} \sin\left(\frac{\pi}{4}(x - 1)\right). \quad \text{Hence in } [1, 2] \quad h''(x) < 0 \text{ implying concavity in } [1, 2].$$

$$\lim_{x \rightarrow 2^-} f(x) = \lim_{x \rightarrow 2^-} h(x) = \frac{1}{\sqrt{2}} = \lim_{x \rightarrow 2^+} g(x) = \lim_{x \rightarrow 2^+} f(x) = f(2).$$

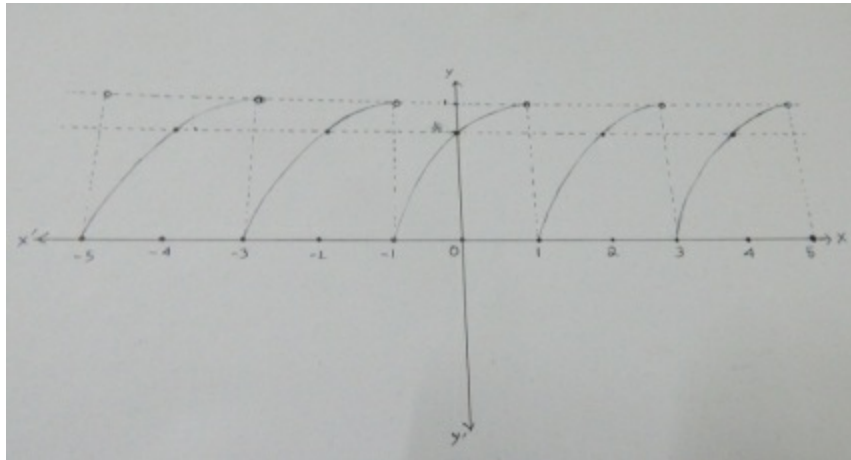
We also see that  $f(x)$  is continuous at  $x = 2$ .

$$\lim_{x \rightarrow 2^-} f'(x) = \lim_{x \rightarrow 2^-} h'(x) = \frac{\pi}{4\sqrt{2}} = \lim_{x \rightarrow 2^+} g'(x) = \lim_{x \rightarrow 2^+} f'(x) = f'(2).$$

Hence  $f(x)$  is differentiable at  $x = 2$ .

It is easily seen that  $f(x)$  is discontinuous and thus obviously non differentiable at  $x = 1$ .

$\therefore$  We conclude  $f(x)$  is discontinuous and non differentiable only at odd integral points.



171. A function  $f$  from set  $A$  into set  $B$  is a rule which assigns each element  $x$  in  $A$ , a unique (one and only one) element (denoted by  $f(x)$ ) in  $B$ . A function of set from  $A$  into  $B$  is called an *onto* function, if for each element  $y$  in  $B$  there is some element  $x$  in  $A$ , such that  $f(x) = y$ . Now suppose that  $A = \{1, 2, 3, \dots, n\}$  and  $B = \{1, 2, 3\}$ . Determine the total number of onto functions of  $A$  into  $B$ .

Ans: We know that the total number of functions from  $A$  to  $B$  are  $3^n$ .

Now, number of functions with exactly two elements in the range is  $3(2^n - 2)$  and the number of functions with exactly one element in range is 3. So by Principle of Inclusion and Exclusion, our answer turns out to be  $3^n - 3(2^n - 1) \dots$

172. If the coefficients of a quadratic equation  $ax^2 + bx + c = 0$ , ( $a \neq 0$ ), are

all odd integers, show that the roots cannot be rational.

Ans: Suppose, if possible, we assume that the equation  $ax^2 + bx + c = 0$   $a \neq 0$  has rational roots.

Clearly then the discriminant would be a perfect square of an integer. The discriminant is  $b^2 - 4ac$ .

We notice that the discriminant is odd. Hence it is a perfect square of an odd integer. We know that any odd perfect square can be expressed in the form of  $8K + 1$  where  $K \in \mathbb{Z}$ .

Since  $b$  is an odd integer, we can write,

$$b^2 = 8n + 1 \text{ where } n \in \mathbb{Z}.$$

Since  $a, c$  are odd,  $ac$  is also odd. We can express  $ac$  in the form of

$$ac = 2m + 1 \text{ where } m \in \mathbb{Z}.$$

Hence,

$$b^2 - 4ac = 8n + 1 - 4(2m + 1) = 8(m + n) - 3 = 8L - 3 \text{ where } L = m + n \text{ and clearly } L \in \mathbb{Z}.$$

Contradiction!

173. i) Prove, from first principles, that,

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta, \text{ for every positive integer } n.$$

ii) Prove that,

$$(\cos\theta + i\sin\theta)^n = \cos(n\theta) + i\sin(n\theta), \text{ for every positive integer } n.$$

Ans: i) Here **1st principles** is actually 1st principle of mathematical induction.

For  $n = 1$  we have:

$$\begin{aligned} & (\cos(\theta) + i\sin(\theta))^1 \\ &= \cos 1\theta + i\sin 1\theta. \end{aligned}$$

So De Moivre holds for  $n = 1$ . Assume now it's true for  $n = k$ . I.e.:

$$(\cos(\theta) + i\sin(\theta))^k = \cos k\theta + i\sin k\theta. \quad (1)$$

We want to show it is also true for  $n = k + 1$ , and then we'll be done.

Consider

then:

$$(\cos(\theta) + i\sin(\theta))^{k+1} = (\cos(\theta) + i\sin(\theta))(\cos(\theta) + i\sin(\theta))^k$$

.

We now use equation (1), and substitute in for  $(\cos(\theta) + i\sin(\theta))^k$

on the right hand

side. Then we get

$$\begin{aligned} & (\cos(\theta) + i\sin(\theta))^{k+1} \\ &= (\cos(\theta) + i\sin(\theta))(\cos(k\theta) + i\sin(k\theta)) \\ &= \cos(\theta)\cos(k\theta) + i^2\sin(\theta)\sin(k\theta) + i\sin(\theta)\cos(k\theta) + i\sin(k\theta)\cos(\theta) \\ &= \cos(\theta)\cos(k\theta) - \sin(\theta)\sin(k\theta) + i(\sin(\theta)\cos(k\theta) + \sin(k\theta)\cos(\theta)) \end{aligned}$$

after expanding the brackets.

Now, as we all know  $\sin(a+b) = \sin a \cos b + \sin b \cos a$  and  $\cos(a+b) = \cos a \cos b - \sin a \sin b$ . So we have

$$(\cos(\theta) + i\sin(\theta))^{k+1} = \cos(k+1)\theta + i\sin(k+1)\theta.$$

Hence we are done by induction.

ii) From Euler's formula, we know:

$$\cos\theta + i\sin\theta = e^{i\theta}$$

.

Taking both sides to the power  $n$  gives us:

$$\begin{aligned} & (\cos\theta + i\sin\theta)^n \\ &= (e^{i\theta})^n \end{aligned}$$

we get:

$$(\cos\theta + i\sin\theta)^n = e^{i(n\theta)}$$

.

We use Euler's formula again:

$$e^{i(n\theta)} = \cos n\theta + i\sin n\theta,$$

and therefore we have De Moivre's Theorem:

$$(\cos\theta + i\sin\theta)^n = \cos n\theta + i\sin n\theta.$$

174. If  $A, B, C$  are the angles of a triangle, then show that

$$\sin A + \sin B - \cos C \leq \frac{3}{2}$$

Ans: We are to show

$$\begin{aligned}\sin A + \sin B - \cos C &\leq \frac{3}{2} \\ \Leftrightarrow 2 \cos \frac{C}{2} \cos \frac{A-B}{2} - 2 \cos^2 \frac{C}{2} + 1 &\leq \frac{3}{2} \\ \Leftrightarrow \cos^2 \frac{C}{2} - \cos \frac{C}{2} \cos \frac{A-B}{2} + \frac{1}{4} &\geq 0\end{aligned}$$

Now,

$$\cos^2 \frac{C}{2} - \cos \frac{C}{2} \cos \frac{A-B}{2} + \frac{1}{4} \geq \cos^2 \frac{C}{2} - \cos \frac{C}{2} + \frac{1}{4} = \left( \cos \frac{C}{2} - \frac{1}{2} \right)^2 \geq 0$$

and hence we are done.

Equality holds when  $C = \frac{2\pi}{3}$  and  $A = B = \frac{\pi}{6}$

175. Let  $a$  and  $b$  be real numbers such that the equations  $2x + 3y = 4$  and  $ax - by = 7$  have exactly one solution. Then, show that the equations  $12x - 8y = 9$  and  $bx + ay = 0$  also have exactly one solution.

Ans:

$$2x + 3y = 4 \dots\dots\dots i \quad ax - by = 7 \dots\dots\dots ii \quad 12x - 8y = 9 \dots\dots\dots iii$$

$$bx + ay = 0 \dots\dots\dots iv$$

$$\text{Condition of } i, ii \text{ to have one solution is } \frac{2}{a} \neq \frac{3}{-b} \implies -2b \neq 3a \Leftrightarrow -8b \neq 12a \implies \frac{b}{12} \neq \frac{a}{-8}$$

which is the condition for  $iii, iv$  to have one solution.

Hence, the equations  $12x - 8y = 9$  and  $bx + ay = 0$  also have exactly one solution.

Geometrically, it is easy to see that  $i$  and  $iii$  are perpendicular and  $ii$  and  $iv$  are perpendicular. Since  $i, ii$  have exactly one solution, they intersect at precisely one point, hence their perpendiculars, i.e.,  $iii, iv$  also intersect at precisely one point, hence have only one solution.

176. Suppose  $k$  and  $n$  are positive integers greater than or equal to 1. Show that  $(nk)!$  is divisible by  $(k!)^n$ .

Ans: And here is my solution:-

We have:-

$(nk)! = \frac{(nk!)}{((n-1)k)!} \cdot \frac{((n-1)k!)}{((n-2)k)!} \cdot \dots \cdot k!$  since cancelling out terms from *RHS* gives *LHS*. There are  $n$  terms on *RHS*. Now, we multiply and divide by  $(k!)^n$

. This gives:-  

$$(nk)! = ((k!)^n) \binom{nk}{k} \cdot \binom{(n-1)k}{k} \dots \binom{k}{k} = ((k!)^n) \cdot L$$
 where  $L$  is a positive integer since the binomial coefficient is a positive integer. Thus  $(nk)! = ((k!)^n)L$ ...which implies that  $(nk)!$  is divisible by  $(k!)^n$ .

177. Let  $n \geq 2$  be an integer. Let  $m$  be the largest integer which is less than or equal to  $n$ , and which is a power of 2. Put  $l_n$  = the least common multiple of  $1, 2, \dots, n$ . Show that,  $l_n/m$  is odd, and that for every integer  $k \leq n$ ,  $k \neq m$ ,  $l_n/k$  is even. Hence prove that

$1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n}$  is not an integer.

Ans: Think of the sequence,  $1, 2, 3, 2^2, \dots, 7, 2^3, 9, \dots, 2^r, \dots, n$

their LCM = product of the all primes which had occurred in the prime factorisations of numbers of above sequence, along with the highest multiplicity with which they have occurred

$= 2^r 3^{a_1} 5^{a_2} \dots [r, a_1, a_2, \dots \text{ are positive integers}]$

$= 2^r s$  where  $s$  is odd.

So  $l_n = ms$  [by question  $\text{LCM} = l_n, m = 2^r$ ]

or  $l_n/m = s$  which is odd.

Again  $l_n/k = \frac{2^r s}{k}$  but  $k \neq 2^r = m$   
 $l_n/k = 2^{r-h} g [h = 0, 1, \dots, r-1 \text{ and } g \text{ is integral}]$

So  $l_n/k$  is even.

$$H = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \frac{2^r s + 2^{r-1} s + 2^r (s/3) + \dots + s + \dots}{2^r s}$$

All numbers in numerator are even except  $s$ . Hence  $H = \frac{\text{odd}}{\text{even}}$

So  $H$  cannot be integral.

178. Suppose  $f(x)$  is a continuous function such that  $f(x) = \int_0^x f(t) dt$ . Prove that  $f(x)$  is identically equal to zero.

Ans:

$$f(x) = \int_0^x f(t) dt$$

differentiate to get  $f' = f$  so either  $f' = f = 0$  as required or

$$(f'/f) dx = 1 dx$$

integrating from 0 to  $x$ ,  $\log(f(x)/f(0)) = x$  or  $f(x) = f(0) * e^x$

but

$$\text{we had } f(x) = \int_0^x f(t) dt$$

$$\text{hence } f(0) = \int_0^0 f(t) dt = 0$$

hence putting this in above we have  $f(x) = 0$

179. Show that, number of ways one can choose a distinct positive integers, each smaller than or equal to 50 such that their sum is odd, is  $2^{49}$ .

Ans: The given condition is possible if we choose an odd number of odd integers, with any number of even integers, less than 50

Now odd number of odd integers less than 50 can be chosen in  $25C1 + 25C3 + \dots + 25C25$  ways as there are 25 odd numbers less than 50

Along with them, no even integer can be chosen in  $25C0$  ways, one in  $25C1$  ways, ..., 25 in  $25C25$  ways.

**Required number=**

$$25C0(25C1 + 25C3 + \dots + 25C25) + 25C1(25C1 + \dots + 25C25) + \dots + 25C25(25C1 + \dots + 25C25)$$

$$= (25C0 + 25C1 + \dots + 25C25)(25C1 + 25C3 + \dots + 25C25)$$

$$= \boxed{2^{49}} \text{ [Proved]}$$

180. Let  $P(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0$  be a polynomial with integer coefficients, such that,  $P(0)$  and  $P(1)$  are odd integers. Show that:

(a)  $P(x)$  does not have any even integer roots.

(b)  $P(x)$  does not have any odd integer roots. Let

$P(x) = x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x + a_0$  be a polynomial with integer coefficients, such that,  $P(0)$  and  $P(1)$  are odd integers. Show that:

(a)  $P(x)$  does not have any even integer roots.

(b)  $P(x)$  does not have any odd integer roots.

Ans: a) Clearly,  $P(0) = a_0$  is odd. Now for even integer  $x$  we have

$x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x$  is even. Also since  $a_0$  is odd, we have  $P(x)$  is odd  $\forall$  even  $x$ . Hence  $P(x) \neq 0 \forall$  even  $x$ .

$\therefore P(x)$  does not have any even integer roots.

(b) We have  $a_0$  is odd and  $1 + a_{n-1} + a_{n-2} + \dots + a_1 + a_0$  is odd. Thus  $1 + a_{n-1} + a_{n-2} + \dots + a_1$  is even. For odd  $x$  we know that  $x^n, x^{n-1}, x^{n-2}, \dots, x$  are odd. We also know that if we multiply an integer by an odd integer, the parity of the original integer does not change. Hence  $a_i x^i$  has the same parity as  $a_i$  for  $i = 1, 2, 3, \dots, n-1$ . Thus  $x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_1x$  has the same parity as  $1 + a_{n-1} + a_{n-2} + \dots + a_1 \forall$  odd  $x$  which is even. Hence  $P(x) - a_0$  is even for all odd  $x$  and therefore  $P(x)$  is odd. Hence it does not have any odd integer roots.

181. If  $a, b, c$  are positive numbers, show that

$$\frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} + \frac{a^2 + b^2}{a + b} \geq a + b + c$$

Ans: We have  $(b - c)^2 \geq 0$  or  $b^2 + c^2 \geq 2bc$  or  $2(b^2 + c^2) \geq 2bc + b^2 + c^2 = (b + c)^2$

$$\text{so } \frac{2(b^2 + c^2)}{(b + c)} \geq b + c$$

applying similar inequalities for pairs  $c, a$  and  $a, b$  and adding and simplifying we get

$$\frac{b^2 + c^2}{b + c} + \frac{c^2 + a^2}{c + a} + \frac{a^2 + b^2}{a + b} \geq a + b + c$$

182. Find the set of all values of  $m$  such that  $y = \frac{x^2 - x}{1 - mx}$  can take all real values.

Ans:

$$y = \frac{x^2 - x}{1 - mx}$$

$$\Leftrightarrow y - myx = x^2 - x$$

$$\Leftrightarrow x^2 + x(my - 1) - y = 0$$

$$\Leftrightarrow x = \frac{1 - my \pm \sqrt{(my - 1)^2 + 4y}}{2}$$

Now  $y$  takes all real values if discriminant  $(my - 1)^2 + 4y$  is always  $> 0$ .

So now we have to find the all values of  $\{m\}$  such that  $(my - 1)^2 + 4y > 0$  for all  $y \in \mathbb{R}$ .

$$\Leftrightarrow m^2 y^2 - 2my + 1 + 4y > 0$$

$$\Leftrightarrow m^2 y^2 + y(4 - 2m) + 1 > 0 \dots (i)$$

Now this is a equation of upside open parabola. If the discriminant is  $\leq 0$  of equation (i) then  $(my - 1)^2 + 4y$  will always positive.

$$\Rightarrow (4 - 2m)^2 - 4m^2 \leq 0$$

$$\Rightarrow 16m + 16 \leq 0$$

$$\Rightarrow m + 1 \leq 0$$

$$\Rightarrow m \leq -1$$

Conclusion: If  $m \leq -1$ , then  $y = \frac{x^2 - x}{1 - mx}$  can take all the values as  $x$  varies over  $\mathbb{R}$

183. Consider the function  $f(t) = e^{-\frac{1}{t}}, t > 0$ . Let for each positive integer  $n$ ,  $P_n$  be the polynomial such that  $\frac{d^n}{dt^n} f(t) = P_n\left(\frac{1}{t}\right) e^{-\frac{1}{t}}$  for all  $t > 0$ . Show that  $P_{n+1}(x) = x^2(P_n(x) - \frac{d}{dx} P_n(x))$

Ans:

$$\begin{aligned}\frac{d^{n+1}}{dt^{n+1}} f(t) &= \frac{d}{dt} \left\{ P_n\left(\frac{1}{t}\right) e^{-\frac{1}{t}} \right\} \\ \Rightarrow P_{n+1}\left(\frac{1}{t}\right) e^{-\frac{1}{t}} &= e^{-\frac{1}{t}} \frac{d}{dt} P_n\left(\frac{1}{t}\right) + \frac{1}{t^2} P_n\left(\frac{1}{t}\right) e^{-\frac{1}{t}} \\ \Rightarrow P_{n+1}\left(\frac{1}{t}\right) &= \frac{d}{dt} P_n\left(\frac{1}{t}\right) + \frac{1}{t^2} P_n\left(\frac{1}{t}\right)\end{aligned}$$

Replace  $x = \frac{1}{t} \Rightarrow \frac{dx}{dt} = -\frac{1}{t^2} = -x^2$

$$\begin{aligned}\therefore P_{n+1}(x) &= \frac{d}{dx} P_n(x) - x^2 + x^2 P_n(x) \\ &= x^2 \left( P_n(x) - \frac{d}{dx} P_n(x) \right)\end{aligned}$$

184. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a differentiable function such that its derivative  $f'$  is a continuous function. Moreover, assume that for all  $x \in \mathbb{R}$ ,  $0 \leq |f'(x)| \leq \frac{1}{2}$ . Define a sequence of real numbers  $\{a_n\}_{n \in \mathbb{N}}$  by :  $a_1 = 1$  and  $a_{n+1} = f(a_n)$  for all  $n \in \mathbb{N}$ . Prove that there exists a positive real number  $M$  such that for all  $n \in \mathbb{N}$ ,  $|a_n| \leq M$ .

Ans. Notice that

$$\begin{aligned}|a_n - a_{n-1}| &= |f(a_{n-1}) - f(a_{n-2})| = |f'(\xi)| |a_{n-1} - a_{n-2}| \leq \frac{1}{2} |a_{n-1} - a_{n-2}|, \text{ where} \\ n &= 3, 4, \dots, \text{ and } a_{n-1} \geq \xi \geq a_{n-2}. \text{ Hence,} \\ |a_n - a_{n-1}| &\leq \frac{1}{2} |a_{n-1} - a_{n-2}| \leq \frac{1}{2^2} |a_{n-2} - a_{n-3}| \leq \dots \leq \frac{1}{2^{n-2}} |a_2 - a_1|. \text{ Therefore,}\end{aligned}$$

$$\begin{aligned}
|a_n - a_1| &= |(a_n - a_{n-1}) + (a_{n-1} - a_{n-2}) + \cdots + (a_2 - a_1)| \\
&\leq |a_n - a_{n-1}| + |a_{n-1} - a_{n-2}| + \cdots + |a_2 - a_1| \\
&\leq \left(1 + \frac{1}{2} + \frac{1}{2^2} + \cdots + \frac{1}{2^{n-2}}\right) |a_2 - a_1| \\
&\leq 2|a_2 - a_1|.
\end{aligned}$$

Hence,

$|a_n| \leq 2|a_2 - a_1| + |a_1|$  for  $n = 3, 4, \dots$ . But we can verify that it also holds for  $n = 1, 2$ . Now, take  $M = 2|a_2 - a_1| + |a_1|$ . We obtain  $|a_n| \leq M$  for  $n = 1, 2, \dots$ . This is exactly what we want to prove

185. Let  $f : \mathbb{R} \rightarrow \mathbb{R}$  be a continuous function such that for all  $x \in \mathbb{R}$  and for all  $t \geq 0$ ,  $f(x) = f(e^t x)$ . Show that  $f$  is a constant function

Ans. First choose  $a > 0$ , then  $f$  is constant on  $[a, \infty)$ . Let  $a \downarrow 0$  to see that  $f$  is constant on  $\mathbb{R}^+$ . By continuity  $f$  is constant on  $[0, \infty)$ . In the same way, with  $a < 0$  one shows that  $f$  is constant on  $(-\infty, 0]$ . Hence  $f$  is constant.

186. Let  $n \geq 3$ . Let  $A = ((a_{ij}))_{1 \leq i, j \leq n}$  be an  $n \times n$  matrix such that  $a_{ij} \in \{-1, 1\}$  for all  $1 \leq i, j \leq n$ . Suppose that  $a_{k1} = 1$  for all  $1 \leq k \leq n$  and

$$\sum_{k=1}^n a_{ki} a_{kj} = 0 \quad \text{for all } i \neq j.$$

Show that  $n$  is a multiple of 4.

Ans. First note that

$0 = \sum_{k=1}^n a_{k1} a_{k2} = \sum_{k=1}^n a_{k2}$  Since the summands are  $\pm 1$ , so half of them should be  $-1$  and half of them are  $+1$ . This proves that  $n$  is even. Say  $n = 2m$ .

So from  $0 = \sum_{k=1}^n a_{kj} \quad \forall j \geq 2$

we can say that  $\prod_{k=1}^n a_{kj} = (-1)^m \forall j \geq 2$ , since half ( $m$ ) of the summands are  $(-1)$ .

Similarly, from  $0 = \sum_{k=1}^n a_{k2} a_{k3}$  we can say  $\prod_{k=1}^n a_{k2} a_{k3} = (-1)^m$  But  
 $\prod_{k=1}^n a_{k2} a_{k3} = \prod_{k=1}^n a_{k2} \prod_{k=1}^n a_{k3} = (-1)^m (-1)^m = 1$  Hence we get that  $(-1)^m = 1$   
 which completes the proof.

187. Let  $f : [0, 1] \rightarrow \mathbb{R}$  be a real-valued continuous function which is differentiable on  $(0, 1)$  and satisfies  $f(0) = 0$ . Suppose there exists a constant  $c \in (0, 1)$  such that  $|f'(x)| \leq c|f(x)|$  for all  $x \in (0, 1)$ . Show that  $f(x) = 0$  for all  $x \in [0, 1]$

Ans.

$|f'(x)| \leq c|f(x)| = c \left| \int_0^x f'(t) dt \right| \leq c \int_0^x |f'(t)| dt \leq c \int_0^x c|f(t)| dt$ . Now,  $f$  is continuous in  $[0, 1]$  so there exists  $M > 0$  such that  $|f(t)| \leq M$  for all  $t \in [0, 1]$ .

This gives us  $|f'(x)| \leq c^2 M \int_0^x dt = c^2 M x$ .

Again,  $|f(x)| = \left| \int_0^x f'(t) dt \right| \leq \int_0^x |f'(t)| dt \leq \int_0^x c^2 M t dt = M(c^2 x)^2 / 2$ . Giving

$|f'(x)| \leq c M(c^2 x)^2 / 2$ . Inductively, we get

$|f'(x)| \leq c M \frac{(cx)^n}{n!}$ ,  $|f(x)| \leq M \frac{(cx)^n}{n!} \forall n \geq 1, x \in [0, 1]$  Finally, from

$|f(x)| \leq M \frac{(cx)^n}{n!} \leq M \frac{1}{n!}$  we get, by letting  $n \rightarrow \infty$  that  $f(x) = 0$  for all  $x \in [0, 1]$ .

188. Consider all permutations of the integers  $1, 2, \dots, 100$ . In how many of

these permutations will the 25th number be the minimum of the first 25 numbers and the 50th number be the minimum of the first 50 numbers?

Ans. Let us count all such permutations  $\pi$ . We can choose freely  $\pi(100), \pi(99), \dots, \pi(51)$ , and then for  $\pi(50)$  we are forced to choose the minimum of the remaining 50 images for  $\pi(50)$ , then again we can choose freely  $\pi(49), \pi(48), \dots, \pi(26)$  and again we are forced to choose the minimum of the remaining images for  $\pi(25)$ , then we can freely choose the remaining images  $\pi(24), \dots, \pi(1)$ . This gives a total of

$$100 \times 99 \times \dots \times 51 \times 49 \times \dots \times 26 \times 24 \dots \times 1 = \frac{100!}{25 \times 50} = \frac{100!}{1250}$$

189. A urn contain  $r$  red balls and  $b$  black balls. A ball is drawn from urn and it's colour is noted, returned to the urn. Further  $c > 0$  additional balls are added to the urn of same colour. This process is continued. Define  $X_i = 1$ , if at the  $i$ th drawn ball is red, 0 otherwise. compute  $E(\sum_{i=1}^n X_i)$ .

Ans. This is called Polya's Urn Scheme. Observe that  $E(X_i) = P(X_i = 1)$

. Claim:  $P(X_i = 1) = \frac{r}{r+b} \forall i = 1, 2, 3, \dots$ . For  $i = 1$ , this is trivial. For  $i = 2$ ,  
 $P(X_2 = 1) = P(X_1 = 1).P(X_2 = 1|X_1 = 1) + P(X_1 = 0).P(X_2 = 1|X_1 = 0) = \frac{r}{r+b} \frac{r+c}{r+b+c} + \frac{b}{r+b} \frac{r}{r+b+c} = \frac{r}{r+b}$ . If

this is true for some  $i > 2$ , then

$$P(X_{i+1} = 1) = P(X_i = 1).P(X_{i+1} = 1|X_i = 1) + P(X_i = 0).P(X_{i+1} = 1|X_i = 0)$$

will imply the same for  $i + 1$ . Now, using linearity of expectation,  $E(\sum X_i) = \sum E(X_i) = \frac{nr}{r+b}$

190. Determine all  $n \in \mathbb{N}$  for which the equation  $x^3 + y^3 + z^3 = n + 3xyz$  have not solutions in  $\mathbb{Z}$ .

Ans BY AM-GM

$$x^3 + y^3 + z^3 \geq 3xyz.$$

This implies  $n + 3xyz \geq 3xyz$

$$n \geq 0$$

So all  $n \in N$ .